Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

3130 FOUNDATION DESIGN OF CONVEYOR 3130-FDR-003 & FDR-004

DESIGN CALCULATION

AS BUILT

MD-216-3130-EG-CV-CAL-1004

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
D01	01-11-2023	AS BUILT	APL	SBY	PRI	
C01	29-04-2020	ISSUED FOR CONSTRUCTION	APL	SBY	PRI	
A01	04-04-2020	FIRST ISSUE	APL	SBY	PRI	

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Ma'aden Document Number:

MD-216-3130-EG-CV-CAL-1004 D01

Outotec Document Number:

900707 GAU01 ZC01 CED01 31304 OU100668041 2

L&T Document Number:

O19045 100 C K 24058 007 2

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Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	04-04-2020	29-04-2020	01-11-2023				Date	04-04-2020	29-04-2020	01-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTIO	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTIO	AS BUILT		
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2007	Saudi Code for Concrete Structures

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
Sub-Supplier	:	A sub entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
QA	:	Quality Assurance
QC	:	Quality Control
Works	:	Full scope of the EPC Contract including the engineering, procurement, construction, commissioning and associated activities
RL	:	Reduced Level
C.L.	:	Centre Line
CJ	:	Construction Joint
FDN	:	Foundation
TOC	:	Top of concrete
FGL	:	Finished Ground Level
FFL	:	Finished Floor Level
BOF	:	Bottom of footing
TYP	:	Typical
TOC	:	Top of concrete
IP	:	Insert Plate
RCC	:	Reinforced Cement Concrete
MKD.	:	Marked
THK.	:	Thick

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1. INTENT

The document is prepared for Concrete design calculation of Conveyors 3130-FDR-003 & FDR-004.
Conveyor 3130-FDR-003 is between Tertiary Crusher House and TP-4. Conveyor 3130-FDR-004 is between Screen building and TP-3. Belt width of these conveyors is 1200 mm.

2. EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.
MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.
The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

This document is prepared for foundation design of Conveyor 3130-CNV-001. All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
The column and beam reinforcement design have been performed in Staad-Pro Software. The foundation of Ground Portal GR-1 to GR-4; TR-4 & VGTU have been performed in MS Excel.
Foundation of TR-1, TR-2 & TR-3 has been designed by FEM analysis in Staad-Pro Software.

4. CONCRETE STRUCTURE

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304 and ACI 318M.

4.2 Materials

Concrete Strength of 28 days
· Foundation and Structural Members: 35 MPa
· Grade slab and pavement: 30 MPa
· Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade
· Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
· Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

4.3 Foundation for Structures

Foundation shall be Isolated and combined Shallow Concrete foundation. The design calculations are prepared as per recommendation of standard SBC-304 Clause 13.3.

4.4 Minimum Concrete Cover

Items	Minimum Cover(mm)
Foundation (Side & Bottom)	75
Foundation (Top)	50
Column & Pedestal	50
Beam	50
Slab	40

4.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 500 kN/m2 at 1m depth below FGL (ie. RL 911.500m).
Refer Table 4 - BH-17,18,19,25,29 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

4.6 Reference

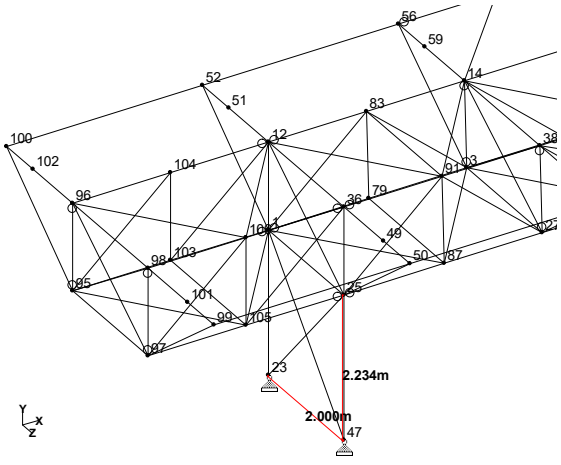
Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

5 INPUT LOADS FROM SUPERSTRUCTURE

For load calculation of superstructure refer Document No: MD-216-3130-EG-ST-CAL-1004-A01

5.1 Support Reaction from TR-1

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
23	1 DL	0.00	61.77	4.36
	2 LL	0.00	33.86	3.94
	3 ML - CONVEYOR MATERIAL LO/	0.00	23.90	2.52
	4 WL (+Z)	0.00	-38.55	-11.62
	5 WL (-Z)	0.00	38.55	11.61
	6 TL1	0.00	-0.07	-22.40
	7 TL2	0.00	0.07	22.40
	8 EARTHQUAKE LOAD (+X)	-1.96	-0.81	-0.11
	9 EARTHQUAKE LOAD (+Z)	0.00	-4.10	-1.16
47	1 DL	0.00	55.74	-10.69
	2 LL	0.00	34.15	-3.94
	3 ML - CONVEYOR MATERIAL LO/	0.00	24.10	-2.52
	4 WL (+Z)	0.00	38.55	-12.18
	5 WL (-Z)	0.00	-38.55	12.18
	6 TL1	0.00	0.07	22.40
	7 TL2	0.00	-0.07	-22.40
	8 EARTHQUAKE LOAD (+X)	-1.96	-0.78	0.08
	9 EARTHQUAKE LOAD (+Z)	0.00	4.10	-1.21



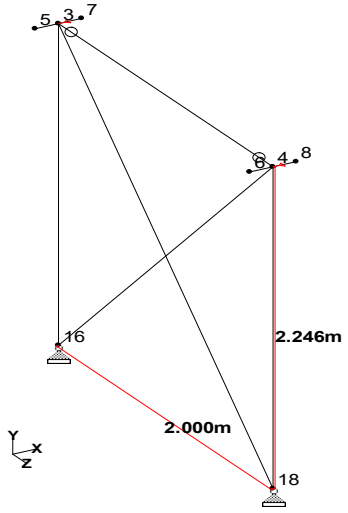
Staad model of (9+3) m Gallery mkd. GR-1 End portal which is acting as TR1

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5.2 Support Reaction from TR-2

Node	L/C	Horizontal Fx kN	Vertical Fy kN	Horizontal Fz kN
16	1 DEAD LOAD	0.29	70.07	2.55
	2 LIVE LOAD	0.22	32.10	2.46
	3 MATERIAL LOAD	0.13	25.60	1.55
	4 WL (+Z)	-0.03	-46.65	-13.25
	5 WL (-Z)	0.03	47.67	14.10
	6 WL (+X)	-0.56	0.00	0.00
	7 WL (-X)	0.56	0.00	0.00
	8 TL1	-0.03	-0.50	6.39
	9 TL2	0.03	0.50	-6.39
	10 EL (+X)	-0.06	0.50	0.04
	11 EL (+Z)	0.00	-8.88	-2.66
18	1 DEAD LOAD	0.34	57.70	-9.55
	2 LIVE LOAD	0.23	31.90	-2.46
	3 MATERIAL LOAD	0.15	25.40	-1.55
	4 WL (+Z)	0.03	46.65	-13.52
	5 WL (-Z)	-0.03	-47.67	14.39
	6 WL (+X)	-0.56	0.00	0.00
	7 WL (-X)	0.56	0.00	0.00
	8 TL1	0.03	0.50	-6.39
	9 TL2	-0.03	-0.50	6.39
	10 EL (+X)	-0.06	0.50	-0.04
	11 EL (+Z)	0.00	8.88	-2.76



Staad model of TR2 superstructure (along with dimension)

5.3 Load Combinations for Design of combined footing

5.3.1 Load Combinations for Base pressure calculation

(As per Section 2.4 of SBC 301 -2007)

Load Cases	Reference
LOAD COMB 301 1.0 DL	(Eq. 2.4.1-1 of SBC 301)
LOAD COMB 302 1.0 (DL + TL1) + 1.0 (LL+ML)	(Eq. 2.4.1-2 of SBC 301)
LOAD COMB 303 1.0 (DL + TL1) + 1.0 (LL+ML)	
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75 (LL+ML)	(Eq. 2.4.1-6 of SBC 301)
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75 (LL+ML)	
LOAD COMB 306 1.0 DL+ 0.75 WL(+X) + 0.75 (LL+ML)	
LOAD COMB 307 1.0 DL+ 0.75 WL(-X) + 0.75 (LL+ML)	
LOAD COMB 308 1.0 DL + 0.525 EL(+X) + 0.75 (LL+ML)	
LOAD COMB 309 1.0 DL + 0.525 EL(-X) + 0.75 (LL+ML)	
LOAD COMB 310 1.0 DL + 0.525 EL(+Z) + 0.75 (LL+ML)	
LOAD COMB 311 1.0 DL + 0.525 EL(-Z) + 0.75 (LL+ML)	
LOAD COMB 312 0.6 DL + 1.0 WL(+Z)	(Eq. 2.4.1-7 of SBC 301)
LOAD COMB 313 0.6 DL + 1.0 WL(-Z)	
LOAD COMB 314 0.6 DL + 1.0 WL(+X)	
LOAD COMB 315 0.6 DL + 1.0 WL(-X)	
LOAD COMB 316 0.6 DL + 0.7 EL(+X)	(Eq. 2.4.1-8 of SBC 301)
LOAD COMB 317 0.6 DL + 0.7 EL(-X)	
LOAD COMB 318 0.6 DL + 0.7 EL(+Z)	
LOAD COMB 319 0.6 DL + 0.7 EL(-Z)	

5.3.2 Load Combinations for Concrete design

Load Case	Reference
LOAD COMB 401 1.4 DL	(Eq. 2.3.2-1 of SBC 301)
LOAD COMB 402 1.4 (DL + TL1) + 1.7 (LL+ML)	(Eq. 2.3.2-2 of SBC 301)
LOAD COMB 403 1.4 (DL + TL1) + 1.7 (LL+ML)	
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5 (LL+ML)	(Eq. 2.3.2-4 of SBC 301)
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5 (LL+ML)	
LOAD COMB 406 1.2 DL+ 1.6 WL(+X) + 0.5 (LL+ML)	
LOAD COMB 407 1.2 DL+ 1.6 WL(-X) + 0.5 (LL+ML)	
LOAD COMB 408 1.2 DL + 1.0 EL(+X) + 0.5 (LL+ML)	(Eq. 2.3.2-5 of SBC 301)
LOAD COMB 409 1.2 DL + 1.0 EL(-X) + 0.5 (LL+ML)	
LOAD COMB 410 1.2 DL + 1.0 EL(+Z) + 0.5 (LL+ML)	
LOAD COMB 411 1.2 DL + 1.0 EL(-Z) + 0.5 (LL+ML)	
LOAD COMB 412 0.9 DL + 1.6 WL(+Z)	(Eq. 2.3.2-6 of SBC 301)
LOAD COMB 413 0.9 DL + 1.6 WL(-Z)	
LOAD COMB 414 0.9 DL + 1.6 WL(+X)	
LOAD COMB 415 0.9 DL + 1.6 WL(-X)	
LOAD COMB 416 0.9 DL + 1.0 EL(+X)	(Eq. 2.3.2-7 of SBC 301)
LOAD COMB 417 0.9 DL + 1.0 EL(-X)	
LOAD COMB 418 0.9 DL + 1.0 EL(+Z)	
LOAD COMB 419 0.9 DL + 1.0 EL(-Z)	

5.4 Load Combinations for Isolated footing of Ground Portals

5.4.1 Load Combination for Base pressure calculation

(As per Section 2.4 of SBC 301 -2007)

Load Cases	Reference
LOAD COMB 301 1.0 DL	(Eq. 2.4.1-1 of SBC 301)
LOAD COMB 302 1.0 (DL + TL1) + 1.0 (LL+ML)	(Eq. 2.4.1-2 of SBC 301)
LOAD COMB 303 1.0 (DL + TL1) + 1.0 (LL+ML)	
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75 (LL+ML)	(Eq. 2.4.1-6 of SBC 301)
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75 (LL+ML)	
LOAD COMB 306 0.6 DL + 1.0 WL(+Z)	(Eq. 2.4.1-7 of SBC 301)
LOAD COMB 307 0.6 DL + 1.0 WL(-Z)	

5.4.2 Load Combination for Concrete design

Load Case	Reference
LOAD COMB 401 1.4 DL	(Eq. 2.3.2-1 of SBC 301)
LOAD COMB 402 1.4 (DL + TL1) + 1.7 (LL+ML)	(Eq. 2.3.2-2 of SBC 301)
LOAD COMB 403 1.4 (DL + TL1) + 1.7 (LL+ML)	
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5 (LL+ML)	(Eq. 2.3.2-4 of SBC 301)
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5 (LL+ML)	
LOAD COMB 406 0.9 DL + 1.6 WL(+Z)	(Eq. 2.3.2-6 of SBC 301)
LOAD COMB 407 0.9 DL + 1.6 WL(-Z)	

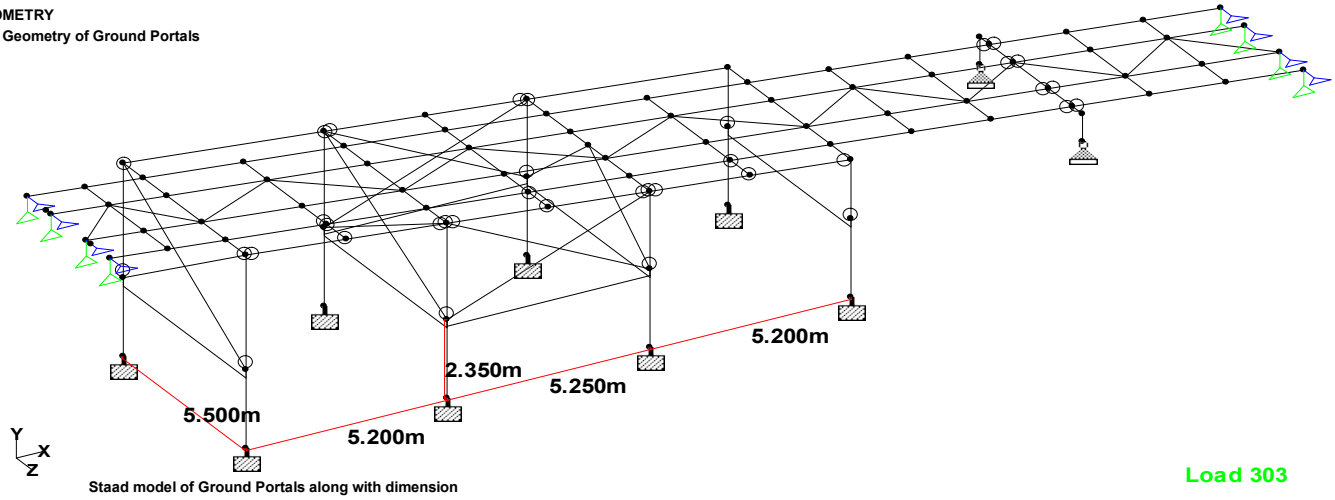
Eq. 2.3.2-3 of SBC 301 - 2007 has not been considered as it would not produce worst load combination in the absence of Roof Live Load & Rain Load

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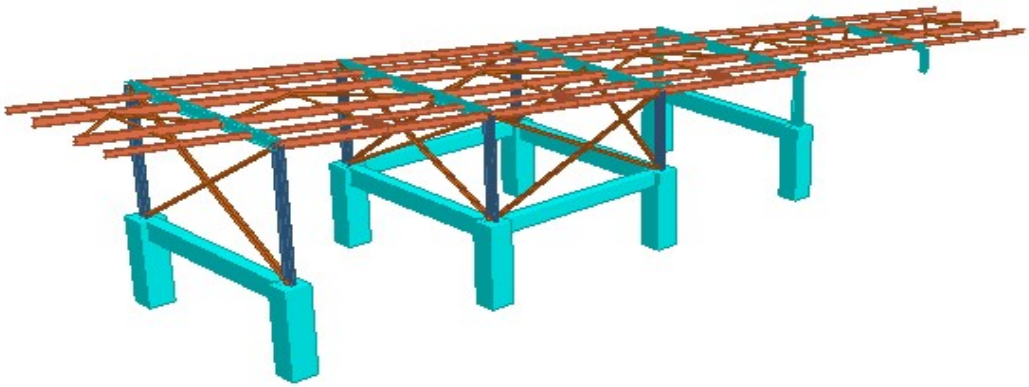
6 GEOMETRY

6.1 Geometry of Ground Portals

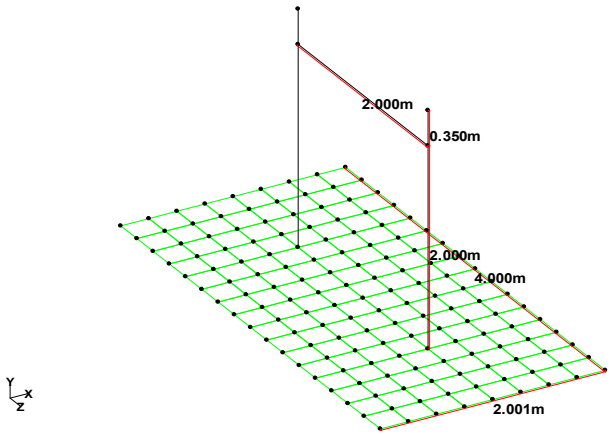


Staad model of Ground Portals along with dimension

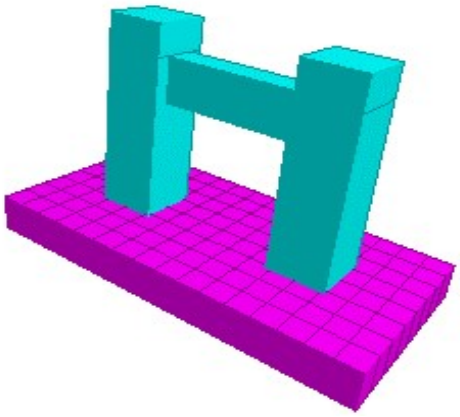
Load 303



6.2 Geometry of combined footing mkd. F2 for trestle mkd. TR-1 & TR-2 of Conveyor 3130-FDR-004.



Staad Model of combined footing mkd. F2



Isometric view of combined footing mkd. F2

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7. DESIGN OF CONCRETE COMPONENTS

All foundations have been designed using worksheet in accordance to SBC-304.

All beams & columns have been designed in Staad-Pro by ACI 318M.

For Design Results, Refer Attachemnt 3 & 4 - Staad Output file.

7.1 Design Summary of Column

7.1.1 Column Reinforcement Summary

Column mark	Width (Along conv)	Depth (Across conv)	Reinforcement	Stirrup	Remarks	Numbers
P1	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP1 of 3130-FDR-003	2
P2	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP2 & GP3 of 3130-FDR-003	4
P3	600	600	12 No.- D20	3 Nos D10 @ 200c/c	GP4 of 3130-FDR-003	2
P3	600	600	12 No.- D20	3 Nos D10 @ 200c/c	TR1 of 3130-FDR-004	2
P3	600	600	12 No.- D20	3 Nos D10 @ 200c/c	TR2 of 3130-FDR-004	2

7.2 Design Summary of Beam

7.2.1 Beam Reinforcement Summary

Beam mark	Width (mm)	Depth (mm)	Top Reinforcement	Bottom Reinforcement	Stirrup	Numbers
TB1	300	500	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	4
TB2	300	400	3 Nos D12	3 Nos D12	2L D8 STRPS @ 200c/c	2
TB3	300	500	3 Nos D16	3 Nos D16	2L D10 STRPS @ 200c/c	2

7.3 Design of Foundation

7.3.1 Foundation marking, Max. Base pressure and Reinforcement Summary

Structure Name	Foundation mark	Length (Along conv) (mm)	Width (Across conv) (mm)	Thickness (mm)	Max pressure (kN/m^2)	Top reinforcement	Bottom reinforcement	F.O.S.	Numbers
GP1 of 3130-FDR-003	F1	1500	2000	300	83	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.16	2
GP2 of 3130-FDR-003	**F1**	**1500**	**2000**	**300**	**128**	**D12 @ 200c/c B/W**	**D16 @ 200c/c B/W**	**0.25**	**2**
GP3 of 3130-FDR-003	F1	1500	2000	300	115	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.22	2
GP4 of 3130-FDR-003	F1	1500	2000	300	89	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.17	2
TR1 of 3130-FDR-004	F2	1500	2000	300	88	D12 @ 200c/c B/W	D16 @ 200c/c B/W	0.17	1
TR2 of 3130-FDR-004	**F2**	**2000**	**4000**	**400**	**91**	**D12 @ 200c/c B/W**	**D16 @ 200c/c B/W**	**0.18**	**1**

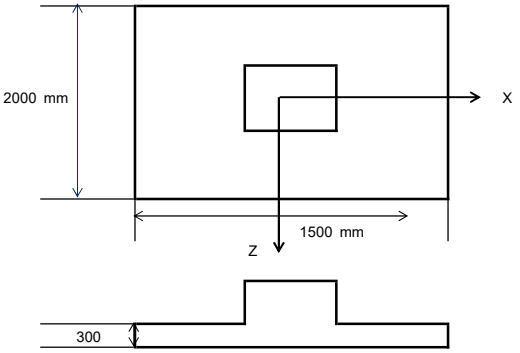
The design of the most critical foundation of the group is attached in the following pages.

Global Project ID	900707
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Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T	0	1	2
Date	03-04-2020	29-04-2020	01-11-2023	Date	03-04-2020	29-04-2020	01-11-2023
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Checked	SBY	SBY	SBY	Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI

7.3.2 Design of foundation F1 for supporting trestle mkd. GP1 to GP4

Finished Ground level	F.G.L	=	-0.500	m	Staad Node no	=	179 & 180	*inputs
Bottom of foundation	B.O.F	=	-2.500	m	Weight of soil	=	80.78 KN	
Staad support reaction level		=	-2.500	m	Weight of foundation	=	21.60 KN	
Ground Water Level	G.W.L	=	-10.000	m	Uplift	=	0.00 KN	
Surcharge	Ws	=	0.000	KN/m^2	Surcharge	=	0 KN	
Length of footing	Lz	=	2.00	m	Pressure on top for tension	=	34.1 KN/m^2	
Width of footing	Lx	=	1.50	m				
Thickness of footing	D	=	0.30	m				
Depth of pedestal	Bx	=	0.60	m				
Width of pedestal	Bz	=	0.60	m				
Height of Load point from bottom of footing	H	=	0.000	m				
Section Modulus of Pedestal	Zxx	=	1.00	m^3				
Section Modulus of Pedestal	Zzz	=	0.75	m^3				
Allowable Net Bearing Capacity		=	500	KN/m^2				
Gross Bearing Capacity		=	536	KN/m^2				
Grade of concrete		=	35	N/mm^2				
Grade of steel		=	420	N/mm^2				
Clear cover		=	75	mm				
Top Reinfocement		=	12	mm				
Bottom Reinforcement		=	16	mm				
Co-efficient of friction between soil & concrete		=	0.3					
Unit wt of soil		=	18	KN/m^3				
Unit wt of concrete		=	24	KN/m^3				



$$\beta_1 = 0.85 - \frac{0.05(f'_c - 28)}{7}$$

β_1 = 0.8 *Table 22.2.2.4.3 SBC 304*

CHECK FOR BEARING PRESSURE CAPACITY

L/C	Fx	Fy	Fz	Mx	My	Mz	Sum Fy	Pmax	Pmin	B.C Factor	F.O.S B.P
301 1.0 DL	13.8	101.518	6.6	8.03	-0.1	-22.1	203.9	105.45	30.48	1	0.20
302 1.0 (DL + TL1) +	11.9	136.46	7.4	14.91	-0.266	-24.9	238.8	127.66	31.57	1	0.24
303 1.0 (DL + TL2) +	15.6	130.65	1.7	4.15	0.014	-19.8	233.0	108.24	47.12	1	0.20
304 1.0 DL+ 0.75 W	13.7	130.342	9.9	17.92	-0.142	-22.2	232.7	125.13	30.02	1	0.23
305 1.0 DL+ 0.75 W	13.8	120.749	0.2	0.40	-0.076	-22.3	223.1	104.52	44.24	1	0.19
306 0.6 DL + 1.0 WL	8.2	67.306	10.4	16.50	-0.1	-13.2	169.7	90.67	22.46	1	0.17
307 0.6 DL + 1.0 WL	8.3	54.515	-2.5	-6.86	0.009	-13.3	156.9	76.90	27.70	1	0.14
301 1.0 DL	13.7	74.556	-1.9	0.69	0.019	-21.6	176.9	88.53	29.43	1	0.17
302 1.0 (DL + TL1) +	11.9	103.275	3.2	4.71	-0.27	-24.5	205.7	105.99	31.12	1	0.20
303 1.0 (DL + TL2) +	15.5	109.844	-3.0	-6.25	0.444	-19.2	212.2	102.58	38.91	1	0.19
304 1.0 DL+ 0.75 W	13.7	93.457	4.3	8.46	0.0	-21.9	195.8	102.89	27.68	1	0.19
305 1.0 DL+ 0.75 W	13.7	103.66	-5.1	-9.28	0.108	-21.8	206.0	106.98	30.38	1	0.20
306 0.6 DL + 1.0 WL	8.3	37.931	5.1	12.24	-0.039	-13.0	140.3	76.40	17.14	1	0.14
307 0.6 DL + 1.0 WL	8.2	51.536	-7.4	-11.41	0.1	-12.9	153.9	79.95	22.66	1	0.15

Max P_{max} = 127.66 KN/m^2
Max P_{min} = 17.14 KN/m^2
Resultant F.O.S = 0.24

BEARING PRESSURE CALCULATION FOR BENDING MOMENT AND SHEAR CHECK

L/C	Fx	Fy	Fz	Mx	My	Mz	Pmax	Pmin
401 1.4 DL	19.3	142.1	9.2	11.25	-0.1	-30.92	99.85	-5.10
402 1.4 (DL + TL1) +	16.7	200.7	9.7	21.32	-0.393	-34.87	134.70	-0.93
403 1.4 (DL + TL2) +	21.9	192.5	1.8	6.26	-0.001	-27.80	107.51	20.84
404 1.2 DL+ 1.6 WL	16.5	148.1	17.1	29.08	-0.174	-26.55	113.83	-15.11
405 1.2 DL+ 1.6 WL	16.6	127.6	-3.4	-8.30	-0.033	-26.71	86.45	-1.38
406 0.9 DL + 1.6 WL	12.3	101.6	16.2	25.92	-0.123	-19.80	86.18	-18.45
407 0.9 DL + 1.6 WL	12.5	81.1	-4.4	-11.46	0.018	-19.96	65.12	-11.03
401 1.4 DL	19.2	104.4	-2.7	0.96	0.027	-30.30	76.16	-6.57
402 1.4 (DL + TL1) +	16.6	154.2	5.2	6.15	-0.357	-34.44	103.46	-0.67
403 1.4 (DL + TL2) +	21.6	163.4	-3.6	-9.19	0.641	-26.93	99.56	9.36
404 1.2 DL+ 1.6 WL	16.5	94.6	8.7	19.01	-0.024	-26.18	85.45	-22.39
405 1.2 DL+ 1.6 WL	16.4	116.4	-11.2	-18.82	0.138	-25.99	92.26	-14.69
406 0.9 DL + 1.6 WL	12.4	56.2	8.2	19.54	-0.064	-19.58	64.38	-26.90
407 0.9 DL + 1.6 WL	12.3	78.0	-11.7	-18.30	0.098	-19.38	70.14	-18.15

Max Net factored B.P = 134.70 KN/m^2

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DETERMINATION OF REINFORCEMENT

Bottom Reinforcement

Length of cantilever along X = 0.45 m
Moment about Z: Muz = 13.64 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 321.51 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 167.0 mm^2

Effective Depth Along X = 217 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
$A_{stmin} = 0.0014bd$

A_{stmin} = 390.6 mm^2
A_{req} = 167.0 mm^2
Required Spacing = **514.8** mm

Length of cantilever along Z = 0.7 m
Moment about X: Mux = 33.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 275.84 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 440.6 mm^2

Effective Depth Along X = 201 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
$A_{stmin} = \frac{0.0018 \times 420}{f_y} bd$ *Table 24.4.3.2 SBC 304*
or
$A_{stmin} = 0.0014bd$

A_{stmin} = 361.8 mm^2
A_{req} = 440.6 mm^2
Required Spacing = **456.3** mm

Provide Bottom Reinforcement = ***16 mm dia @ 200 c/c***
Ast prov = 1005.3 mm^2

Top Reinforcement

Length of cantilever along X = 0.45 m
Moment about Z: Muz = 0.00 KNm

$M_{uzlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 327.46 KNm

As $M_{uzlim} > M_{uz}$ Tension controlled section

Therefore φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along X = 219 mm

For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd
Table 24.4.3.2 SBC 304

A_{stmin} = 197.1 mm^2
A_{req} = 0.0 mm^2
Required Spacing = **573.8** mm

Length of cantilever along Z = 0.7 m
Moment about X: Mux = 0.00 KNm

$M_{uxlim} = \emptyset\ 0.85\ f'_c \beta_1 \left(\frac{c}{d}\right)_{max} \left(1 - \frac{\beta_1}{2}\left(\frac{c}{d}\right)_{max}\right) bd^2$

= 292.56 KNm

As $M_{uxlim} > M_{ux}$ Tension controlled section

Therefore Φ = 0.9 *Table 21.2.2 SBC 304*

$A_{st} = \frac{0.85\ f'_c bd}{f_y}\left(1 - \sqrt{1 - \frac{2.35 M_u}{\emptyset f'_c bd^2}}\right)$

= 0.0 mm^2

Effective Depth Along Z = 207 mm

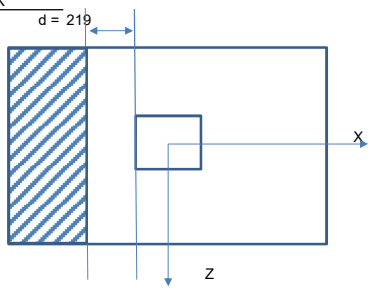
For tension controlled section $\left(\frac{c}{d}\right)_{max} = \frac{0.003}{0.003 + 0.005} = 0.375$

Min reinforcement greater of
A_{stmin} = 50% of $\frac{0.0018 \times 420}{f_y} bd$ since 50% or more prov in bottom
or
A_{stmin} = 50% of 0.0014bd
Table 24.4.3.2 SBC 304

A_{stmin} = 186.3 mm^2
A_{req} = 0.0 mm^2
Required Spacing = **607.1** mm

Provide top Reinforcement = ***12 mm dia @ 200 c/c***
Ast prov = 565.5 mm^2

ONE WAY SHEAR CKECK



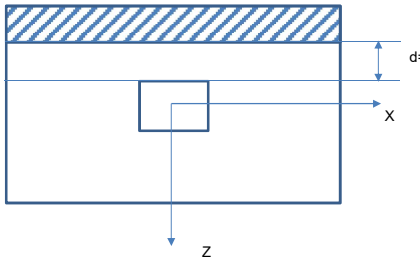
Global Project ID	900707
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Along X Direction	Cantilever	L	=	0.231	m		Least of	i)	V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0046					*Table 22.5.5.1 SBC 304*
Effective Depth Along X		d	=	219	mm				λ = 1 Normal concrete
Shear Force at d distance		V_u	=	31.12	KN				Mu = 3.59 KNm
Allowable shear strength		φVc	=	168.29	KN	O.K			φ = 0.75 for shear
Utility ratio			=	0.18					φVc = 179.8 KN
								ii)	V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
									φVc = 168.3 KN *Table 22.5.5.1 SBC 304*
								iii)	V_c = $0.29\lambda\sqrt{f_c'}b_w d$
									= 281.8 KN

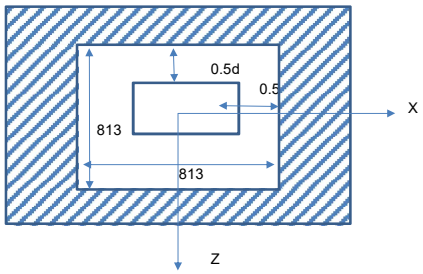
ONE WAY SHEAR CKECK



Along Z Direction	Cantilever	L	=	0.493	m		Least of	i)	V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\left(\frac{V_u d}{M_u}\right)\right)b_w d$
Reinforcement ratio		ρ_w	=	0.0049					*Table 22.5.5.1 SBC 304*
Effective Depth Along Z		d	=	207	mm				λ = 1 Normal concrete
Shear Force at d distance		V_u	=	66.41	KN				Mu = 16.37 KNm
Allowable shear strength		φVc	=	157.72	KN	O.K			φ = 0.75 for shear
Utility ratio			=	0.42					φVc = 157.7 KN
								ii)	V_c = $\left(0.16\lambda\sqrt{f_c'}+17\rho_w\right)b_w d$
									φVc = 159.8 KN
								iii)	V_c = $0.29\lambda\sqrt{f_c'}b_w d$
									φVc = 266.4 KN

TWO WAY SHEAR CKECK

Punching at 0.5d from face of column



Total area	=	3	m2	
Punching area	=	0.66	m2	
λ	=	1		
β	=	1		
αs	=	40		
b0	=	3252.0	mm	

Allowable shear stress — least of — *Table 22.6.5.2 SBC 304*

v_c = $0.33\lambda\sqrt{f_c'}$ = 1.95 N/mm^2

v_c = $0.17\left(1+\frac{2}{\beta}\right)\lambda\sqrt{f_c'}$ = 3.02 N/mm^2

v_c = $0.083\left(2+\frac{\alpha_s d}{b_0}\right)\lambda\sqrt{f_c'}$ = 2.27 N/mm^2

Design vc = 1.95 N/mm^2

Check at 0.5d from column face

Perimeter at 0.5d from column face	u	=	3.25	m	Vc	=	$v_c b_0 d$
Depth at section	d	=	213	mm		=	1352.32 KN
Shear Force at section	Vu	=	315.1	KN	φ	=	0.75 for shear
Allowable shear strength	φVc	=	1014.24	KN	φV_n	=	φVc
Utility ratio			0.31	O.K	φVc	=	1014.2 KN

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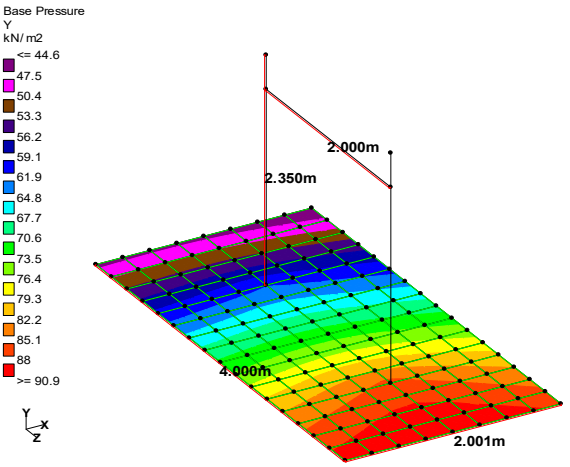
Ma'aden Revision No:	A01	C01	D01
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7.3.3 Design of combined footing for Trestle mkd. TR2 and foundation mkd. F2

For Calculation of Base pressure and design of beam, column and raft, Refer STAAD File: 3130-CNV-001_TR3 FDN MKD F2

Staad Supoort Nodes are 5 & 6



Finished Floor level	F.F.L	=	0.0	m	(Corresponds to RL +913.0m)
Finished Ground level	F.G.L	=	-0.5	m	
Bottom of foundation	B.O.F	=	-2.5	m	
Top of pedestal (staad support)	T.O.F	=	-0.15	m	
Ground Water Level	G.W.L	=	-10	m	
Length of footing	Lz	=	2.000	m	
Width of footing	Lx	=	4.000	m	
Thickness of footing	D	=	0.40	m	
Depth of pedestal	Bx	=	0.60	m	
Width of pedestal	Bz	=	0.60	m	
Unit wt of soil		=	18.0	KN/m^3	
Allowable Net Bearing Capacity		=	500	KN/m^2	
Gross Bearing Capacity		=	536	KN/m^2	
Soil load on footing in staad		=	28.8	KN/m^2	staad input

1) Check for Gross Base Pressure :

Maximum Base Pressure		=	90.9	KN/m^2	Load Case 304
Minimum Base Pressure		=	6.1	KN/m^2	Load Case 313

2) Design of foundation Raft :

Slab Thickness =	400 mm	Yield Strength of Main Reinforcement , fy =	420 N/mm^2
Width =	1000 mm	Yield Strength of Shear Reinforcement , fyt =	420 N/mm^2
Clear Cover =	75 mm	Strength Reduction Factor of Main Reinforcement , Φ =	0.9
Compressive Strength of Concrete , fc' =	35 N/mm^2	Strength Reduction Factor of Shear Reinforcement , Φt =	0.75

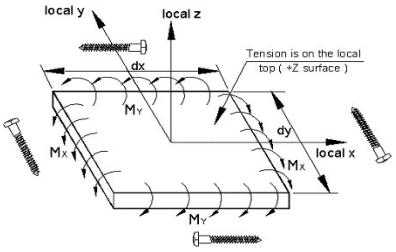
STAAD OUTPUT FROM FACTORED LOAD COMBINATION :

(Staad Group : MX,MY)

MOMENT	VALUE	UNIT	TORSIONAL MOMENT	VALUE	UNIT	
Max. Mx	2.296	kN-m	Corresponding Mxy	1.435	kN-m	*(Plate No. - 115 , L/C No. - 412)*
Min. Mx	-13.955	kN-m	Corresponding Mxy	3.867	kN-m	*(Plate No. - 24 , L/C No. - 404)*
Max. My	8.980	kN-m	Corresponding Mxy	0.726	kN-m	*(Plate No. - 114 , L/C No. - 412)*
Min. My	-20.478	kN-m	Corresponding Mxy	2.089	kN-m	*(Plate No. - 25 , L/C No. - 404)*

FOR TOP FACE REINFORCEMRNT :

Mx1 =	Mx + abs(Mxy) =	2.296 + 1.435 =	3.731 kN-m
My1 =	My + abs(Mxy) =	8.98 + 0.726 =	9.706 kN-m
Mx2 =	Mx+abs(Mxy^2/My) =	2.296 + 0.229 =	2.525 kN-m
My2 =	My+abs(Mxy^2/Mx) =	8.98 + 0.13 =	9.11 kN-m



DESIGN MOMENT CALCULATION FOR TOP FACE REINFORCEMENT :

Mxd =	3.731 kN-m
Myd =	9.706 kN-m

ALONG LOCAL X - DIRECTION :

Mu =	3.731 kN-m						
Φ*Kn =	Mu/bd^2				deff. =	319 mm	
Mu/bd^2 =	3.731X10^6/(1000X319^2) =	0.037	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0001	Reqd. Ast =	0.0001X1000X319/100 =	0.32		cm^2	
Provided	12 mm @	200 C/C	(Provide Ast =	5.65	)cm^2		
			Total Provide Ast =	5.65	cm^2	O.K.	
			Total Provide Pt =	0.0018			

ALONG LOCAL Y - DIRECTION :

Mu =	9.706 kN-m						
Φ*Kn =	Mu/bd^2				deff. =	307 mm	
Mu/bd^2 =	9.706X10^6/(1000X307^2) =	0.095	N/mm^2	<	6.9	N/mm^2	
Pt =	0.0003	Reqd. Ast =	0.0003X1000X319/100 =	0.92		cm^2	
Provided	12 mm @	200 C/C	(Provide Ast =	5.65	)cm^2		
			Total Provide Ast =	5.65	cm^2	O.K.	
			Total Provide Pt =	0.0018			

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	03-04-2020	29-04-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2
Date	03-04-2020	29-04-2020	01-11-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	APL	APL	APL
Checked	SBY	SBY	SBY
Approved	PRI	PRI	PRI

FOR BOTTOM FACE REINFORCEMRNT :

Mx1 = Mx - abs(Mxy) = -13.955 - 3.867 = -17.822 kN-m
My1 = My - abs(Mxy) = -20.478 - 2.089 = -22.567 kN-m
Mx2 = Mx-abs(Mxy^2/My) = -13.955 - 0.73 = -14.685 kN-m
My2 = My-abs(Mxy^2/Mx) = -20.478 - 0.313 = -20.165 kN-m

DESIGN MOMENT CALCULATION FOR BOTTOM FACE REINFORCEMENT :

Mxd = -17.822 kN-m
Myd = -22.567 kN-m

ALONG LOCAL X - DIRECTION :

Mu = 17.822 kN-m
Φ*Kn = Mu/bd^2 deff. = 317 mm
Mu/bd^2 = 17.822X10^6/(1000X317^2) = 0.177 N/mm^2 < 6.9 N/mm^2
Pt = 0.0005 Reqd. Ast =)0.0005X1000X317/100 = 1.59 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 **O.K.**
Total Provide Pt = 0.0032

ALONG LOCAL Y - DIRECTION :

Mu = 22.567 kN-m
Φ*Kn = Mu/bd^2 deff. = 301 mm
Mu/bd^2 = 22.567X10^6/(1000X301^2) = 0.249 N/mm^2 < 6.9 N/mm^2
Pt = 0.0007 Reqd. Ast =)0.0007X1000X301/100 = 2.11 cm^2
Provided **16** mm @ **200** C/C (Provide Ast = 10.05)cm^2

Total Provide Ast = 10.05 cm^2 **O.K.**
Total Provide Pt = 0.0033

SHEAR CHECK

(Staad Group : SHEAR)
(FROM FACTORED LOAD COMBINATION)

ALONG LOCAL X - DIRECTION :

SQx = -0.03 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 166 , L/C No. - 404)*
Vu = 9.51 kN

Provided Pt = 0.0032
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0032X9.51X10^3X317/17.822/10^6)X1000X317/10^3
= 302.981 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0032)X1000X317/10^3
= 317.308 kN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X317/10^3
= 543.865 kN
HENCE, Vc = 302.981 kN
Φt*Vc = 0.75X302.981 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 227.236 kN **> 9.51T , Hence O.K.**

ALONG LOCAL Y - DIRECTION :

SQy = -0.035 N/mm^2 *(At d-distance from face of the Wall)* *(Plate No. - 73 , L/C No. - 403)*
Vu = 10.535 kN

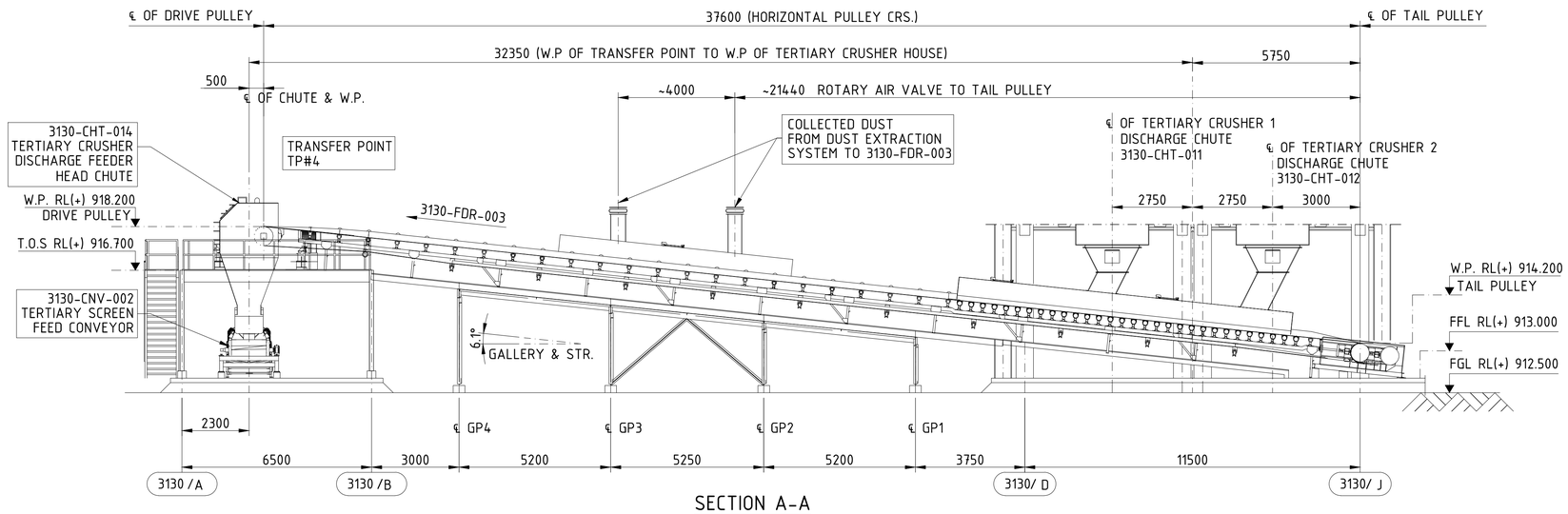
Pt provided = 0.0033
Nominal Shear Strength, Vc = Min [(a) , (b) , (c)] *[SBC:304-CR-18, Table 22.5.5.1, Page-174]*
[a] = (0.16*λ*fc'^0.5 + 17*Pt*Vu*d.eff/Mu)*b*d.eff λ = 1
= (0.16X1X35^0.5+17X0.0033X10.535X10^3X301/22.567/10^3)X1000X301/10^3
= 287.291 kN
[b] = (0.16*λ*fc'^0.5 + 17*Pt)*b*d.eff
= (0.16X1X35^0.5+17X0.0033)X1000X301/10^3
= 301.805 KN
[c] = 0.29*λ*fc'^0.5*b*d.eff
= 0.29X1X5.92X1000X301/10^3
= 516.415 kN
HENCE, Vc = 287.291 kN
Φt*Vc = 0.75X287.291 *[SBC:304-CR-18, CL. No. 7.6.3.1, Page-42]*
= 215.468 kN **> 10.535T , Hence O.K.**

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

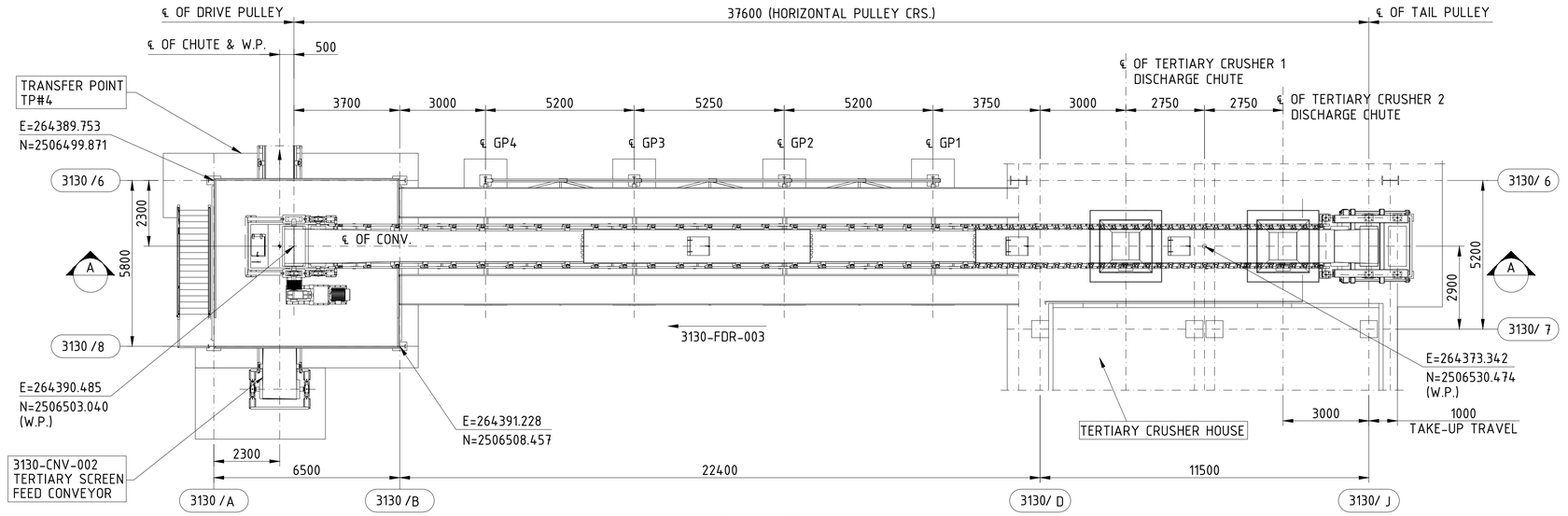
Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	04-04-2020	29-04-2020	01-11-2023				Date	04-04-2020	29-04-2020	01-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 1 **MECHANICAL GENERAL ARRANGEMENT DRAWINGS**

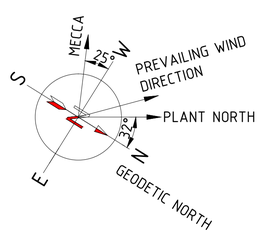
DRAWING NUMBER MD-216-3130-EG-ME-LPX-1007 REV A01



SECTION A-A
SCALE 1:100



PLAN VIEW
SCALE 1:100



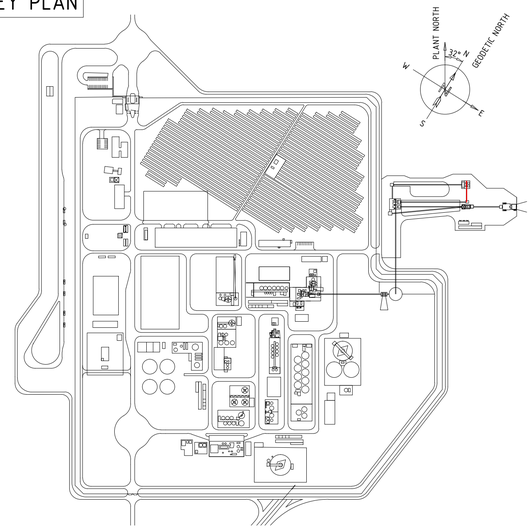
ABBREVIATION

CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
BRG.:	BEARING	OPNG.:	OPENING

NOTE:
1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+)913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+)912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN INDICATIVE.

CONCRETE FLOOR
CHECKERED PLATE
GRATING
BRICK WALL
CONCRETE WALL

KEY PLAN



REV	DOCUMENT NUMBER	REFERENCE DOCUMENTS	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
A01	MD-216-3130-EG-ME-ELV-1013	FEEDER 3130-FDR-003-ELEVATION	06/03/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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Saudi Arabian Mining Company شركة معادن العربية السعودية
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LARSEN & TOUBRO DRAWING NUMBER	019045_100_H_G_16119_001	REVISION 0
OUTOTEC DRAWING NUMBER	900707__GAU01__ZC01__MTL01__31329	DOCUMENT ID OU100669070 / REVISION 0

DRAWN	HKD	02/03/2020
CHECKED	BRD	03/03/2020
ENGINEER	BD	25/02/2020
APPROVED	BGR	06/03/2020
EMPLOYER		
SCALE	1:100	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
TERTIARY CRUSHING
FEEDER 3130-FDR-003 - PLAN

DRAWING NUMBER MD-216-3130-EG-ME-LPX-1007 REV A01

DRAWING NUMBER: MD-216-3130-EG-ME-ELV-1013
REV: A01

ABBREVIATION

CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
BRG.:	BEARING	OPNG.:	OPENING

TABLE-1 : LOAD FOR CONVEYOR SHORT POST

FOR CONV. 3130-CNV-001

POINT	VERTICAL LOAD IN "t"		HORIZONTAL LOAD IN "t"	
	DEAD LOAD	LIVE LOAD	ALONG	ACROSS
F1	(+)0.65	(+)0.80	(±)0.15	(±)0.08

SIGN CONVENTION

UPWARD (-)
DOWNWARD (+)
TOWARDS DRIVE PULLEY=(+)
TOWARDS TAIL PULLEY=(-)
REVERSIBLE LOAD=(±)

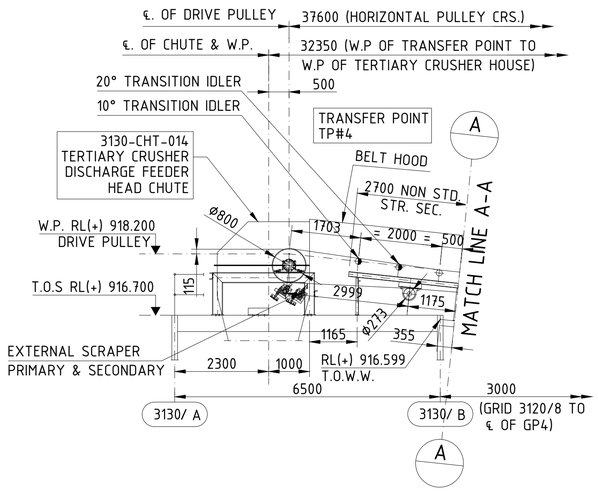
Note: Wind load to be considered during civil & structural design

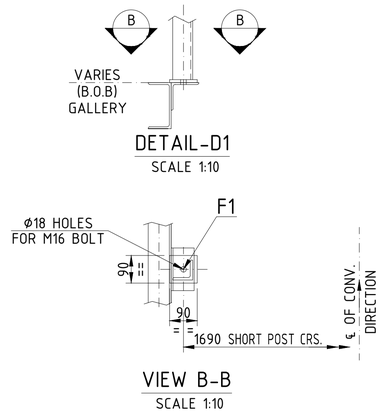
CONVEYOR CHARACTERISTICS

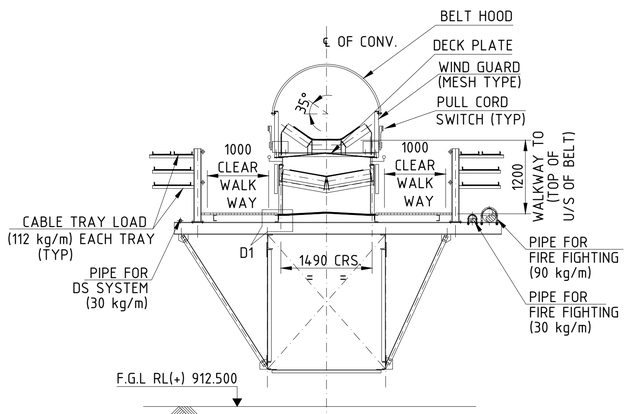
	PARTICULARS	SPECIFICATION	QUANTITY
MISCELLANEOUS EQUIPMENT	TAKE-UP / TRAVEL	SCREW / 1000 mm	
	TAKE-UP LOAD	N.A.	
	SCRAPER AT DISCHARGE PULLEY	EXTERNAL / MAIN & PRE CLEANER	1 NO. + 1 NO.
	SCRAPER AT TAKE-UP	N.A.	
	SCRAPER AT TAIL PULLEY	INTERNAL V-PLOUGH	1 NO.
	BELT SWAY SWITCH (ZS)	@ 50 m INTERVAL	1 PAIRS
	PULL CORD SWITCH (HS)	@ 30 m INTERVAL	2 PAIRS
	ZERO SPEED SWITCH (SS)	AT TAIL PULLEY	1 NO.
	BELT RIP SWITCH (YS)	AFTER FEEDING ZONE	1 NO.
	DECK PLATE	3.15 mm thk. CONTINUOUS	1 LOT.
	BELT HOOD	GALVANIZED CORRUGATED SHEET	1 LOT.

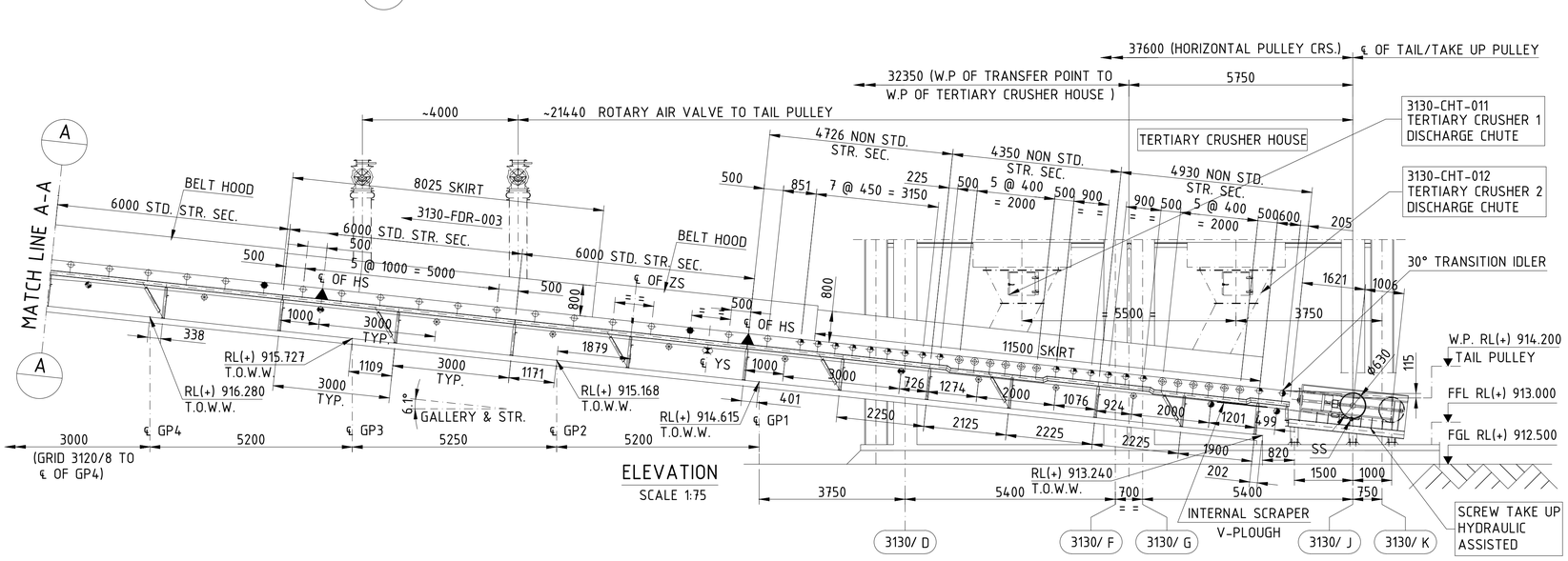
CONVEYOR CHARACTERISTICS

	PARTICULARS		SPECIFICATION	QUANTITY
CONVEYOR	EQUIPMENT TAG NO.		3130-FDR-003	1 NO.
	EQUIPMENT DESCRIPTION		TERTIARY CRUSHER DISCHARGE FEEDER	
	LOCATION		TERTIARY CRUSHER HOUSE TO TRANSFER POINT TP#4	
	CAPACITY RATED / DESIGN	(t/h)	782 / 870	
	BELT WIDTH	(mm)	1200	
	BELT SPEED	(m/sec)	1.537	
	TRIPPER FIXED / MOVABLE		N.A.	
	CONVEYOR LENGTH / LIFT	(mm)	37600 / 4000	
MATERIAL	MATERIAL		GOLD ORE : FRESH ORE / OXIDE ORE	
	BULK DENSITY	(t/m³)	1.6	
	MAX. LUMP SIZE	(mm)	(-)21.5	
	MOISTURE CONTENT	(%)	2	
	SURCHARGE ANGLE	(deg.)	20	
BELTING	MAX. RUNNING TENSION	(T1 kN)	63.105	
	SLACK SIDE TENSION	(T2 kN)	48.986	
	TAIL TENSION	(T0 kN)	51.127	
	BELT DESIGNATION		EP-1000 / 4 (HEAVY DUTY)	
	TOP / BOTTOM COVER THICKNESS	(mm)	6 / 3	
	COVER GRADE / FABRIC		DIN-X / EP	
PULLEYS	DISCHARGE WITH DRIVE	(MKD. x DIA.)	D-5 x 800 mm	1 NO.
	DISCHARGE WITHOUT DRIVE	(MKD. x DIA.)	N.A.	
	SNUB	(MKD. x DIA.)	N.A.	
	TAIL	(MKD. x DIA.)	ND-6 x 630 mm	1 NO.
	TAKE-UP	(MKD. x DIA.)	N.A.	
	DEFLECTOR ROLLER	(MKD. x DIA.)	DR-3 x 273 mm	1 NO.
	TRIPPER DISCHARGE	(MKD. x DIA.)	N.A.	
	TRIPPER BEND	(MKD. x DIA.)	N.A.	
	BEND	(MKD. x DIA.)	N.A.	
	LAGGING	(mm)	DRIVE: 16thk CERAMIC	
			NON-DRIVE: 16thk PLAIN RUBBER	
BEARING/SLEEVE PLUMMER BLOCK	DISCHARGE WITH DRIVE		TBA	2 SETS
	DISCHARGE WITHOUT DRIVE		N.A.	
	SNUB		TBA	
	TAIL		TBA	2 SETS
	TAKE-UP		N.A.	
	DEFLECTOR ROLLER		TBA	2 SETS
	BEND		TBA	
	TRIPPER DISCHARGE		N.A.	
	TRIPPER BEND		N.A.	
IDLER	TROUGHING	(DIA. x deg.)	152.4 x 35°	18 NOS.
	TROUGHING RETRACTABLE	(DIA. x deg.)	152.4 x 45°	16 NOS.
	TROUGHING TRAINER	(DIA. x deg.)	152.4 x 35°	2 NOS.
	RETURN	(DIA. x deg.)	152.4 x 10°	8 NOS.
	SELF ALIGNING RETURN TRAINER INVERTED IDLER	(DIA. x deg.)	152.4 x 10°	2 NOS.
	IMPACT RETRACTABLE	(DIA. x deg.)	190 x 45°	12 NOS.
	TRANSITION (ADJUSTABLE)	(DIA. x deg.)	152.4 x 10°/20°/30°	(1+1+1) NOS.
	RETURN SELF CLEANING	(DIA. x deg.)	190 x 0°	1 NO.
	NON MAGNETIC TROUGHING	(DIA. x deg.)	152.4 x 35°	
	WEIGHING IDLER	(DIA. x deg.)	152.4 x 35°	
	SPECIAL MACHINED IDLER	(DIA. x deg.)	152.4 x 35°	
	RETURN FLAT IDLER	(DIA. x deg.)	152.4 x 0°	2 NOS.
DRIVE	MOTOR: RATING	kW/rpm (SYN)	45 / 1800	1 NO.
	MOTOR: FRAME SIZE		TBA	
	MOTOR: TYPE		SQUIRREL CAGE	
	MOTOR: SUPPLY		400V, 60Hz	
	REDUCER: TYPE		HELICAL	1 NO.
	REDUCER: MODEL NO./COOLING/RATIO		TBA	
	REDUCER: RATIO		50	
	REDUCER: HOLD BACK		YES	
	BRAKE: TYPE		N.A.	
	BRAKE: MODEL NO.		TBA	
	BRAKE: DRAM DIAMETER		TBA	
	COUPLING: LOW SPEED: TYPE		RESILIENT COUPLING	1 NO.
	COUPLING: LOW SPEED: MODEL NO.		TBA	
	COUPLING: HIGH SPEED: TYPE		GEAR	1 NO.
	COUPLING: HIGH SPEED: MODEL NO.		TBA	









DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-LPX-1007	FEEDER 3130-FDR-003-PLAN	A01	06/03/2020	ISSUED FOR APPROVAL	HKD	BRD	BD	BGR	

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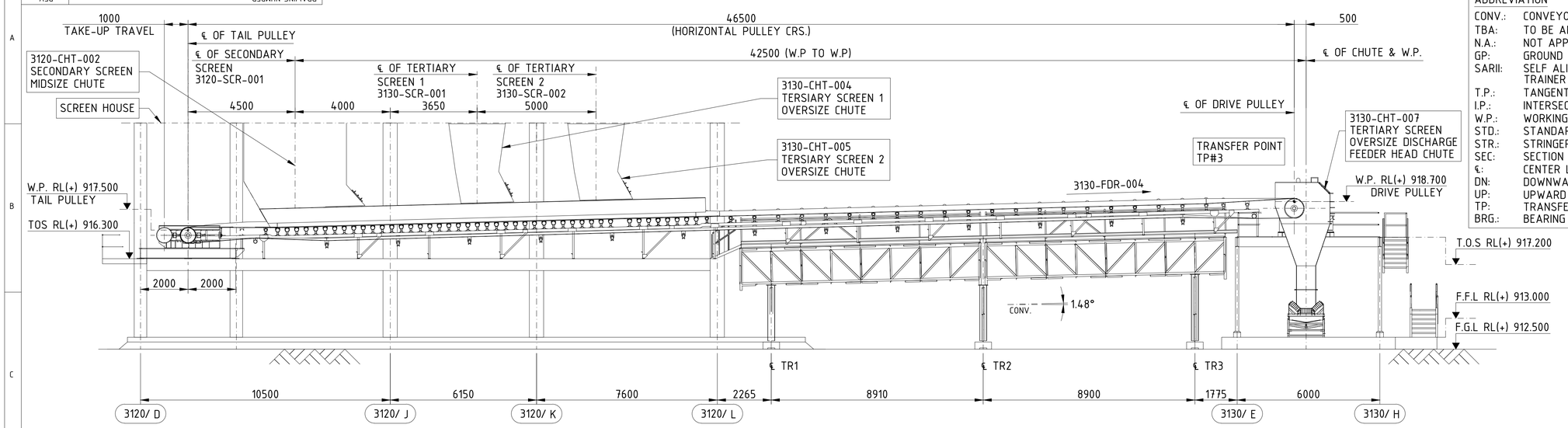
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DRAWN	HKD	02/03/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	BRD	03/03/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	BD	25/02/2020	TERTIARY CRUSHING
APPROVED	BGR	06/03/2020	FEEDER 3130-FDR-003 -ELEVATION
EMPLOYER			
SCALE	1:75, 1:50, 1:10	A1	

LARSEN & TOUBRO DRAWING NUMBER: 019045_100_H_G_16119_002 — REVISION: 0
OUTOTEC DRAWING NUMBER: 900707__GAU01__ZZ01__MTL01__31348 — DOCUMENT ID: OU100669111 — REVISION: 0
DRAWING NUMBER: MD-216-3130-EG-ME-ELV-1013 — REV: A01

DRAWING NUMBER MD-216-3130-EG-ME-LPX-1008 REV A01



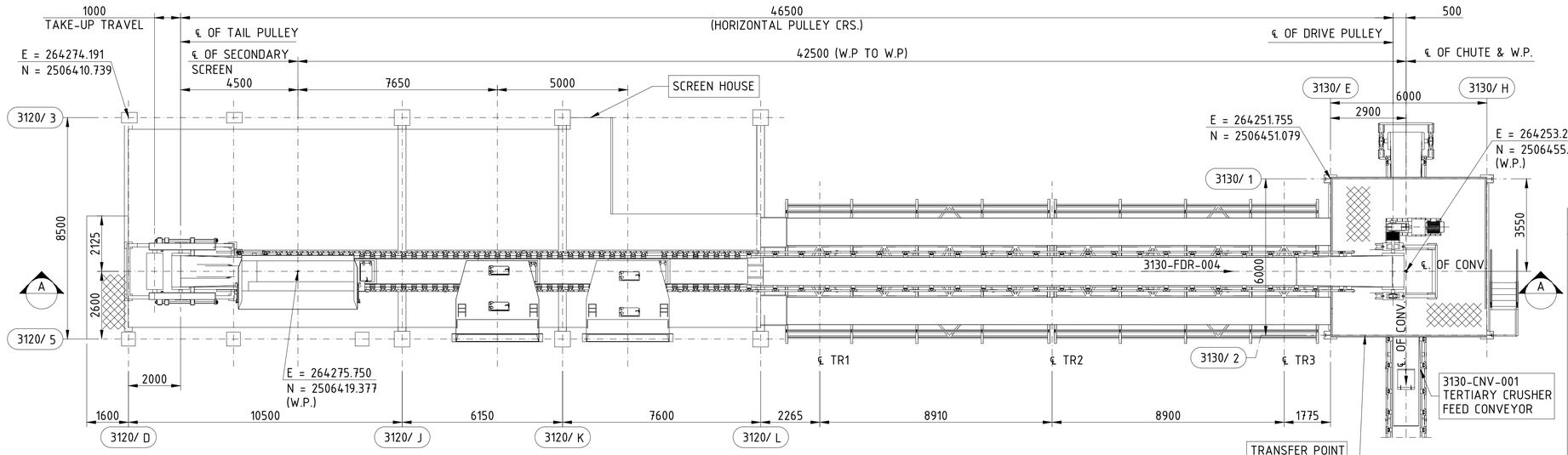
SECTION A-A

SCALE 1:000

ABBREVIATION			
CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
BRG.:	BEARING	OPNG.:	OPENING

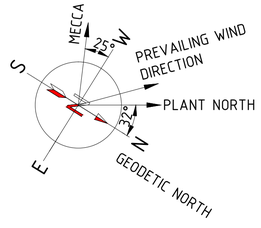
NOTE:

1. ALL DIMENSIONS ARE IN MILLIMETER, CO-ORDINATES AND ELEVATION ARE IN METER, UNLESS OTHERWISE STATED.
2. FINISHED FLOOR LEVEL (F.F.L) FOR CRUSHING AREA = RL(+) 913.000 m
3. FINISHED GROUND LEVEL (F.G.L) FOR CRUSHING AREA = RL(+) 912.500 m
4. F.F.L, F.G.L, T.O.C, = FINISH FLOOR LEVEL, FINISH GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B= TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE ARE SHOWN TENTATIVE.



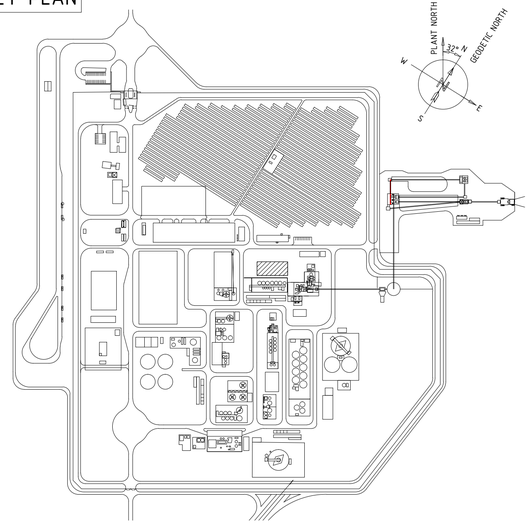
PLAN VIEW

SCALE 1:100



- CONCRETE FLOOR
- CHECKERED PLATE
- GRATING
- BRICK WALL
- CONCRETE WALL

KEY PLAN



REV	DATE	DESCRIPTION	DRN	CHK	ENG	APP	EMP	DOCUMENT NUMBER	REFERENCE DOCUMENTS
A01	04/03/2020	ISSUED FOR APPROVAL	AFO	BRD	BD	BGR		MD-216-3130-EG-ME-ELV-1015	FEEDER 3130-FDR-004 - ELEVATION

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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Saudi Arabian Mining Company شركة التعدين العربية السعودية

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LARSEN & TOUBRO DRAWING NUMBER			REVISION
O19045_100_H_G_16120_001			0
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707__GAU01__ZC01__MTL01__31330		OU100669071	0
DRAWN	AFO	28/02/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	BRD	02/03/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	BD	26/02/2020	TERTIARY CRUSHING
APPROVED	BGR	04/03/2020	FEEDER 3130-FDR-004 - PLAN
EMPLOYER			
SCALE 1:50	A1		DRAWING NUMBER MD-216-3130-EG-ME-LPX-1008 REV A01

DRAWING NUMBER: MD-216-3130-EG-ME-ELV-1015 | REV: A01

ABBREVIATION

CONV.:	CONVEYOR	CRS.:	CENTERS
TBA:	TO BE ANNOUNCED	DS:	DUST SUPPRESSION
N.A.:	NOT APPLICABLE	VGTU.:	VERTICAL GRAVITY TAKE-UP
GP:	GROUND PORTAL	MKD:	MARKED
SARII:	SELF ALIGNING RETURN TRAINER INVERTED IDLER	DIA:	DIAMETER
		SYN:	SYNCHRONOUS
T.P.:	TANGENT POINT	thk.:	THICKNESS
I.P.:	INTERSECTION POINT	RCC:	REINFORCED CEMENT CONCRETE
W.P.:	WORKING POINT	B/W:	BELT WIDTH
STD.:	STANDARD	MAX:	MAXIMUM
STR.:	STRINGER	deg.:	DEGREE
SEC:	SECTION	U/S.:	UNDERSIDE
℄:	CENTER LINE	B.O.M.R:	BOTTOM OF MONORAIL
DN:	DOWNWARD	T.O.W.W.:	TOP OF WALKWAY
UP:	UPWARD	TR:	TRESTLE
TP:	TRANSFER POINT	TYP.:	TYPICAL
BRG.:	BEARING	OPNG.:	OPENING

TABLE-1 : LOAD FOR CONVEYOR SHORT POST

FOR CONV. 3130-FDR-004

POINT	VERTICAL LOAD IN "t"		HORIZONTAL LOAD IN "t"	
	DEAD LOAD	LIVE LOAD	ALONG	ACROSS
F1	(+)0.65	(+)0.80	(±)0.15	(±)0.08

SIGN CONVENTION

UPWARD (-)
DOWNWARD (+)
TOWARDS DRIVE PULLEY=(+)
TOWARDS TAIL PULLEY=(-)
REVERSIBLE LOAD=(±)

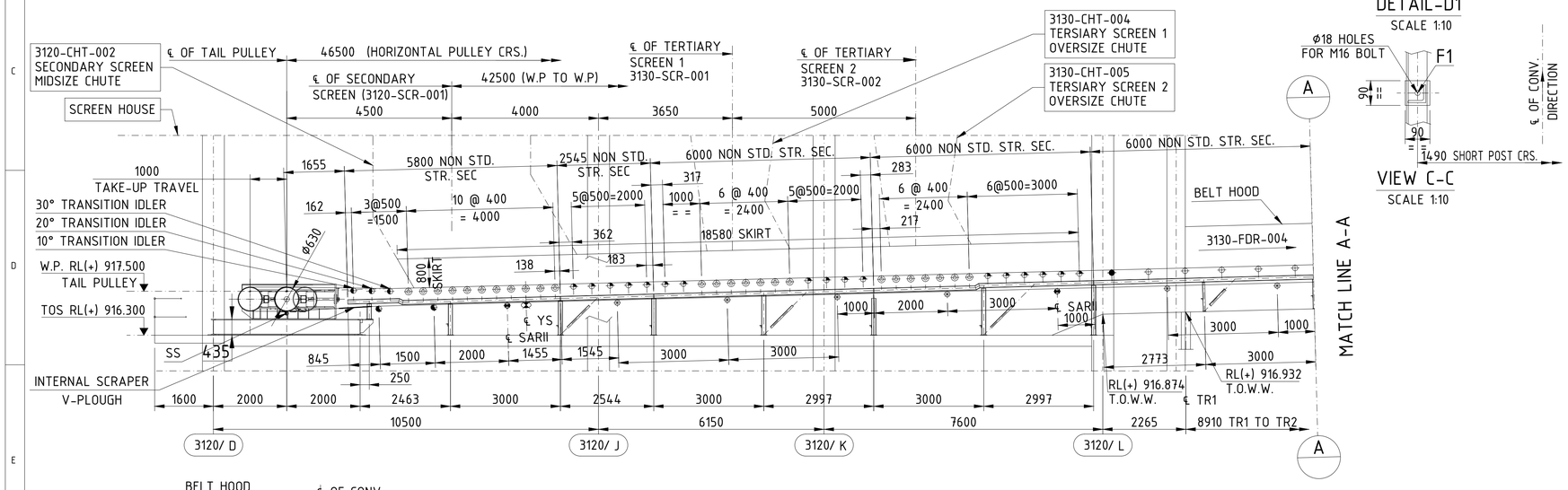
Note: Wind & seismic load to be considered during civil & structural design.

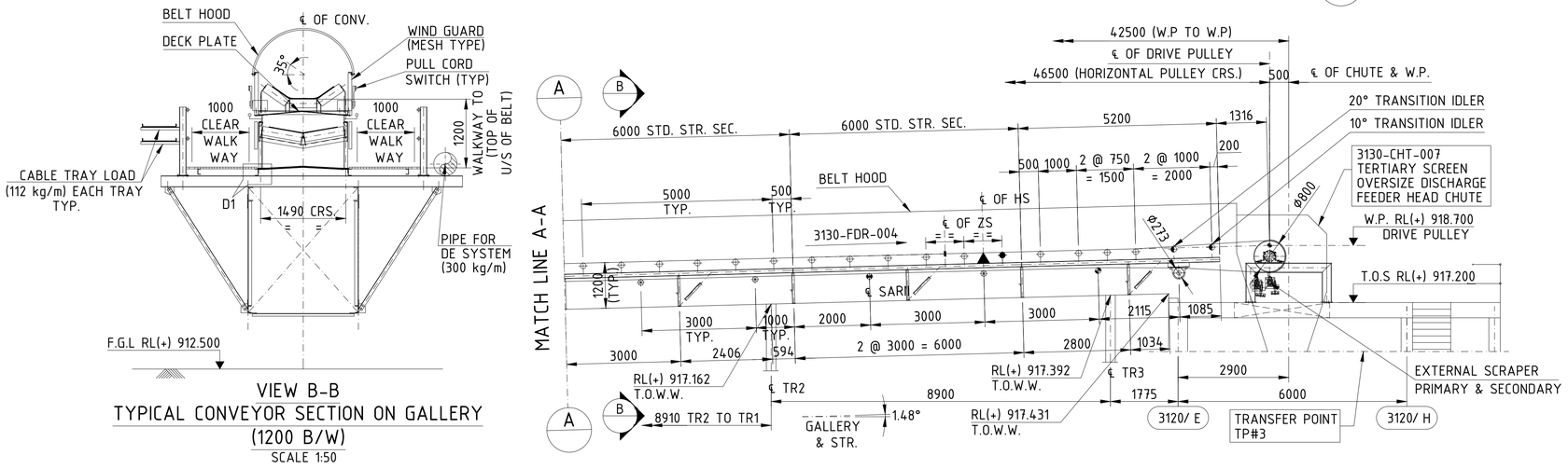
CONVEYOR CHARACTERISTICS

	PARTICULARS	SPECIFICATION	QUANTITY
MISCELLANEOUS EQUIPMENT	TAKE-UP / TRAVEL	SCREW / 1000 mm	
	TAKE-UP LOAD	N.A.	
	SCRAPER AT DISCHARGE PULLEY	EXTERNAL / MAIN & PRE CLEANER	1 NO. + 1 NO.
	SCRAPER AT TAKE-UP	N.A.	
	SCRAPER AT TAIL PULLEY	INTERNAL V-PLOUGH	1 NO.
	BELT SWAY SWITCH (ZS)	@ 50 m INTERVAL	1 PAIR
	PULL CORD SWITCH (HS)	@ 30 m INTERVAL	1 PAIR
	ZERO SPEED SWITCH (SS)	AT TAIL PULLEY	1 NO.
	BELT RIP SWITCH (YS)	AFTER FEEDING ZONE	1 NO.
	DECK PLATE	3.15 mm thk. CONTINUOUS	1 LOT.
	BELT HOOD	GALVANIZED CORRUGATED SHEET	1 LOT.

CONVEYOR CHARACTERISTICS

	PARTICULARS		SPECIFICATION	QUANTITY
CONVEYOR	EQUIPMENT TAG NO.		3130-FDR-004	1 NO.
	EQUIPMENT DESCRIPTION		TERTIARY SCREEN OVERSIZE DISCHARGE FEEDER	
	LOCATION		SCREEN HOUSE TO TRANSFER POINT TP#3	
	CAPACITY RATED / DESIGN	(t/h)	782 / 870	
	BELT WIDTH	(mm)	1200	
	BELT SPEED	(m/sec)	1.506	
	TRIPPER FIXED / MOVABLE		N.A.	
	CONVEYOR LENGTH / LIFT	(mm)	46500 / 1200	
MATERIAL	MATERIAL		GOLD ORE : FRESH ORE / OXIDE ORE	
	BULK DENSITY	(t/m³)	1.6	
	MAX. LUMP SIZE	(mm)	(-)40	
	MOISTURE CONTENT	(%)	2	
	SURCHARGE ANGLE	(deg.)	20	
BELTING	MAX. RUNNING TENSION	(T1 kN)	26.262	
	SLACK SIDE TENSION	(T2 kN)	14.744	
	TAIL TENSION	(T0 kN)	17.410	
	BELT DESIGNATION		EP-1000 / 4 (HEAVY DUTY)	
	TOP / BOTTOM COVER THICKNESS	(mm)	6 / 3	
	COVER GRADE / FABRIC		DIN-X / EP	
PULLEYS	DISCHARGE WITH DRIVE	(MKD. x DIA.)	D-5 x 800 mm	1 NO.
	DISCHARGE WITHOUT DRIVE	(MKD. x DIA.)	N.A.	
	SNUB	(MKD. x DIA.)	N.A.	
	TAIL	(MKD. x DIA.)	ND-6 x 630 mm	1 NO.
	TAKE-UP	(MKD. x DIA.)	N.A.	
	DEFLECTOR ROLLER	(MKD. x DIA.)	DR-3 x 273 mm	1 NO.
	TRIPPER DISCHARGE	(MKD. x DIA.)	N.A.	
	TRIPPER BEND	(MKD. x DIA.)	N.A.	
	BEND	(MKD. x DIA.)	N.A.	
	LAGGING	(mm)	DRIVE: 16thk CERAMIC	
			NON-DRIVE: 16thk PLAIN RUBBER	
BEARING/SLEEVE PLUMMER BLOCK	DISCHARGE WITH DRIVE		TBA	2 SETS
	DISCHARGE WITHOUT DRIVE		N.A.	
	SNUB		TBA	
	TAIL		TBA	2 SETS
	TAKE-UP		N.A.	
	DEFLECTOR ROLLER		TBA	2 SETS
	BEND		TBA	
	TRIPPER DISCHARGE		N.A.	
	TRIPPER BEND		N.A.	
IDLER	TROUGHING	(DIA. x deg.)	152.4 x 35°	20 NOS.
	TROUGHING RETRACTABLE	(DIA. x deg.)	152.4 x 45°	17 NOS.
	TROUGHING TRAINER	(DIA. x deg.)	152.4 x 35°	2 NOS.
	RETURN	(DIA. x deg.)	152.4 x 10°	9 NOS.
	SELF ALIGNING RETURN TRAINER INVERTED IDLER	(DIA. x deg.)	152.4 x 10°	3 NOS.
	IMPACT RETRACTABLE	(DIA. x deg.)	190 x 45°	25 NOS.
	TRANSITION (ADJUSTABLE)	(DIA. x deg.)	152.4 x 10°/20°/30°	(2+2+1) NOS.
	RETURN SELF CLEANING	(DIA. x deg.)	190 x 0°	1 NO.
	NON MAGNETIC TROUGHING	(DIA. x deg.)	152.4 x 35°	
	WEIGHING IDLER	(DIA. x deg.)	152.4 x 35°	
	SPECIAL MACHINED IDLER	(DIA. x deg.)	152.4 x 35°	
	RETURN FLAT IDLER	(DIA. x deg.)	152.4 x 0°	2 NOS.
DRIVE	MOTOR RATING	kW/rpm (SYN)	55 / 1800	1 NO.
	MOTOR FRAME SIZE		TBA	
	MOTOR TYPE		SQUIRREL CAGE	
	MOTOR SUPPLY		400V, 60Hz	
	REDUCER TYPE		HELICAL	1 NO.
	REDUCER MODEL NO./COOLING/RATIO		TBA	
	REDUCER RATIO		50	
	REDUCER HOLD BACK		YES	
	BRAKE TYPE		N.A.	
	BRAKE MODEL NO.		TBA	
	BRAKE DRAM DIAMETER		TBA	
	COUPLING HIGH SPEED TYPE		FLUID COUPLING	1 NO.
	COUPLING HIGH SPEED MODEL NO.		TBA	
	COUPLING LOW SPEED TYPE		GEAR	1 NO.
	COUPLING LOW SPEED MODEL NO.		TBA	





DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-LPX-1008	FEEDER 3130-FDR-004 - PLAN	A01	04/03/2020	ISSUED FOR APPROVAL	AFO	BRD	BD	BGR	

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DRAWN	AFO	28/02/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	BRD	02/03/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	BD	26/02/2020	TERTIARY CRUSHING
APPROVED	BGR	04/03/2020	FEEDER 3130-FDR-004 - ELEVATION
EMPLOYER			
SCALE	1:75, 1:50, 1:10	A1	

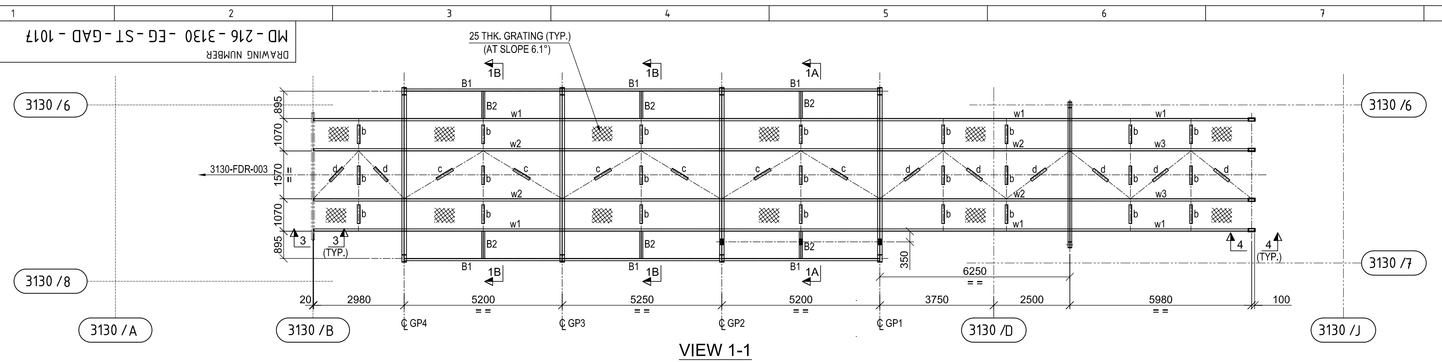
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OUTOTEC DRAWING NUMBER: 900707__GAU01__ZC01__MTL01__31350 | DOCUMENT ID: OU100669113 | REVISION: 0
DRAWING NUMBER: MD-216-3130-EG-ME-ELV-1015 | REV: A01

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

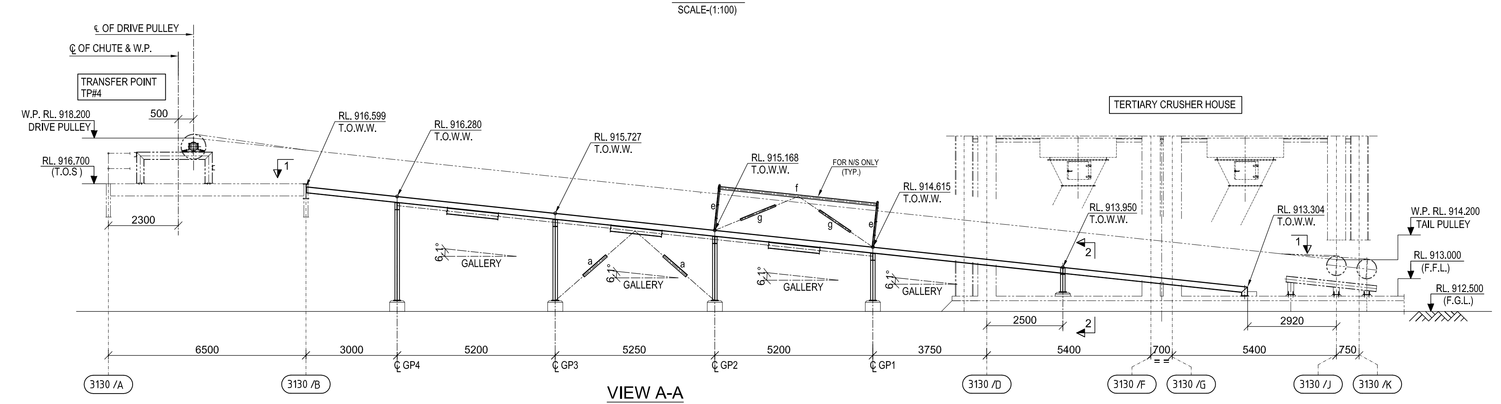
Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	04-04-2020	29-04-2020	01-11-2023				Date	04-04-2020	29-04-2020	01-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 2 **STRUCTURAL GENERAL ARRANGEMENT DRAWINGS CONTAINING BASE PLATE DETAIL**

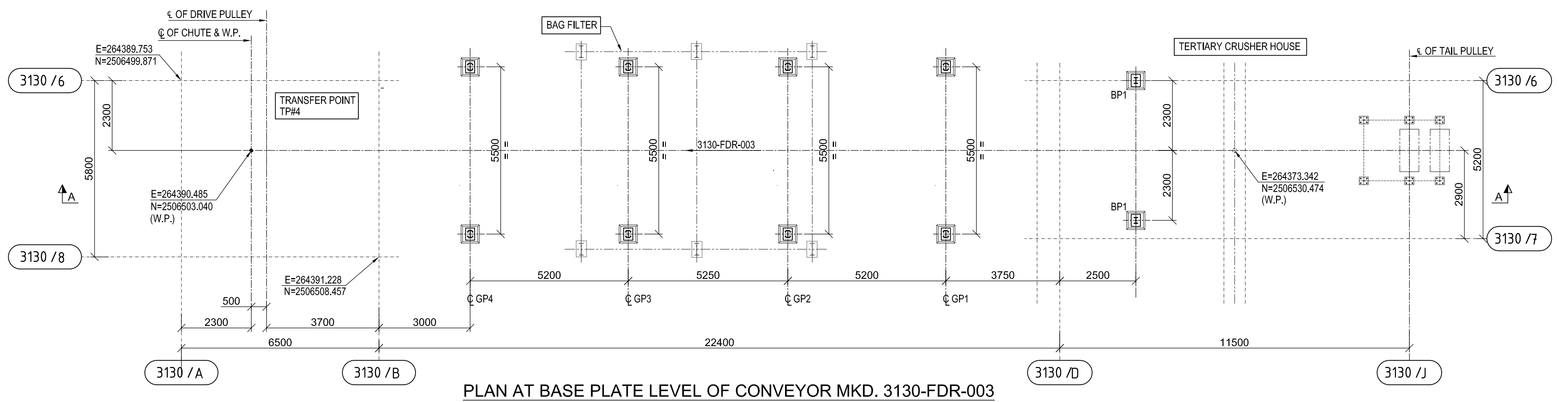
DRAWING NUMBER: MD - 216 - 3130 - EG - ST - GAD - 1017 | REV: A01



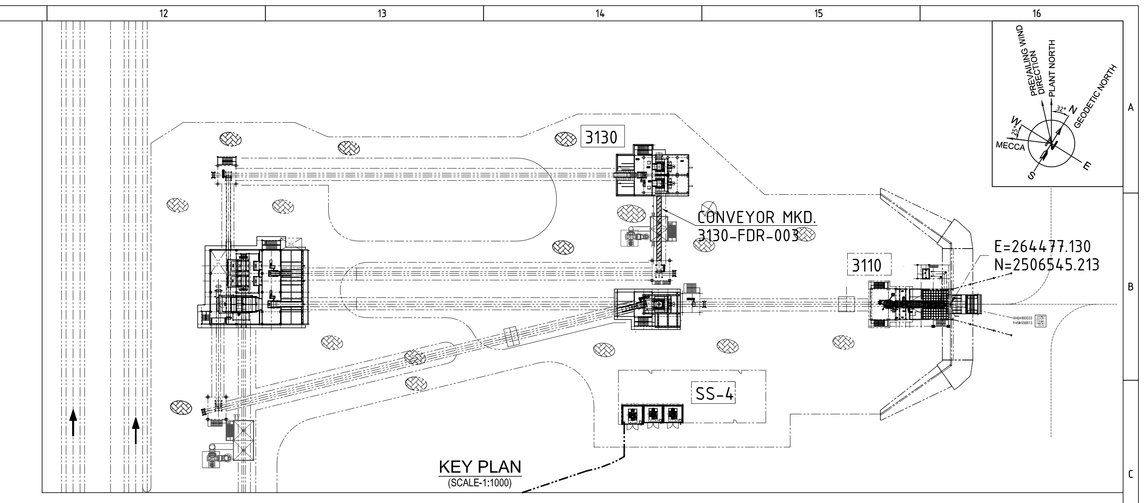
VIEW 1-1
SCALE-(1:100)



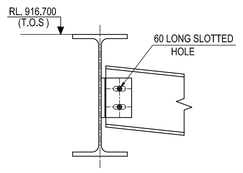
VIEW A-A
ELEVATION OF CONVEYOR MKD. 3130-FDR-003
(SCALE 1:100)



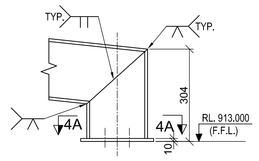
PLAN AT BASE PLATE LEVEL OF CONVEYOR MKD. 3130-FDR-003
(SCALE 1:100)



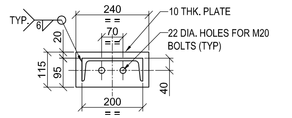
KEY PLAN
(SCALE-1:1000)



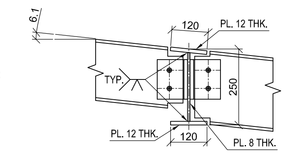
SECTION 3-3
(SCALE 1:10)



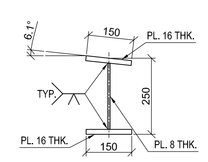
SECTION 4-4
(SCALE 1:10)



SECTION 4A-4A
(SCALE 1:10)



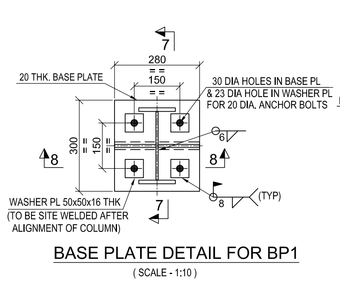
SECTION 5-5
(SCALE 1:10)



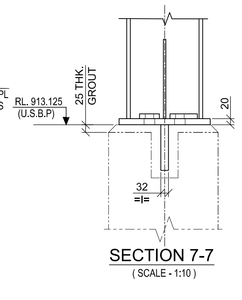
SECTION 6-6
(SCALE 1:10)

SCHEDULE OF MEMBERS					
MEMBER MARKED	MEMBER	SECTON	FACTORED AXIAL FORCE(kN)	FACTORED SHEAR FORCE(kN)	REMARKS
C1	IPE 240	I	-	-	MATERIAL GRADE:S275-HR
C2	1. PL. 150X16 2. PL. 218X8	I-1 I-2 I-1	-	-	
C3	1. PL. 120X12 2. PL. 226X8	I-1 I-2 I-1	-	-	
B1	UPN 200	[	-	-	
B2	IPE 200	I	-	-	
a	2L 75X75X8	T	95	-	
b	L 50X50X6	L	10	-	
c	L 75X75X8	L	20	-	
d	L 65X65X6	L	20	-	
e	UPN 100	[	-	-	
f	L 75X75X8	L	-	-	
g	L 50X50X6	L	-	-	
w3	UPN 240	[	-	-	
w2	UPN 200	[	-	-	
w1	UPN 140	[	-	-	

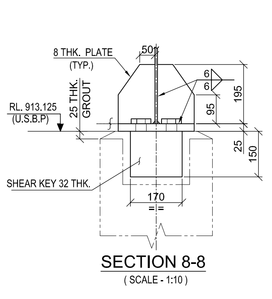
ALL GUSSET PLATE ARE 8mm THK.
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED FOR WELDED CONNECTION. FOR BOLTED CONNECTION, MINIMUM 2 Nos. M20 BOLTS SHALL BE PROVIDED.



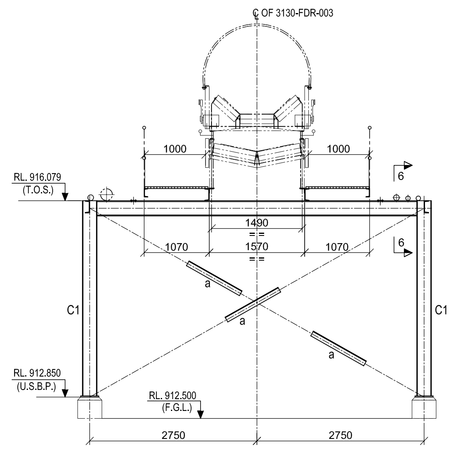
BASE PLATE DETAIL FOR BP1
(SCALE - 1:10)



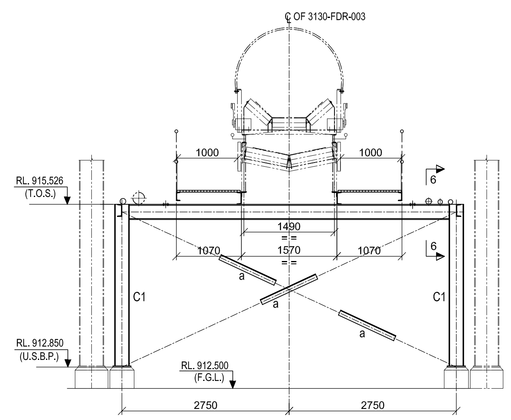
SECTION 7-7
(SCALE - 1:10)



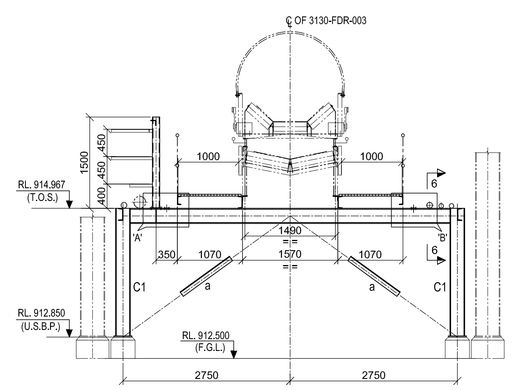
SECTION 8-8
(SCALE - 1:10)



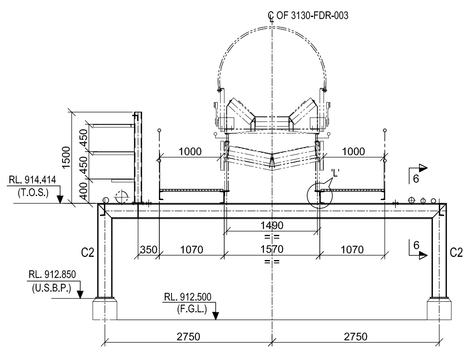
ELEVATION OF GROUND PORTAL MKD. GP4
(SCALE 1:50)



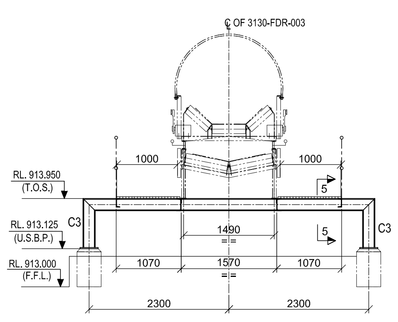
ELEVATION OF GROUND PORTAL MKD. GP3
(SCALE 1:50)



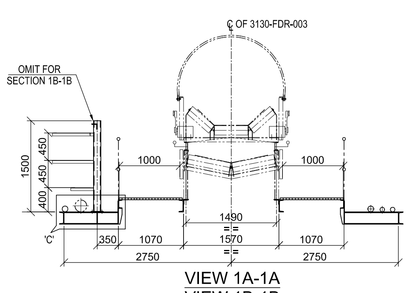
ELEVATION OF GROUND PORTAL MKD. GP2
(SCALE 1:50)



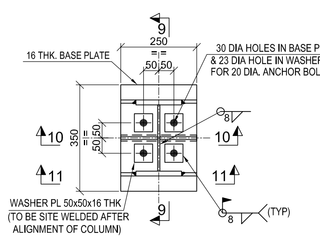
ELEVATION OF GROUND PORTAL MKD. GP1
(SCALE 1:50)



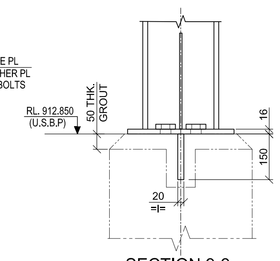
VIEW 2-2
(SCALE 1:50)



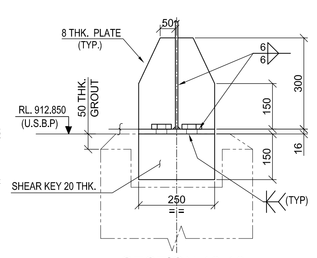
VIEW 1A-1A
VIEW 1B-1B
(SCALE 1:50)



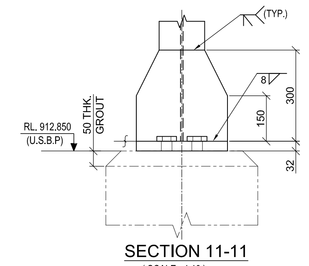
BASE PLATE DETAIL FOR GP1
(SCALE - 1:10)



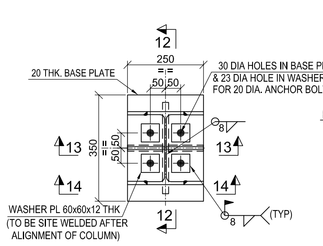
SECTION 9-9
(SCALE - 1:10)



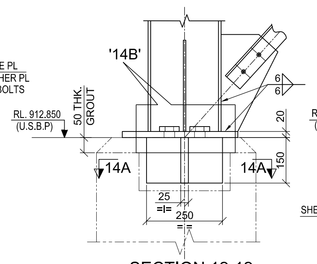
SECTION 10-10
(SCALE - 1:10)



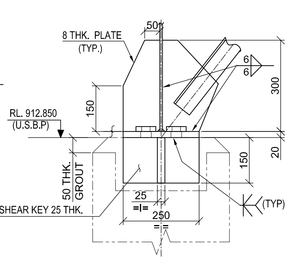
SECTION 11-11
(SCALE - 1:10)



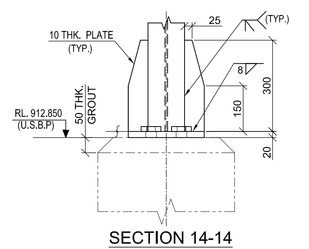
BASE PLATE DETAIL FOR GP2 & GP3
(SCALE - 1:10)



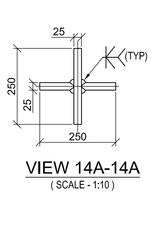
SECTION 12-12
(SCALE - 1:10)



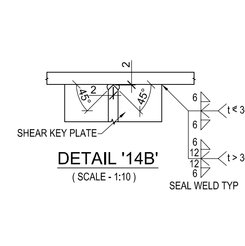
SECTION 13-13
(SCALE - 1:10)



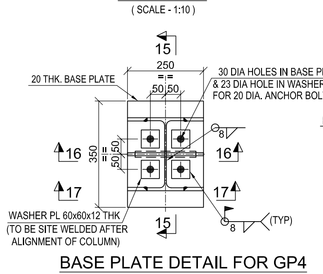
SECTION 14-14
(SCALE - 1:10)



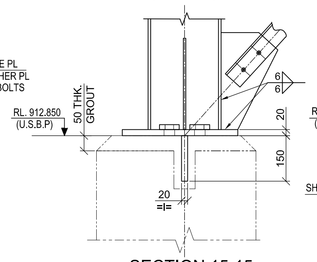
VIEW 14A-14A
(SCALE - 1:10)



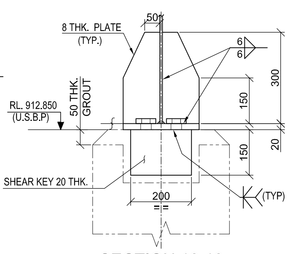
DETAIL '14B'
(SCALE - 1:10)



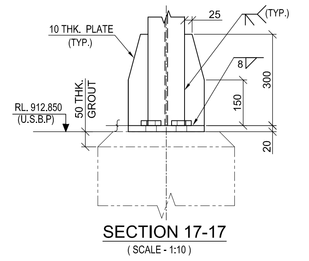
BASE PLATE DETAIL FOR GP4
(SCALE - 1:10)



SECTION 15-15
(SCALE - 1:10)



SECTION 16-16
(SCALE - 1:10)



SECTION 17-17
(SCALE - 1:10)

LEGEND :-
R.L - REDUCED LEVEL
℄ - CENTRE LINE
T.O.S. - TOP OF STEEL,
F.G.L. - FINISHED GROUND LEVEL,
F.F.L - FINISHED FLOOR LEVEL,
TYP. - TYPICAL,
MKD. - MARKED,
THK. - THICK
F/S - FAR SIDE
N/S - NEAR SIDE
W.P.-WORKING POINT
U.S.B.P. - UNDER SIDE OF BASE PLATE.
U.N.O. - UNLESS NOTED OTHERWISE
T.O.W.W. - TOP OF WALKWAY
REF. - REFER
DRG. NO. - DRAWING NUMBER

NOTES :-

1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS & COORDINATES ARE IN METER UNO.
2. GRADE OF STEEL USED IS S275R AS PER EN10025.
3. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD - 216 - 3130 -EG -ST - GAD - 1018 & MD - 216 - 3130 -EG -ST - GAD - 1019.
4. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITARIYA NO MD - 216 - 1000 -EG -ST - CRT - 1001

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-ELV-1015	FEEDER 3130-FDR-004 - ELEVATION								
MD-216-3130-EG-ME-LPX-1008	FEEDER 3130-FDR-004 - PLAN								
MD-216-3130-EG-ME-ELV-1013	FEEDER 3130-FDR-003 -ELEVATION								
MD-216-3130-EG-ME-LPX-1007	FEEDER 3130-FDR-003 - PLAN	A01	03/04/2020	ISSUED FOR REFERENCE	SUS	SBY	APR	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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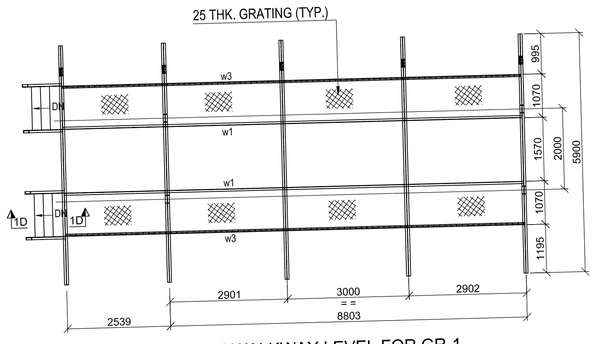
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Saudi Arabian Mining Company شركة معادن العربية السعودية

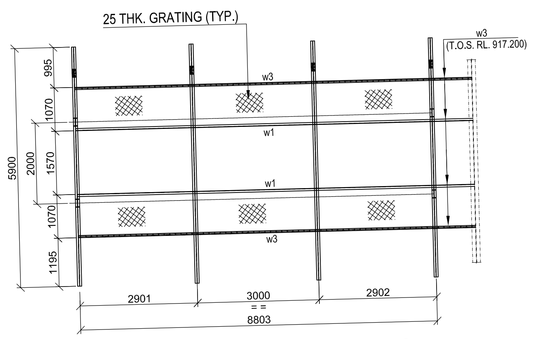
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LARSEN & TOUBRO DRAWING NUMBER			REVISION
019045_100_S_G_26013_029			0
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707 _ GAU01 _ ZC01 _ MTL01 _ 31308		OU100668060	0
DRAWN	SUS	01/04/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SBY	02/04/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	APR	01/04/2020	STRUCTURAL GA FOR 3130-FDR-003 & 004, SH1
APPROVED	PRI	03/04/2020	PLAN, ELEVATIONS & DETAILS
EMPLOYER			
SCALE	1:100,50,20,10	A0	DRAWING NUMBER: MD - 216 - 3130 - EG - ST - GAD - 1017 | REV: A01

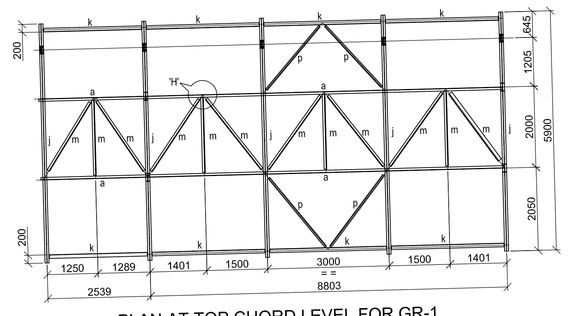
DRAWING NUMBER: MD - 216 - 3130 - EG - ST - GAD - 1018 | REV: A01



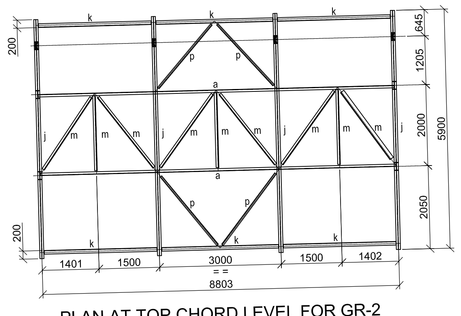
PLAN AT WALKWAY LEVEL FOR GR-1
(SCALE 1:75)



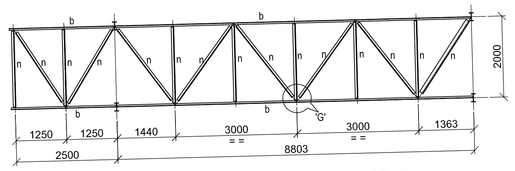
PLAN AT WALKWAY LEVEL FOR GR-2
(SCALE 1:75)



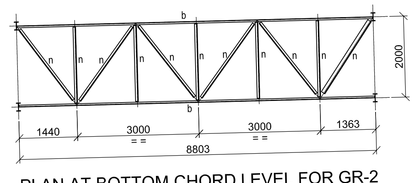
PLAN AT TOP CHORD LEVEL FOR GR-1
(SCALE 1:75)



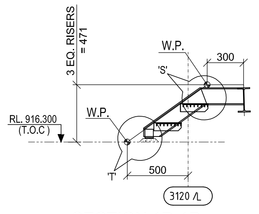
PLAN AT TOP CHORD LEVEL FOR GR-2
(SCALE 1:75)



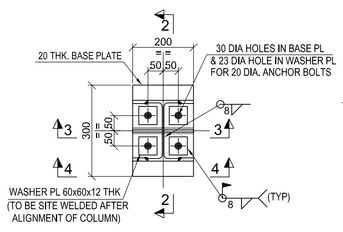
PLAN AT BOTTOM CHORD LEVEL FOR GR-1
(SCALE 1:75)



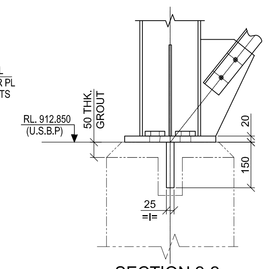
PLAN AT BOTTOM CHORD LEVEL FOR GR-2
(SCALE 1:75)



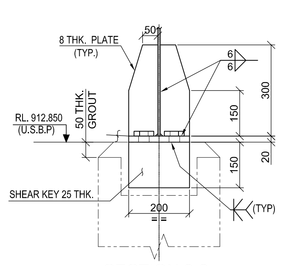
SECTION 1D-1D
(SCALE 1:25)



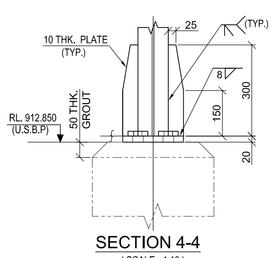
BASE PLATE DETAIL FOR TR1
(SCALE - 1:10)



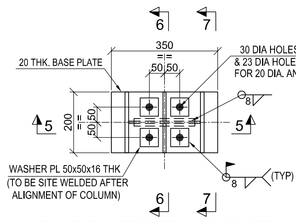
SECTION 2-2
(SCALE - 1:10)



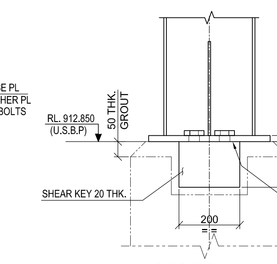
SECTION 3-3
(SCALE - 1:10)



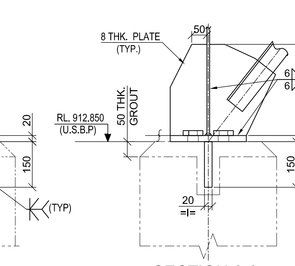
SECTION 4-4
(SCALE - 1:10)



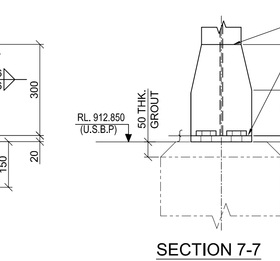
BASE PLATE DETAIL FOR TR2
(SCALE - 1:10)



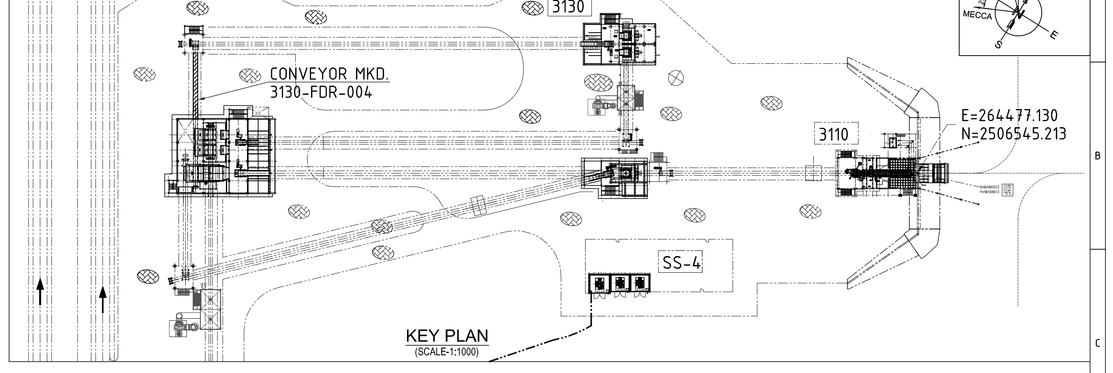
SECTION 5-5
(SCALE - 1:10)



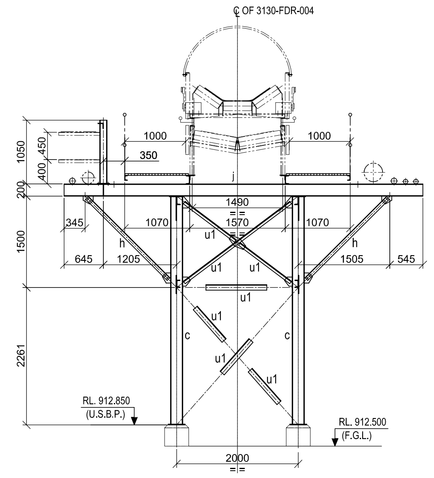
SECTION 6-6
(SCALE - 1:10)



SECTION 7-7
(SCALE - 1:10)



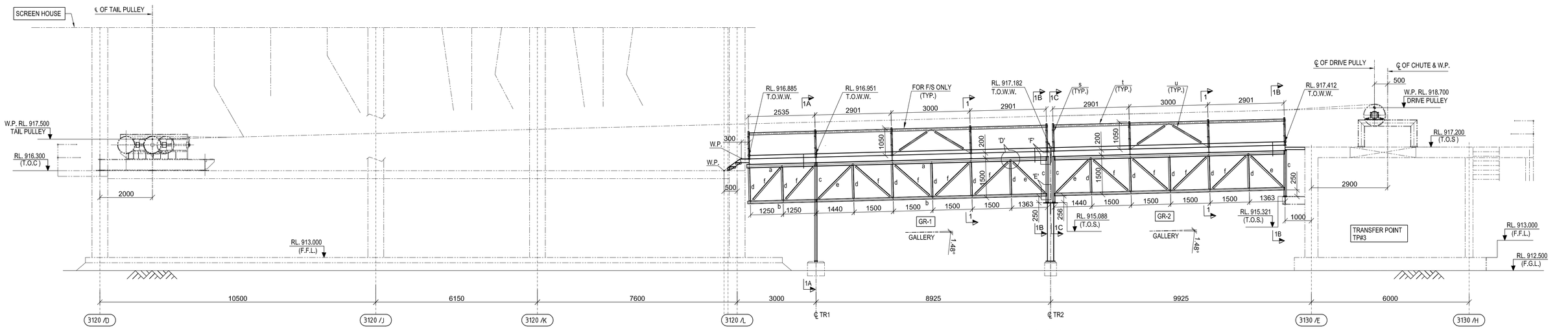
KEY PLAN
(SCALE-1:1000)



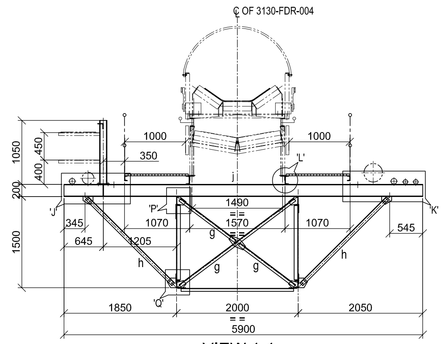
VIEW 1A-1A
ELEVATION OF TRESTLE MKD. TR1
(SCALE 1:50)

SCHEDULE OF MEMBERS					
MEMBER MARKED	MEMBER	SECTON	FACTORED AXIAL FORCE(kN)	FACTORED SHEAR FORCE(kN)	REMARKS
a	UPN 140	[	130	-	GUSSET PLATE 10 THK.
b	UPN 100	[	130	-	
c	IPE 200	I	-	-	
d	L 65X65X6	L	40	-	
e	L 90X90X8	L	90	-	
f	L 50X50X6	L	80	-	
g	L 50X50X6	L	20	-	GUSSET PLATE 8 THK.
h	L 65X65X6	L	20	-	
j	IPE 200	I	-	-	
k	L 90X90X8	L	-	-	
m	L 65X65X6	L	15	-	
n	L 65X65X6	L	15	-	
p	L 65X65X6	L	-	-	
s	UPN 100	[	-	-	
t	L 75X75X8	L	-	-	
u	L 50X50X6	L	-	-	
u1	2L 50X50X6	⊓	75	-	
w1	UPN 140	[	-	-	
w3	UPN 100	[	-	-	
C1	1. PL. 130X12 2. PL. 236X8	I 1 2 1	-	-	

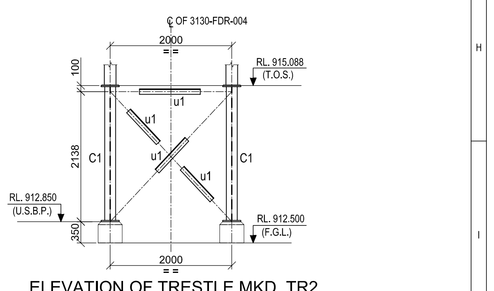
GRADE OF MATERIAL IS S275 JR
FOR DEVELOPING CONNECTIONS MEMBER END FORCES MENTIONED IN SCHEDULE OF MEMBERS SHALL BE FOLLOWED.
MINIMUM 6mm FILLET WELD OF 150 mm LENGTH SHALL BE PROVIDED FOR WELDED CONNECTION. FOR BOLTED CONNECTION, MINIMUM 2 Nos. M20 BOLTS SHALL BE PROVIDED.



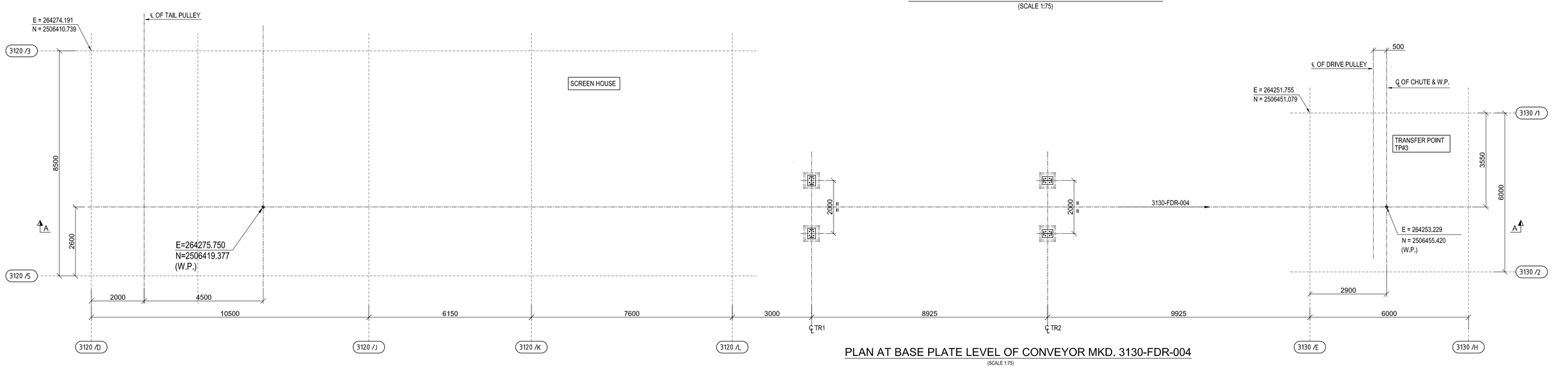
VIEW A-A
ELEVATION OF CONVEYOR MKD. 3130-FDR-004
(SCALE 1:75)



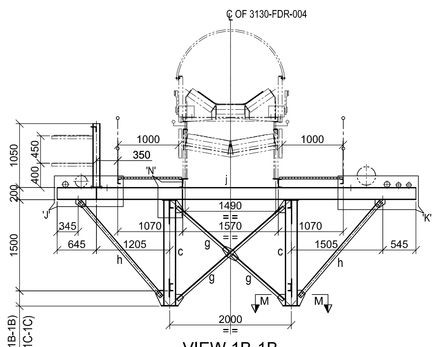
VIEW 1-1
(SCALE 1:50)



ELEVATION OF TRESTLE MKD. TR2
(SCALE 1:50)



PLAN AT BASE PLATE LEVEL OF CONVEYOR MKD. 3130-FDR-004
(SCALE 1:75)



VIEW 1B-1B
VIEW 1C-1C
(SCALE 1:50)

LEGEND :-
R.L. - REDUCED LEVEL
℄ - CENTRE LINE
T.O.S. - TOP OF STEEL.
F.G.L. - FINISHED GROUND LEVEL.
F.F.L. - FINISHED FLOOR LEVEL.
TYP. - TYPICAL
MKD. - MARKED.
THK. - THICK
F/S - FAR SIDE
N/S - NEAR SIDE
W.P.-WORKING POINT
U.S.B.P. - UNDER SIDE OF BASE PLATE
U.N.O. - UNLESS NOTED OTHERWISE
T.O.W.W. - TOP OF WALKWAY
REF. - REFER
DRG. NO. - DRAWING NUMBER

NOTES :-
1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS & COORDINATES ARE IN METER UNO.
2. GRADE OF STEEL USED IS S275R AS PER EN10025.
3. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING MD - 216 - 3130 -EG -ST - GAD - 1017 & MD - 216 - 3130 -EG -ST - GAD - 1019.
4. ALL STRUCTURAL WORK SHALL CONFORM TO CIVIL STRUCTURAL DESIGN CRITARIYA NO MD - 216 - 1000 -EG -ST - CRT - 1001

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3130-EG-ME-ELV-1015	FEEDER 3130-FDR-004 - ELEVATION								
MD-216-3130-EG-ME-LPX-1008	FEEDER 3130-FDR-004 - PLAN								
MD-216-3130-EG-ME-ELV-1013	FEEDER 3130-FDR-003 -ELEVATION								
MD-216-3130-EG-ME-LPX-1007	FEEDER 3130-FDR-003 - PLAN	A01	03/04/2020	ISSUED FOR REFERENCE	SUS	SBY	APR	PRI	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER			REVISION
019045_100_S_G_26013_030			0
OUTOTEC DRAWING NUMBER		DOCUMENT ID	REVISION
900707 _ GAU01 _ ZC01 _ MTL01 _ 31309		OU100668061	0
DRAWN	SUS	01/04/2020	MANSOURAH - MASSARAH GOLD PROJECT
CHECKED	SBY	02/04/2020	GENERAL ARRANGEMENT DRAWING
ENGINEER	APR	01/04/2020	STRUCTURAL GA FOR 3130-FDR-003 & 004, SH2
APPROVED	PRI	03/04/2020	PLAN, ELEVATIONS & DETAILS
EMPLOYER			
SCALE	1:100,50,20,10	A0	DRAWING NUMBER: MD - 216 - 3130 - EG - ST - GAD - 1018
			REV: A01

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	04-04-2020	29-04-2020	01-11-2023				Date	04-04-2020	29-04-2020	01-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 3 **STAAD INPUT & OUTPUT FILES OF FOUNDATION FOR GROUND PORTALS**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 03-Apr-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART GROUND PORTAL FOUNDATIONS
JOB REF FOUNDATIONS FOR CONVEYOR 3130-FDR-003
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 24.9 0.125 -0.35; 4 24.9 0.95 -0.35; 7 24.9 0.125 -4.95; 13 24.9 0.95 -0.795;
14 24.9 0.95 -1.865; 15 24.9 0.95 -3.435; 16 24.9 0.95 -4.95;
33 30.88 0.314 -0.795; 34 30.88 0.314 -1.865; 35 30.88 0.314 -3.435;
36 30.88 0.314 -4.505; 53 18.65 -0.15 0.1; 54 18.65 1.615 0.1;
55 18.65 -0.15 -5.4; 56 18.65 1.615 -5.4; 57 18.65 1.615 -0.795;
58 18.65 1.615 -1.865; 59 18.65 1.615 -3.435; 60 18.65 1.615 -4.505;
61 13.45 -0.15 0.1; 62 13.45 2.168 0.1; 63 13.45 -0.15 -5.4;
64 13.45 2.168 -5.4; 65 13.45 2.168 -0.795; 66 13.45 2.168 -1.865;
67 13.45 2.168 -3.435; 68 13.45 2.168 -4.505; 69 8.2 -0.15 0.1;
70 8.2 2.727 0.1; 71 8.2 -0.15 -5.4; 72 8.2 2.727 -5.4; 73 8.2 2.727 -0.795;
74 8.2 2.727 -1.865; 75 8.2 2.727 -3.435; 76 8.2 2.727 -4.505; 77 3 -0.15 0.1;
78 3 3.28 0.1; 79 3 -0.15 -5.4; 80 3 3.28 -5.4; 81 3 3.28 -0.795;
82 3 3.28 -1.865; 83 3 3.28 -3.435; 84 3 3.28 -4.505; 89 0 3.599 -0.795;
90 0 3.599 -1.865; 91 0 3.599 -3.435; 92 0 3.599 -4.505; 93 0 3.599 -1.65;
94 0 3.599 -3.65; 100 16.0505 1.892 -0.795; 101 10.8255 2.4475 -0.795;
103 5.6 3.003 0.1; 104 10.8255 2.4475 0.1; 105 16.0505 1.892 0.1;
108 1.5005 3.44 -3.435; 109 1.5005 3.44 -1.865; 111 5.6 3.003 -3.435;
113 5.6 3.003 -1.865; 114 10.8255 2.4475 -3.435; 115 10.8255 2.4475 -1.865;
116 16.0505 1.892 -3.435; 117 16.0505 1.892 -1.865; 134 22.8 1.17367 -1.865;
135 20.75 1.39233 -1.865; 136 22.8 1.17367 -3.435; 137 20.75 1.39233 -3.435;
138 26.8995 0.7385 -1.865; 139 26.8995 0.7385 -3.435;
140 26.8995 0.7385 -0.795; 141 26.8995 0.7385 -4.505; 142 16.0505 1.892 -4.505;
143 10.8255 2.4475 -4.505; 144 1.5005 3.44 -4.505; 145 1.5005 3.44 -0.795;
146 22.8 1.17367 -0.795; 147 20.75 1.39233 -0.795; 150 22.8 1.17367 -4.505;
151 20.75 1.39233 -4.505; 163 5.6 3.003 -4.505; 169 5.6 3.003 -0.795;
170 10.8255 2.4475 -5.4; 171 28.8798 0.52525 -1.865;
172 28.8798 0.52525 -3.435; 173 24.9 0.95 -4.505; 174 28.8798 0.52525 -0.795;
175 28.8798 0.52525 -4.505; 176 13.45 2.168 -2.65; 177 18.65 -2.5 0.1;
178 18.65 -2.5 -5.4; 179 13.45 -2.5 0.1; 180 13.45 -2.5 -5.4; 181 8.2 -2.5 0.1;
182 8.2 -2.5 -5.4; 183 3 -2.5 0.1; 184 3 -2.5 -5.4;
MEMBER INCIDENCES
1 1 4; 4 7 16; 7 4 13; 10 13 14; 11 14 15; 12 15 173; 15 13 146; 24 14 134;
30 15 136; 36 173 150; 41 33 174; 42 34 171; 43 35 172; 44 36 175; 67 53 54;
68 55 56; 69 54 57; 70 57 58; 71 58 59; 72 59 60; 73 60 56; 74 61 62; 75 63 64;
76 62 65; 77 65 66; 78 66 176; 79 67 68; 80 68 64; 81 69 70; 82 71 72;
83 70 73; 84 73 74; 85 74 75; 86 75 76; 87 76 72; 88 77 78; 89 79 80; 90 78 81;
91 81 82; 92 82 83; 93 83 84; 94 84 80; 100 91 94; 103 60 142; 104 68 143;
106 84 144; 108 59 116; 109 67 114; 111 83 108; 113 58 117; 114 66 115;
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116 82 109; 118 57 100; 119 65 101; 121 81 145; 122 78 103; 123 70 104;
124 62 105; 127 80 72; 128 72 170; 129 64 56; 132 69 62; 133 61 70; 134 71 64;
135 63 72; 136 93 90; 150 100 65; 151 101 73; 152 73 169; 153 103 70;
154 104 62; 155 105 54; 158 103 169; 159 104 101; 160 105 100; 163 108 91;
164 109 90; 165 75 111; 166 111 83; 167 74 113; 168 113 82; 169 114 75;
170 115 74; 171 116 67; 172 117 66; 189 134 135; 190 135 58; 191 136 137;
192 137 59; 193 138 14; 194 139 15; 195 108 109; 196 111 113; 198 114 115;
199 116 117; 208 137 135; 209 136 134; 210 139 138; 211 90 108; 212 108 82;
213 82 111; 214 111 74; 216 74 114; 217 114 66; 218 66 116; 219 116 58;
232 58 137; 233 137 134; 234 134 15; 235 15 138; 236 138 172; 237 140 13;
238 141 173; 239 142 68; 240 143 76; 241 144 92; 242 145 89; 243 146 147;
244 147 57; 247 150 151; 248 151 60; 259 76 163; 260 163 84; 261 144 108;
262 109 145; 263 163 111; 265 143 114; 266 142 116; 275 150 136; 276 151 137;
278 134 146; 279 135 147; 284 141 139; 285 138 140; 294 117 100; 295 115 101;
298 169 81; 299 113 169; 302 170 64; 303 170 143; 307 77 80; 308 71 70;
309 69 72; 310 61 176; 312 171 138; 313 172 139; 314 172 34; 315 172 171;
316 173 16; 317 174 140; 318 175 141; 319 175 172; 320 171 174; 325 176 67;
326 63 176; 327 79 78; 328 184 79; 329 183 77; 330 182 71; 331 181 69;
332 180 63; 333 179 61; 334 178 55; 335 177 53; 336 79 77; 337 71 69;
338 63 61; 339 55 53; 340 69 61; 341 71 63;
DEFINE MATERIAL START
ISOTROPIC STEEL
E 2.05e+008
POISSON 0.3
DENSITY 76.8195
ALPHA 1.2e-005
DAMP 0.03
TYPE STEEL
STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY EUROPEAN
74 75 81 82 88 89 TABLE ST IPE240
100 136 TABLE ST IPE200
1 4 7 10 TO 12 316 TAPERED 0.2 0.008 0.2 0.12 0.012 0.12 0.012
67 TO 73 76 TO 80 83 TO 87 90 TO 94 -
325 TAPERED 0.25 0.008 0.25 0.15 0.016 0.15 0.016
15 36 41 44 103 104 106 118 119 121 150 TO 152 237 TO 244 247 248 259 260 -
298 317 318 TABLE ST UPN140
24 30 108 109 111 113 114 116 163 TO 172 189 TO 192 TABLE ST UPN200
122 TO 124 127 TO 129 153 TO 155 158 TO 160 302 TABLE ST UPN200
132 TO 135 307 TO 310 326 327 TABLE LD L75X75X8 SP 0.01
198 211 212 232 TO 236 265 295 314 TABLE ST L65X65X6
195 196 199 208 TO 210 261 TO 263 266 275 276 278 279 284 285 294 299 303 -
315 319 320 TABLE ST L50X50X6
42 43 193 194 312 313 TABLE ST UPN240
MEMBER PROPERTY FRENCH
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213 214 216 TO 219 TABLE ST L75X75X8
MEMBER PROPERTY FRENCH
328 TO 335 PRIS YD 0.6 ZD 0.6
336 TO 339 PRIS YD 0.5 ZD 0.3
MEMBER PROPERTY
340 341 PRIS YD 0.4 ZD 0.3
CONSTANTS
BETA 90 MEMB 1 4 67 68 74 75 81 82 88 89
MATERIAL STEEL MEMB 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 -
106 108 109 111 113 114 116 118 119 121 TO 124 127 TO 129 132 TO 136 150 -
151 TO 155 158 TO 160 163 TO 172 189 TO 196 198 199 208 TO 214 216 TO 219 -
232 TO 244 247 248 259 TO 263 265 266 275 276 278 279 284 285 294 295 298 -
299 302 303 307 TO 310 312 TO 320 325 TO 327
MATERIAL CONCRETE MEMB 328 TO 341
SUPPORTS
33 TO 36 89 TO 92 FIXED BUT FX MX MY MZ
177 TO 184 FIXED
1 7 PINNED
MEMBER TRUSS
127 129 132 TO 135 195 196 198 199 208 TO 214 216 TO 219 232 TO 236 -
261 TO 263 265 266 275 276 278 279 284 285 294 295 299 303 307 TO 310 326
MEMBER RELEASE
15 24 30 36 67 68 74 TO 76 81 TO 83 88 TO 90 122 TO 124 128 158 TO 159 -
160 START MP 0.99
80 87 94 153 TO 155 158 TO 160 193 194 237 238 302 END MP 0.99
MEMBER OFFSET
340 341 START 0 -0.2 0
340 341 END 0 -0.2 0
336 TO 339 START 0 -0.25 0
336 TO 339 END 0 -0.25 0
LOAD 1 DL
SELFWEIGHT Y -1.1
*WALKWAY LOAD @50KG/SQ. M Including Seal Plate & H/R
MEMBER LOAD
15 36 41 44 103 104 106 118 119 121 150 TO 152 237 TO 244 247 248 259 260 -
298 317 318 UNI GY -0.32
24 30 42 43 108 109 111 113 114 116 163 TO 172 189 TO 194 312 313 UNI GY -0.47
*PIPE DEAD LOAD
MEMBER LOAD
69 76 83 90 CON GY -2.03
73 80 87 94 CON GY -3.57
*CONVEYOR DEAD LOAD
MEMBER LOAD
42 43 CON GY -5 0.2
42 43 CON GY -5 1.5
312 313 CON GY -5 1.714
193 194 CON GY -5 1.845
24 30 CON GY -5 1.96
190 192 CON GY -5 0.036
108 113 CON GY -5 0.401
171 172 CON GY -5 0.736
109 114 CON GY -5 1.171
169 170 CON GY -5 1.532
165 167 CON GY -5 1.891
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166 168 CON GY -5 2.277
163 164 CON GY -5 1.154
*
MEMBER LOAD
42 43 CON GX 1.5 0.2
42 43 CON GX 1.5 1.5
312 313 CON GX 1.5 1.714
193 194 CON GX 1.5 1.845
24 30 CON GX 1.5 1.96
190 192 CON GX 1.5 0.036
108 113 CON GX 1.5 0.401
171 172 CON GX 1.5 0.736
109 114 CON GX 1.5 1.171
169 170 CON GX 1.5 1.532
165 167 CON GX 1.5 1.891
166 168 CON GX 1.5 2.277
163 164 CON GX 1.5 1.154
*
MEMBER LOAD
42 43 CON GZ 0.8 0.2
42 43 CON GZ 0.8 1.5
312 313 CON GZ 0.8 1.714
193 194 CON GZ 0.8 1.845
24 30 CON GZ 0.8 1.96
190 192 CON GZ 0.8 0.036
108 113 CON GZ 0.8 0.401
171 172 CON GZ 0.8 0.736
109 114 CON GZ 0.8 1.171
169 170 CON GZ 0.8 1.532
165 167 CON GZ 0.8 1.891
166 168 CON GZ 0.8 2.277
163 164 CON GZ 0.8 1.154
*CABLE TRAY LOAD
69 76 83 90 158 TO 160 CON GY -11.76
69 76 83 90 158 TO 160 CMOM GX 4.7
LOAD 2 LL
*WALKWAY LOAD @400KG/SQ. MTER(FOR GALLERY)
MEMBER LOAD
15 24 30 36 41 TO 44 103 104 106 108 109 111 113 114 116 118 119 121 -
150 TO 152 163 TO 172 189 TO 194 237 TO 244 247 248 259 260 298 312 313 317 -
318 UNI GY -2
LOAD 3 ML
*CONVEYOR MATERIAL LOAD
MEMBER LOAD
42 43 CON GY -5.5 0.2
42 43 CON GY -5.5 1.5
312 313 CON GY -5.5 1.714
193 194 CON GY -5.5 1.845
24 30 CON GY -5.5 1.96
190 192 CON GY -5.5 0.036
108 113 CON GY -5.5 0.401
171 172 CON GY -5.5 0.736
109 114 CON GY -5.5 1.171
169 170 CON GY -5.5 1.532
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165 167 CON GY -5.5 1.891
166 168 CON GY -5.5 2.277
163 164 CON GY -5.5 1.154
** Spillage Load @75kg/ Sq. m on seal plate
24 30 42 43 108 109 111 113 114 116 163 TO 172 189 TO 194 312 313 UNI GY -0.375
LOAD 4 TL1
TEMPERATURE LOAD
1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 106 108 109 111 113 -
114 116 118 119 121 TO 124 127 TO 129 132 TO 136 150 TO 155 158 TO 160 163 -
164 TO 172 189 TO 196 198 199 208 TO 214 216 TO 219 232 TO 244 247 248 259 -
260 TO 263 265 266 275 276 278 279 284 285 294 295 298 299 302 303 -
307 TO 310 312 TO 320 TEMP 25
LOAD 5 TL2
TEMPERATURE LOAD
1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 106 108 109 111 113 -
114 116 118 119 121 TO 124 127 TO 129 132 TO 136 150 TO 155 158 TO 160 163 -
164 TO 172 189 TO 196 198 199 208 TO 214 216 TO 219 232 TO 244 247 248 259 -
260 TO 263 265 266 275 276 278 279 284 285 294 295 298 299 302 303 -
307 TO 310 312 TO 320 TEMP -25
LOAD 6 WL +Z
MEMBER LOAD
1 4 67 68 74 75 81 82 88 89 UNI GZ 0.5
*DUE TO CONVEYOR HOOD
42 43 CON GZ 0.58 0.2
42 43 CON GZ 0.58 1.5
312 313 CON GZ 0.58 1.714
193 194 CON GZ 0.58 1.845
24 30 CON GZ 0.58 1.96
190 192 CON GZ 0.58 0.036
108 113 CON GZ 0.58 0.401
171 172 CON GZ 0.58 0.736
109 114 CON GZ 0.58 1.171
169 170 CON GZ 0.58 1.532
165 167 CON GZ 0.58 1.891
166 168 CON GZ 0.58 2.277
163 164 CON GZ 0.58 1.154
*DUE TO CABLE TRAY
69 76 83 90 158 TO 160 CON GZ 2.38
69 76 83 90 158 TO 160 CMOM GX 2
LOAD 7 WL -Z
MEMBER LOAD
1 4 67 68 74 75 81 82 88 89 UNI GZ -0.5
*DUE TO CONVEYOR HOOD
42 43 CON GZ -0.58 0.2
42 43 CON GZ -0.58 1.5
312 313 CON GZ -0.58 1.714
193 194 CON GZ -0.58 1.845
24 30 CON GZ -0.58 1.96
190 192 CON GZ -0.58 0.036
108 113 CON GZ -0.58 0.401
171 172 CON GZ -0.58 0.736
109 114 CON GZ -0.58 1.171
169 170 CON GZ -0.58 1.532
165 167 CON GZ -0.58 1.891
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166 168 CON GZ -0.58 2.277
163 164 CON GZ -0.58 1.154
*DUE TO CABLE TRAY
69 76 83 90 158 TO 160 CON GZ -2.38
69 76 83 90 158 TO 160 CMOM GX -2
LOAD 11 G
REPEAT LOAD
1 1.0
LOAD 12 G + QN + MN
REPEAT LOAD
1 1.0 2 1.0 3 1.0
****LOAD COMBINATION for structural design
*LOAD COMB 201 1.4 DL
*1 1.4
*LOAD COMB 202 1.2 (DL + TL1) + 1.6  (LL+ML)
*1 1.2 4 1.2 2 1.6 3 1.6
*LOAD COMB 203 1.2 (DL + TL1) + 1.6  (LL+ML)
*1 1.2 5 1.2 2 1.6 3 1.6
*LOAD COMB 204 1.2 DL + 1.6 WL + 0.5 (LL+ML)
*1 1.2 6 1.6 2 0.5 3 0.5
*LOAD COMB 205 1.2 (DL + TL1) + 1.6  (LL+ML)
*1 1.2 7 1.6 2 0.5 3 0.5
*LOAD COMB 206 0.9 DL+ 1.6 WL
*1 0.9 6 1.6
*LOAD COMB 207 0.9 DL+ 1.6 WL
*1 0.9 7 1.6
****LOAD COMBINATION FOR BEARING PRESSURE CHECK
LOAD COMB 301 1.0 DL
1 1.0
LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
1 1.0 4 1.0 2 1.0 3 1.0
LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
1 1.0 5 1.0 2 1.0 3 1.0
LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
1 1.0 6 0.75 2 0.75 3 0.75
LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
1 1.0 7 0.75 2 0.75 3 0.75
LOAD COMB 306 0.6 DL + 1.0 WL(+Z)
1 0.6 6 1.0
LOAD COMB 307 0.6 DL + 1.0 WL(-Z)
1 0.6 7 1.0
************************************************************
****LOAD COMBINATION FOR FOUNDATION DESIGN
LOAD COMB 401 1.4 DL
1 1.4
LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
1 1.4 4 1.4 2 1.7 3 1.7
LOAD COMB 403 1.4 (DL + TL2) + 1.7  (LL+ML)
1 1.4 5 1.4 2 1.7 3 1.7
LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
1 1.2 6 1.6 2 0.5 3 0.5
LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
1 1.2 7 1.6 2 0.5 3 0.5
LOAD COMB 406 0.9 DL + 1.6 WL(+Z)
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1 0.9 6 1.6
LOAD COMB 407 0.9 DL + 1.6 WL(-Z)
1 0.9 7 1.6
************************************************************
PERFORM ANALYSIS
*STEEL TAKE OFF LIST 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 -
*106 108 109 111 113 114 116 118 119 121 TO 124 127 TO 129 132 TO 136 150 -
*151 TO 155 158 TO 160 163 TO 172 189 TO 196 198 199 208 TO 214 216 TO 219 -
*232 TO 244 247 248 259 TO 263 265 266 275 276 278 279 284 285 294 295 298 -
*299 302 303 307 TO 310 312 313 316 TO 318 325
*LOAD LIST 201 TO 207
*PARAMETER 1
*CODE LRFD
*FYLD 275000 MEMB 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 -
*106 108 109 111 113 114 116 118 119 121 123 124 127 TO 129 132 TO 136 150 -
*151 TO 152 154 155 158 TO 160 163 TO 172 189 TO 194 237 TO 244 247 248 259 -
*260 298 302 312 313 316 TO 318 325
*FU 410000 MEMB 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 106 -
*108 109 111 113 114 116 118 119 121 123 124 127 TO 129 132 TO 136 -
*150 TO 152 154 155 158 TO 160 163 TO 172 189 TO 194 237 TO 244 247 248 259 -
*260 298 302 312 313 316 TO 318 325
*TRACK 0 MEMB 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 106 -
*108 109 111 113 114 116 118 119 121 123 124 127 TO 129 132 TO 136 -
*150 TO 152 154 155 158 TO 160 163 TO 172 189 TO 194 237 TO 244 247 248 259 -
*260 298 302 312 313 316 TO 318 325
*BEAM 1 MEMB 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 106 108 -
*109 111 113 114 116 118 119 121 123 124 127 TO 129 132 TO 136 150 TO 152 -
*154 155 158 TO 160 163 TO 172 189 TO 194 237 TO 244 247 248 259 260 298 302 -
*312 313 316 TO 318 325
*TMAIN 300 MEMB 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 106 -
*108 109 111 113 114 116 118 119 121 123 124 127 TO 129 132 TO 136 -
*150 TO 152 154 155 158 TO 160 163 TO 172 189 TO 194 237 TO 244 247 248 259 -
*260 298 302 312 313 316 TO 318 325
*MAIN 200 MEMB 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 106 -
*108 109 111 113 114 116 118 119 121 123 124 127 TO 129 132 TO 136 -
*150 TO 152 154 155 158 TO 160 163 TO 172 189 TO 194 237 TO 244 247 248 259 -
*260 298 302 312 313 316 TO 318 325
*NSF 0.8 MEMB 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 106 -
*108 109 111 113 114 116 118 119 121 123 124 127 TO 129 132 TO 136 -
*150 TO 152 154 155 158 TO 160 163 TO 172 189 TO 194 237 TO 244 247 248 259 -
*260 298 302 312 313 316 TO 318 325
*UNT 1.9 MEMB 7 10 TO 12 69 TO 73 76 TO 80 83 TO 87 90 TO 94 100 136 316 325
*LY 1.9 MEMB 7 10 TO 12 69 TO 73 76 TO 80 83 TO 87 90 TO 94 100 136 316 325
*LX 1.9 MEMB 7 10 TO 12 69 TO 73 76 TO 80 83 TO 87 90 TO 94 100 136 316 325
*LZ 5.5 MEMB 7 10 TO 12 69 TO 73 76 TO 80 83 TO 87 90 TO 94 100 136 316 325
*KZ 2 MEMB 103 104 106 108 109 111 113 114 116 118 119 121 150 TO 152 -
*163 TO 172 239 TO 242 259 260 298
*KY 0.5 MEMB 127 129
*KX 0.5 MEMB 127 129
*KZ 3 MEMB 15 24 30 36 41 TO 44 165 TO 168 189 TO 194 237 238 243 244 247 248 -
*259 260 312 313 317 318
*KZ 0.85 MEMB 211 TO 214 216 TO 219 232 TO 236 314
*KZ 0.5 MEMB 132 TO 135 307 TO 309 327
*CHECK CODE MEMB 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 106 -
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*108 109 111 113 114 116 118 119 121 TO 124 127 TO 129 132 TO 136 150 TO 155 -
*158 TO 160 163 TO 172 189 TO 196 198 199 208 TO 214 216 TO 219 232 TO 244 -
*247 248 259 TO 263 265 266 275 276 278 279 284 285 294 295 298 299 302 303 -
*307 TO 310 312 TO 320 325 TO 327
LOAD LIST 401 TO 407
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 328 TO 341
FYMAIN 420000 MEMB 328 TO 341
TRACK 1 MEMB 328 TO 341
CLB 0.05 MEMB 328 TO 341
CLT 0.05 MEMB 328 TO 341
CLS 0.05 MEMB 328 TO 341
MAXMAIN 20 MEMB 328 TO 335
MINMAIN 20 MEMB 328 TO 335
MAXMAIN 16 MEMB 336 TO 341
MINMAIN 12 MEMB 336 TO 341
MINSEC 10 MEMB 328 TO 335
MINSEC 8 MEMB 336 TO 341
REINF 0 MEMB 328 TO 335
DESIGN COLUMN 328 TO 335
DESIGN BEAM 336 TO 341
END CONCRETE DESIGN
FINISH
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**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR  4, 2020                *
*           Time=    12:33:55                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
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    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\13-Maaden-Crushing_n_Screening\Conv-3130-FDR-003 _004\Staad\3130-FDR-0.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 03-APR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 0
    8. JOB PART GROUND PORTAL FOUNDATIONS
    9. JOB REF FOUNDATIONS FOR CONVEYOR 3130-FDR-003
   10. ENGINEER NAME APL
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 24.9 0.125 -0.35; 4 24.9 0.95 -0.35; 7 24.9 0.125 -4.95; 13 24.9 0.95 -0.795
   18. 14 24.9 0.95 -1.865; 15 24.9 0.95 -3.435; 16 24.9 0.95 -4.95
   19. 33 30.88 0.314 -0.795; 34 30.88 0.314 -1.865; 35 30.88 0.314 -3.435
   20. 36 30.88 0.314 -4.505; 53 18.65 -0.15 0.1; 54 18.65 1.615 0.1
   21. 55 18.65 -0.15 -5.4; 56 18.65 1.615 -5.4; 57 18.65 1.615 -0.795
   22. 58 18.65 1.615 -1.865; 59 18.65 1.615 -3.435; 60 18.65 1.615 -4.505
   23. 61 13.45 -0.15 0.1; 62 13.45 2.168 0.1; 63 13.45 -0.15 -5.4
   24. 64 13.45 2.168 -5.4; 65 13.45 2.168 -0.795; 66 13.45 2.168 -1.865
   25. 67 13.45 2.168 -3.435; 68 13.45 2.168 -4.505; 69 8.2 -0.15 0.1
   26. 70 8.2 2.727 0.1; 71 8.2 -0.15 -5.4; 72 8.2 2.727 -5.4; 73 8.2 2.727 -0.795
   27. 74 8.2 2.727 -1.865; 75 8.2 2.727 -3.435; 76 8.2 2.727 -4.505; 77 3 -0.15 0.1
   28. 78 3 3.28 0.1; 79 3 -0.15 -5.4; 80 3 3.28 -5.4; 81 3 3.28 -0.795
   29. 82 3 3.28 -1.865; 83 3 3.28 -3.435; 84 3 3.28 -4.505; 89 0 3.599 -0.795
   30. 90 0 3.599 -1.865; 91 0 3.599 -3.435; 92 0 3.599 -4.505; 93 0 3.599 -1.65
   31. 94 0 3.599 -3.65; 100 16.0505 1.892 -0.795; 101 10.8255 2.4475 -0.795
   32. 103 5.6 3.003 0.1; 104 10.8255 2.4475 0.1; 105 16.0505 1.892 0.1
   33. 108 1.5005 3.44 -3.435; 109 1.5005 3.44 -1.865; 111 5.6 3.003 -3.435
   34. 113 5.6 3.003 -1.865; 114 10.8255 2.4475 -3.435; 115 10.8255 2.4475 -1.865
   35. 116 16.0505 1.892 -3.435; 117 16.0505 1.892 -1.865; 134 22.8 1.17367 -1.865
   36. 135 20.75 1.39233 -1.865; 136 22.8 1.17367 -3.435; 137 20.75 1.39233 -3.435
   37. 138 26.8995 0.7385 -1.865; 139 26.8995 0.7385 -3.435
   38. 140 26.8995 0.7385 -0.795; 141 26.8995 0.7385 -4.505; 142 16.0505 1.892 -4.505
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39. 143 10.8255 2.4475 -4.505; 144 1.5005 3.44 -4.505; 145 1.5005 3.44 -0.795
40. 146 22.8 1.17367 -0.795; 147 20.75 1.39233 -0.795; 150 22.8 1.17367 -4.505
41. 151 20.75 1.39233 -4.505; 163 5.6 3.003 -4.505; 169 5.6 3.003 -0.795
42. 170 10.8255 2.4475 -5.4; 171 28.8798 0.52525 -1.865
43. 172 28.8798 0.52525 -3.435; 173 24.9 0.95 -4.505; 174 28.8798 0.52525 -0.795
44. 175 28.8798 0.52525 -4.505; 176 13.45 2.168 -2.65; 177 18.65 -2.5 0.1
45. 178 18.65 -2.5 -5.4; 179 13.45 -2.5 0.1; 180 13.45 -2.5 -5.4; 181 8.2 -2.5 0.1
46. 182 8.2 -2.5 -5.4; 183 3 -2.5 0.1; 184 3 -2.5 -5.4
47. MEMBER INCIDENCES
48. 1 1 4; 4 7 16; 7 4 13; 10 13 14; 11 14 15; 12 15 173; 15 13 146; 24 14 134
49. 30 15 136; 36 173 150; 41 33 174; 42 34 171; 43 35 172; 44 36 175; 67 53 54
50. 68 55 56; 69 54 57; 70 57 58; 71 58 59; 72 59 60; 73 60 56; 74 61 62; 75 63 64
51. 76 62 65; 77 65 66; 78 66 176; 79 67 68; 80 68 64; 81 69 70; 82 71 72
52. 83 70 73; 84 73 74; 85 74 75; 86 75 76; 87 76 72; 88 77 78; 89 79 80; 90 78 81
53. 91 81 82; 92 82 83; 93 83 84; 94 84 80; 100 91 94; 103 60 142; 104 68 143
54. 106 84 144; 108 59 116; 109 67 114; 111 83 108; 113 58 117; 114 66 115
55. 116 82 109; 118 57 100; 119 65 101; 121 81 145; 122 78 103; 123 70 104
56. 124 62 105; 127 80 72; 128 72 170; 129 64 56; 132 69 62; 133 61 70; 134 71 64
57. 135 63 72; 136 93 90; 150 100 65; 151 101 73; 152 73 169; 153 103 70
58. 154 104 62; 155 105 54; 158 103 169; 159 104 101; 160 105 100; 163 108 91
59. 164 109 90; 165 75 111; 166 111 83; 167 74 113; 168 113 82; 169 114 75
60. 170 115 74; 171 116 67; 172 117 66; 189 134 135; 190 135 58; 191 136 137
61. 192 137 59; 193 138 14; 194 139 15; 195 108 109; 196 111 113; 198 114 115
62. 199 116 117; 208 137 135; 209 136 134; 210 139 138; 211 90 108; 212 108 82
63. 213 82 111; 214 111 74; 216 74 114; 217 114 66; 218 66 116; 219 116 58
64. 232 58 137; 233 137 134; 234 134 15; 235 15 138; 236 138 172; 237 140 13
65. 238 141 173; 239 142 68; 240 143 76; 241 144 92; 242 145 89; 243 146 147
66. 244 147 57; 247 150 151; 248 151 60; 259 76 163; 260 163 84; 261 144 108
67. 262 109 145; 263 163 111; 265 143 114; 266 142 116; 275 150 136; 276 151 137
68. 278 134 146; 279 135 147; 284 141 139; 285 138 140; 294 117 100; 295 115 101
69. 298 169 81; 299 113 169; 302 170 64; 303 170 143; 307 77 80; 308 71 70
70. 309 69 72; 310 61 176; 312 171 138; 313 172 139; 314 172 34; 315 172 171
71. 316 173 16; 317 174 140; 318 175 141; 319 175 172; 320 171 174; 325 176 67
72. 326 63 176; 327 79 78; 328 184 79; 329 183 77; 330 182 71; 331 181 69
73. 332 180 63; 333 179 61; 334 178 55; 335 177 53; 336 79 77; 337 71 69
74. 338 63 61; 339 55 53; 340 69 61; 341 71 63
75. DEFINE MATERIAL START
76. ISOTROPIC STEEL
77. E 2.05E+008
78. POISSON 0.3
79. DENSITY 76.8195
80. ALPHA 1.2E-005
81. DAMP 0.03
82. TYPE STEEL
83. STRENGTH FY 253200 FU 407800 RY 1.5 RT 1.2
84. ISOTROPIC CONCRETE
85. E 2.17185E+007
86. POISSON 0.17
87. DENSITY 23.5616
88. ALPHA 1E-005
89. DAMP 0.05
90. TYPE CONCRETE
91. STRENGTH FCU 27579
92. END DEFINE MATERIAL
93. MEMBER PROPERTY EUROPEAN
94. 74 75 81 82 88 89 TABLE ST IPE240
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 95. 100 136 TABLE ST IPE200
 96. 1 4 7 10 TO 12 316 TAPERED 0.2 0.008 0.2 0.12 0.012 0.12 0.012
 97. 67 TO 73 76 TO 80 83 TO 87 90 TO 94 -
 98. 325 TAPERED 0.25 0.008 0.25 0.15 0.016 0.15 0.016
 99. 15 36 41 44 103 104 106 118 119 121 150 TO 152 237 TO 244 247 248 259 260 -
100. 298 317 318 TABLE ST UPN140
101. 24 30 108 109 111 113 114 116 163 TO 172 189 TO 192 TABLE ST UPN200
102. 122 TO 124 127 TO 129 153 TO 155 158 TO 160 302 TABLE ST UPN200
103. 132 TO 135 307 TO 310 326 327 TABLE LD L75X75X8 SP 0.01
104. 198 211 212 232 TO 236 265 295 314 TABLE ST L65X65X6
105. 195 196 199 208 TO 210 261 TO 263 266 275 276 278 279 284 285 294 299 303 -
106. 315 319 320 TABLE ST L50X50X6
107. 42 43 193 194 312 313 TABLE ST UPN240
108. MEMBER PROPERTY FRENCH
109. 213 214 216 TO 219 TABLE ST L75X75X8
110. MEMBER PROPERTY FRENCH
111. 328 TO 335 PRIS YD 0.6 ZD 0.6
112. 336 TO 339 PRIS YD 0.5 ZD 0.3
113. MEMBER PROPERTY
114. 340 341 PRIS YD 0.4 ZD 0.3
115. CONSTANTS
116. BETA 90 MEMB 1 4 67 68 74 75 81 82 88 89
117. MATERIAL STEEL MEMB 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 -
118. 106 108 109 111 113 114 116 118 119 121 TO 124 127 TO 129 132 TO 136 150 -
119. 151 TO 155 158 TO 160 163 TO 172 189 TO 196 198 199 208 TO 214 216 TO 219 -
120. 232 TO 244 247 248 259 TO 263 265 266 275 276 278 279 284 285 294 295 298 -
121. 299 302 303 307 TO 310 312 TO 320 325 TO 327
122. MATERIAL CONCRETE MEMB 328 TO 341
123. SUPPORTS
124. 33 TO 36 89 TO 92 FIXED BUT FX MX MY MZ
125. 177 TO 184 FIXED
126. 1 7 PINNED
127. MEMBER TRUSS
128. 127 129 132 TO 135 195 196 198 199 208 TO 214 216 TO 219 232 TO 236 -
129. 261 TO 263 265 266 275 276 278 279 284 285 294 295 299 303 307 TO 310 326
130. MEMBER RELEASE
131. 15 24 30 36 67 68 74 TO 76 81 TO 83 88 TO 90 122 TO 124 128 158 TO 159 -
132. 160 START MP 0.99
133. 80 87 94 153 TO 155 158 TO 160 193 194 237 238 302 END MP 0.99
134. MEMBER OFFSET
135. 340 341 START 0 -0.2 0
136. 340 341 END 0 -0.2 0
137. 336 TO 339 START 0 -0.25 0
138. 336 TO 339 END 0 -0.25 0
139. LOAD 1 DL
140. SELFWEIGHT Y -1.1
141. *WALKWAY LOAD @50KG/SQ. M INCLUDING SEAL PLATE _H/R
142. MEMBER LOAD
143. 15 36 41 44 103 104 106 118 119 121 150 TO 152 237 TO 244 247 248 259 260 -
144. 298 317 318 UNI GY -0.32
145. 24 30 42 43 108 109 111 113 114 116 163 TO 172 189 TO 194 312 313 UNI GY -0.47
146. *PIPE DEAD LOAD
147. MEMBER LOAD
148. 69 76 83 90 CON GY -2.03
149. 73 80 87 94 CON GY -3.57
150. *CONVEYOR DEAD LOAD
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151. MEMBER LOAD
152. 42 43 CON GY -5 0.2
153. 42 43 CON GY -5 1.5
154. 312 313 CON GY -5 1.714
155. 193 194 CON GY -5 1.845
156. 24 30 CON GY -5 1.96
157. 190 192 CON GY -5 0.036
158. 108 113 CON GY -5 0.401
159. 171 172 CON GY -5 0.736
160. 109 114 CON GY -5 1.171
161. 169 170 CON GY -5 1.532
162. 165 167 CON GY -5 1.891
163. 166 168 CON GY -5 2.277
164. 163 164 CON GY -5 1.154
165. *
166. MEMBER LOAD
167. 42 43 CON GX 1.5 0.2
168. 42 43 CON GX 1.5 1.5
169. 312 313 CON GX 1.5 1.714
170. 193 194 CON GX 1.5 1.845
171. 24 30 CON GX 1.5 1.96
172. 190 192 CON GX 1.5 0.036
173. 108 113 CON GX 1.5 0.401
174. 171 172 CON GX 1.5 0.736
175. 109 114 CON GX 1.5 1.171
176. 169 170 CON GX 1.5 1.532
177. 165 167 CON GX 1.5 1.891
178. 166 168 CON GX 1.5 2.277
179. 163 164 CON GX 1.5 1.154
180. *
181. MEMBER LOAD
182. 42 43 CON GZ 0.8 0.2
183. 42 43 CON GZ 0.8 1.5
184. 312 313 CON GZ 0.8 1.714
185. 193 194 CON GZ 0.8 1.845
186. 24 30 CON GZ 0.8 1.96
187. 190 192 CON GZ 0.8 0.036
188. 108 113 CON GZ 0.8 0.401
189. 171 172 CON GZ 0.8 0.736
190. 109 114 CON GZ 0.8 1.171
191. 169 170 CON GZ 0.8 1.532
192. 165 167 CON GZ 0.8 1.891
193. 166 168 CON GZ 0.8 2.277
194. 163 164 CON GZ 0.8 1.154
195. *CABLE TRAY LOAD
196. 69 76 83 90 158 TO 160 CON GY -11.76
197. 69 76 83 90 158 TO 160 CMOM GX 4.7
198. LOAD 2 LL
199. *WALKWAY LOAD @400KG/SQ. MTER(FOR GALLERY)
200. MEMBER LOAD
201. 15 24 30 36 41 TO 44 103 104 106 108 109 111 113 114 116 118 119 121 -
202. 150 TO 152 163 TO 172 189 TO 194 237 TO 244 247 248 259 260 298 312 313 317 -
203. 318 UNI GY -2
204. LOAD 3 ML
205. *CONVEYOR MATERIAL LOAD
206. MEMBER LOAD
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207. 42 43 CON GY -5.5 0.2
208. 42 43 CON GY -5.5 1.5
209. 312 313 CON GY -5.5 1.714
210. 193 194 CON GY -5.5 1.845
211. 24 30 CON GY -5.5 1.96
212. 190 192 CON GY -5.5 0.036
213. 108 113 CON GY -5.5 0.401
214. 171 172 CON GY -5.5 0.736
215. 109 114 CON GY -5.5 1.171
216. 169 170 CON GY -5.5 1.532
217. 165 167 CON GY -5.5 1.891
218. 166 168 CON GY -5.5 2.277
219. 163 164 CON GY -5.5 1.154
220. ** SPILLAGE LOAD @75KG/ SQ. M ON SEAL PLATE
221. 24 30 42 43 108 109 111 113 114 116 163 TO 172 189 TO 194 312 313 UNI GY -0.375
222. LOAD 4 TL1
223. TEMPERATURE LOAD
224. 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 106 108 109 111 113 -
225. 114 116 118 119 121 TO 124 127 TO 129 132 TO 136 150 TO 155 158 TO 160 163 -
226. 164 TO 172 189 TO 196 198 199 208 TO 214 216 TO 219 232 TO 244 247 248 259 -
227. 260 TO 263 265 266 275 276 278 279 284 285 294 295 298 299 302 303 -
228. 307 TO 310 312 TO 320 TEMP 25
229. LOAD 5 TL2
230. TEMPERATURE LOAD
231. 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 106 108 109 111 113 -
232. 114 116 118 119 121 TO 124 127 TO 129 132 TO 136 150 TO 155 158 TO 160 163 -
233. 164 TO 172 189 TO 196 198 199 208 TO 214 216 TO 219 232 TO 244 247 248 259 -
234. 260 TO 263 265 266 275 276 278 279 284 285 294 295 298 299 302 303 -
235. 307 TO 310 312 TO 320 TEMP -25
236. LOAD 6 WL +Z
237. MEMBER LOAD
238. 1 4 67 68 74 75 81 82 88 89 UNI GZ 0.5
239. *DUE TO CONVEYOR HOOD
240. 42 43 CON GZ 0.58 0.2
241. 42 43 CON GZ 0.58 1.5
242. 312 313 CON GZ 0.58 1.714
243. 193 194 CON GZ 0.58 1.845
244. 24 30 CON GZ 0.58 1.96
245. 190 192 CON GZ 0.58 0.036
246. 108 113 CON GZ 0.58 0.401
247. 171 172 CON GZ 0.58 0.736
248. 109 114 CON GZ 0.58 1.171
249. 169 170 CON GZ 0.58 1.532
250. 165 167 CON GZ 0.58 1.891
251. 166 168 CON GZ 0.58 2.277
252. 163 164 CON GZ 0.58 1.154
253. *DUE TO CABLE TRAY
254. 69 76 83 90 158 TO 160 CON GZ 2.38
255. 69 76 83 90 158 TO 160 CMOM GX 2
256. LOAD 7 WL -Z
257. MEMBER LOAD
258. 1 4 67 68 74 75 81 82 88 89 UNI GZ -0.5
259. *DUE TO CONVEYOR HOOD
260. 42 43 CON GZ -0.58 0.2
261. 42 43 CON GZ -0.58 1.5
262. 312 313 CON GZ -0.58 1.714
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263. 193 194 CON GZ -0.58 1.845
264. 24 30 CON GZ -0.58 1.96
265. 190 192 CON GZ -0.58 0.036
266. 108 113 CON GZ -0.58 0.401
267. 171 172 CON GZ -0.58 0.736
268. 109 114 CON GZ -0.58 1.171
269. 169 170 CON GZ -0.58 1.532
270. 165 167 CON GZ -0.58 1.891
271. 166 168 CON GZ -0.58 2.277
272. 163 164 CON GZ -0.58 1.154
273. *DUE TO CABLE TRAY
274. 69 76 83 90 158 TO 160 CON GZ -2.38
275. 69 76 83 90 158 TO 160 CMOM GX -2
276. LOAD 11 G
277. REPEAT LOAD
278. 1 1.0
279. LOAD 12 G + QN + MN
280. REPEAT LOAD
281. 1 1.0 2 1.0 3 1.0
282. ****LOAD COMBINATION FOR STRUCTURAL DESIGN
283. *LOAD COMB 201 1.4 DL
284. *1 1.4
285. *LOAD COMB 202 1.2 (DL + TL1) + 1.6  (LL+ML)
286. *1 1.2 4 1.2 2 1.6 3 1.6
287. *LOAD COMB 203 1.2 (DL + TL1) + 1.6  (LL+ML)
288. *1 1.2 5 1.2 2 1.6 3 1.6
289. *LOAD COMB 204 1.2 DL + 1.6 WL + 0.5 (LL+ML)
290. *1 1.2 6 1.6 2 0.5 3 0.5
291. *LOAD COMB 205 1.2 (DL + TL1) + 1.6  (LL+ML)
292. *1 1.2 7 1.6 2 0.5 3 0.5
293. *LOAD COMB 206 0.9 DL+ 1.6 WL
294. *1 0.9 6 1.6
295. *LOAD COMB 207 0.9 DL+ 1.6 WL
296. *1 0.9 7 1.6
297. ****LOAD COMBINATION FOR BEARING PRESSURE CHECK
298. LOAD COMB 301 1.0 DL
299. 1 1.0
300. LOAD COMB 302 1.0 (DL + TL1) + 1.0  (LL+ML)
301. 1 1.0 4 1.0 2 1.0 3 1.0
302. LOAD COMB 303 1.0 (DL + TL2) + 1.0  (LL+ML)
303. 1 1.0 5 1.0 2 1.0 3 1.0
304. LOAD COMB 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
305. 1 1.0 6 0.75 2 0.75 3 0.75
306. LOAD COMB 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
307. 1 1.0 7 0.75 2 0.75 3 0.75
308. LOAD COMB 306 0.6 DL + 1.0 WL(+Z)
309. 1 0.6 6 1.0
310. LOAD COMB 307 0.6 DL + 1.0 WL(-Z)
311. 1 0.6 7 1.0
312. ************************************************************
313. ****LOAD COMBINATION FOR FOUNDATION DESIGN
314. LOAD COMB 401 1.4 DL
315. 1 1.4
316. LOAD COMB 402 1.4 (DL + TL1) + 1.7  (LL+ML)
317. 1 1.4 4 1.4 2 1.7 3 1.7
318. LOAD COMB 403 1.4 (DL + TL2) + 1.7  (LL+ML)
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319. 1 1.4 5 1.4 2 1.7 3 1.7
320. LOAD COMB 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
321. 1 1.2 6 1.6 2 0.5 3 0.5
322. LOAD COMB 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
323. 1 1.2 7 1.6 2 0.5 3 0.5
324. LOAD COMB 406 0.9 DL + 1.6 WL(+Z)
325. 1 0.9 6 1.6
326. LOAD COMB 407 0.9 DL + 1.6 WL(-Z)
327. 1 0.9 7 1.6
328. ************************************************************
329. PERFORM ANALYSIS
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P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**95**	**NUMBER OF MEMBERS**	**170**
NUMBER OF PLATES	**0**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**18**

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 9, TOTAL DEGREES OF FREEDOM = 500
TOTAL LOAD COMBINATION CASES = 14 SO FAR.

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330. *STEEL TAKE OFF LIST 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 -
331. *106 108 109 111 113 114 116 118 119 121 TO 124 127 TO 129 132 TO 136 150 -
332. *151 TO 155 158 TO 160 163 TO 172 189 TO 196 198 199 208 TO 214 216 TO 219 -
333. *232 TO 244 247 248 259 TO 263 265 266 275 276 278 279 284 285 294 295 298 -
334. *299 302 303 307 TO 310 312 313 316 TO 318 325
335. *LOAD LIST 201 TO 207
336. *PARAMETER 1
337. *CODE LRFD
338. *FYLD 275000 MEMB 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 -
339. *106 108 109 111 113 114 116 118 119 121 123 124 127 TO 129 132 TO 136 150 -
340. *151 TO 152 154 155 158 TO 160 163 TO 172 189 TO 194 237 TO 244 247 248 259 -
341. *260 298 302 312 313 316 TO 318 325
342. *FU 410000 MEMB 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 106 -
343. *108 109 111 113 114 116 118 119 121 123 124 127 TO 129 132 TO 136 -
344. *150 TO 152 154 155 158 TO 160 163 TO 172 189 TO 194 237 TO 244 247 248 259 -
345. *260 298 302 312 313 316 TO 318 325
346. *TRACK 0 MEMB 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 106 -
347. *108 109 111 113 114 116 118 119 121 123 124 127 TO 129 132 TO 136 -
348. *150 TO 152 154 155 158 TO 160 163 TO 172 189 TO 194 237 TO 244 247 248 259 -
349. *260 298 302 312 313 316 TO 318 325
350. *BEAM 1 MEMB 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 106 108 -
351. *109 111 113 114 116 118 119 121 123 124 127 TO 129 132 TO 136 150 TO 152 -
352. *154 155 158 TO 160 163 TO 172 189 TO 194 237 TO 244 247 248 259 260 298 302 -
353. *312 313 316 TO 318 325
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354. *TMAIN 300 MEMB 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 106 -
355. *108 109 111 113 114 116 118 119 121 123 124 127 TO 129 132 TO 136 -
356. *150 TO 152 154 155 158 TO 160 163 TO 172 189 TO 194 237 TO 244 247 248 259 -
357. *260 298 302 312 313 316 TO 318 325
358. *MAIN 200 MEMB 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 106 -
359. *108 109 111 113 114 116 118 119 121 123 124 127 TO 129 132 TO 136 -
360. *150 TO 152 154 155 158 TO 160 163 TO 172 189 TO 194 237 TO 244 247 248 259 -
361. *260 298 302 312 313 316 TO 318 325
362. *NSF 0.8 MEMB 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 106 -
363. *108 109 111 113 114 116 118 119 121 123 124 127 TO 129 132 TO 136 -
364. *150 TO 152 154 155 158 TO 160 163 TO 172 189 TO 194 237 TO 244 247 248 259 -
365. *260 298 302 312 313 316 TO 318 325
366. *UNT 1.9 MEMB 7 10 TO 12 69 TO 73 76 TO 80 83 TO 87 90 TO 94 100 136 316 325
367. *LY 1.9 MEMB 7 10 TO 12 69 TO 73 76 TO 80 83 TO 87 90 TO 94 100 136 316 325
368. *LX 1.9 MEMB 7 10 TO 12 69 TO 73 76 TO 80 83 TO 87 90 TO 94 100 136 316 325
369. *LZ 5.5 MEMB 7 10 TO 12 69 TO 73 76 TO 80 83 TO 87 90 TO 94 100 136 316 325
370. *KZ 2 MEMB 103 104 106 108 109 111 113 114 116 118 119 121 150 TO 152 -
371. *163 TO 172 239 TO 242 259 260 298
372. *KY 0.5 MEMB 127 129
373. *KX 0.5 MEMB 127 129
374. *KZ 3 MEMB 15 24 30 36 41 TO 44 165 TO 168 189 TO 194 237 238 243 244 247 248 -
375. *259 260 312 313 317 318
376. *KZ 0.85 MEMB 211 TO 214 216 TO 219 232 TO 236 314
377. *KZ 0.5 MEMB 132 TO 135 307 TO 309 327
378. *CHECK CODE MEMB 1 4 7 10 TO 12 15 24 30 36 41 TO 44 67 TO 94 100 103 104 106 -
379. *108 109 111 113 114 116 118 119 121 TO 124 127 TO 129 132 TO 136 150 TO 155 -
380. *158 TO 160 163 TO 172 189 TO 196 198 199 208 TO 214 216 TO 219 232 TO 244 -
381. *247 248 259 TO 263 265 266 275 276 278 279 284 285 294 295 298 299 302 303 -
382. *307 TO 310 312 TO 320 325 TO 327
383. LOAD LIST 401 TO 407
384. START CONCRETE DESIGN
385. CODE ACI 2008
386. FC 35000 MEMB 328 TO 341
387. FYMAIN 420000 MEMB 328 TO 341
388. TRACK 1 MEMB 328 TO 341
389. CLB 0.05 MEMB 328 TO 341
390. CLT 0.05 MEMB 328 TO 341
391. CLS 0.05 MEMB 328 TO 341
392. MAXMAIN 20 MEMB 328 TO 335
393. MINMAIN 20 MEMB 328 TO 335
394. MAXMAIN 16 MEMB 336 TO 341
395. MINMAIN 12 MEMB 336 TO 341
396. MINSEC 10 MEMB 328 TO 335
397. MINSEC 8 MEMB 336 TO 341
398. REINF 0 MEMB 328 TO 335
399. DESIGN COLUMN 328 TO 335
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COLUMN NO. 328 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	65.48	0.19	0.00126

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	65.48	20.38	0.13244

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COLUMN NO. 329 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	104.54	0.02	0.00008

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	104.54	21.06	0.08572

==========

COLUMN NO. 330 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	70.97	9.14	0.05480

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	70.97	18.78	0.11263

==========

COLUMN NO. 331 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	124.81	8.37	0.02855

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	124.81	21.25	0.07245

==========

COLUMN NO. 332 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	113.36	22.67	0.08509

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	113.36	11.23	0.04215

==========

COLUMN NO. 333 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	171.43	22.13	0.05494

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	171.43	15.95	0.03960

==========

COLUMN NO. 334 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	80.03	0.03	0.00018

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	80.03	14.36	0.07636

==========

COLUMN NO. 335 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION REINF PCT. LOAD LOCATION PHI

12 - 20 MM 1.047 401 END 0.650
(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 10 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	118.73	0.00	0.00001

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	118.73	16.46	0.05898

**********END OF COLUMN DESIGN RESULTS**********

400. DESIGN BEAM 336 TO 341

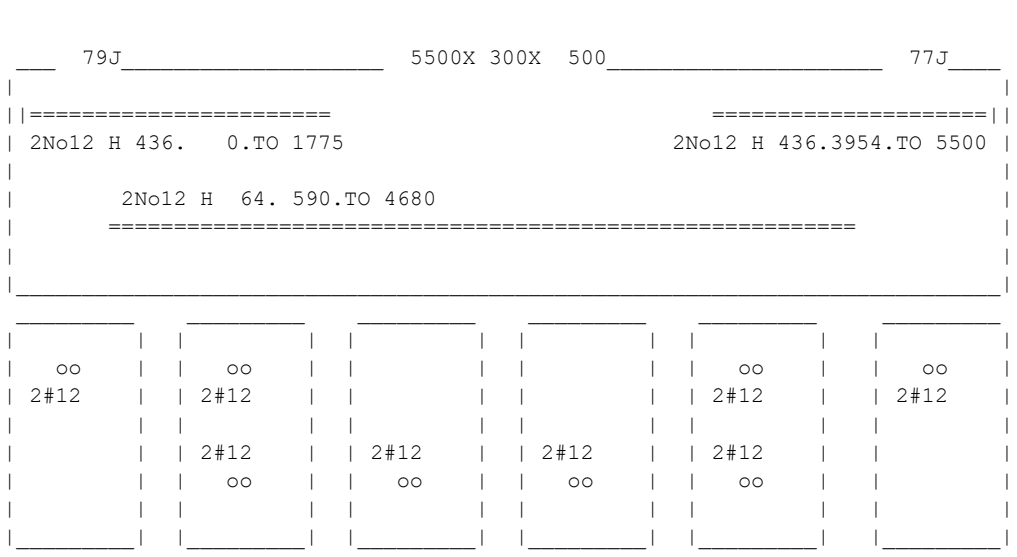
BEAM NO. 336 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	590.	4680.	NO	NO
	CRITICAL POS MOMENT= 8.82 KN-MET AT 2292.MM, LOAD 402					
	REQD STEEL= 71.MM2, RHO=0.0005, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 172. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 24968.6 cm^4					
2	436.	2 - 12MM	0.	1775.	YES	NO
	CRITICAL NEG MOMENT= 17.88 KN-MET AT 0.MM, LOAD 403					
	REQD STEEL= 145.MM2, RHO=0.0011, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 172. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 24968.6 cm^4					
3	436.	2 - 12MM	3954.	5500.	NO	YES
	CRITICAL NEG MOMENT= 17.81 KN-MET AT 5500.MM, LOAD 402					
	REQD STEEL= 145.MM2, RHO=0.0011, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 172. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 24968.6 cm^4					

B E A M N O. 336 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   12.37 KNS  Vc=  123.25 KNS  Vs=    0.00 KNS
Tu=    0.02 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                    STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   12.84 KNS  Vc=  123.10 KNS  Vs=    0.00 KNS
Tu=    0.02 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                    STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 337 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

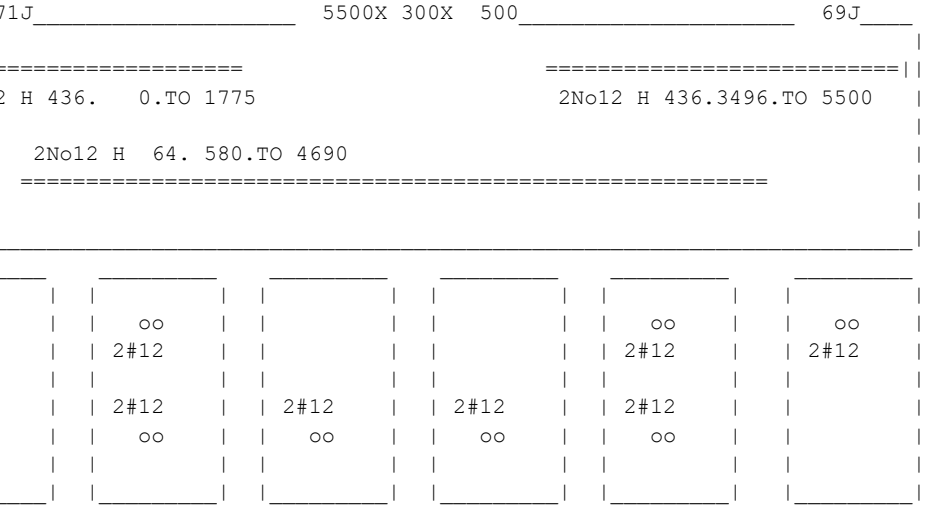
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	580.	4690.	NO	NO
	CRITICAL POS MOMENT= 9.72 KN-MET AT 2292.MM, LOAD 403					
	REQD STEEL= 79.MM2, RHO=0.0006, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 172. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 24968.6 cm^4					
2	436.	2 - 12MM	0.	1775.	YES	NO
	CRITICAL NEG MOMENT= 17.36 KN-MET AT 0.MM, LOAD 402					
	REQD STEEL= 141.MM2, RHO=0.0011, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 172. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 24968.6 cm^4					
3	436.	2 - 12MM	3496.	5500.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     15.36 KN-MET  AT  5500.MM, LOAD  404|
|   REQD STEEL=   125.MM2, RHO=0.0010, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   37./  172. MMS        |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    24968.6 cm^4
```

B E A M N O. 337 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=   12.30 KNS  Vc=  123.39 KNS  Vs=    0.00 KNS
 Tu=    0.01 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   12.91 KNS  Vc=  123.18 KNS  Vs=    0.00 KNS
 Tu=    0.01 KN-MET  Tc=    5.2 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 338 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	149.	4664.	NO	NO

```
|--------------------------------------------------------------|
|   CRITICAL POS MOMENT=      7.32 KN-MET  AT  2750.MM, LOAD  401|
|   REQD STEEL=    59.MM2, RHO=0.0005, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   37./  172. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|--------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 24968.6 cm^4

```
 2        436.     2 - 12MM         0.         1775.        YES  NO
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     15.56 KN-MET  AT     0.MM, LOAD  405|
|   REQD STEEL=   126.MM2, RHO=0.0010, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   37./  172. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|--------------------------------------------------------------|
```

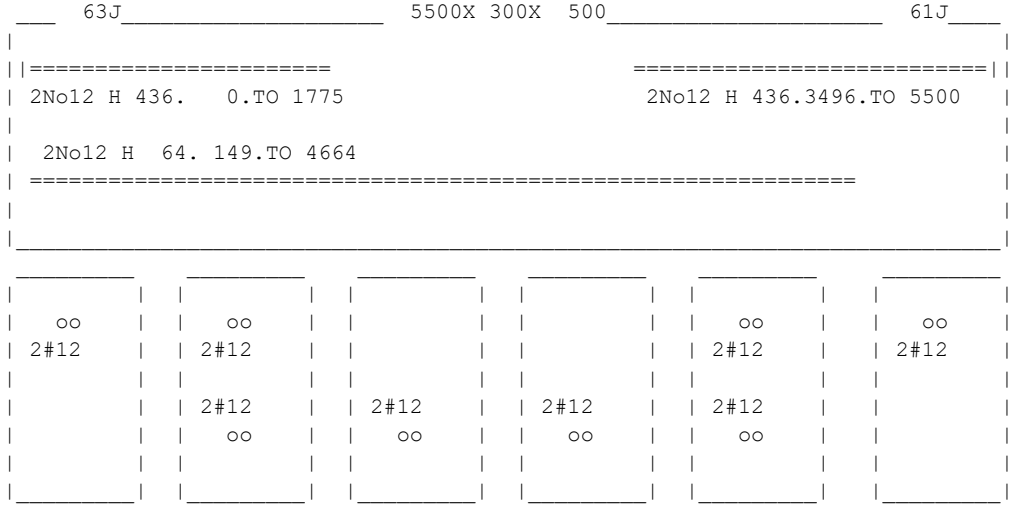
Cracked Moment of Inertia Iz at above location = 24968.6 cm^4

```
 3        436.     2 - 12MM      3496.         5500.        NO   YES
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     18.37 KN-MET  AT  5500.MM, LOAD  402|
|   REQD STEEL=   149.MM2, RHO=0.0011, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   37./  172. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|--------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 24968.6 cm^4

B E A M N O. 338 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=   12.06 KNS  Vc=  124.02 KNS  Vs=    0.00 KNS
Tu=    0.02 KN-MET  Tc=    5.0 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   13.15 KNS  Vc=  124.02 KNS  Vs=    0.00 KNS
Tu=    0.02 KN-MET  Tc=    5.0 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 339 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5500. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

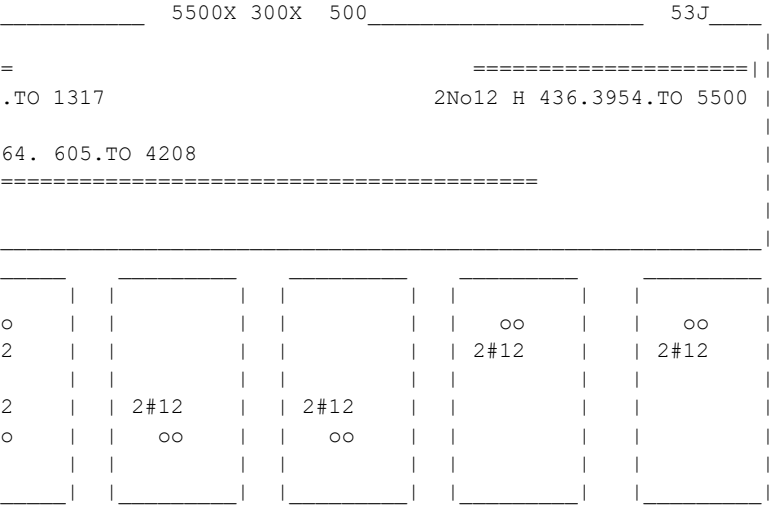
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	605.	4208.	NO	NO
	CRITICAL POS MOMENT= 7.51 KN-MET AT 2750.MM, LOAD 401					
	REQD STEEL= 61.MM2, RHO=0.0005, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 172. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 24968.6 cm^4					
2	436.	2 - 12MM	0.	1317.	YES	NO
	CRITICAL NEG MOMENT= 14.26 KN-MET AT 0.MM, LOAD 402					
	REQD STEEL= 116.MM2, RHO=0.0009, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 172. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 24968.6 cm^4					
3	436.	2 - 12MM	3954.	5500.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     15.70 KN-MET  AT  5500.MM, LOAD  403|
|   REQD STEEL=   127.MM2, RHO=0.0010, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   37./  172. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|-------------------------------------------------------------|
```

Cracked Moment of Inertia Iz at above location = 24968.6 cm^4

B E A M N O. 339 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 12.33 KNS Vc= 125.61 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 12.88 KNS Vc= 125.61 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



BEAM NO. 340 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5250. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	561.	4032.	NO	NO

```
|--------------------------------------------------------------|
|   CRITICAL POS MOMENT=      6.64 KN-MET  AT  2188.MM, LOAD  403|
|   REQD STEEL=    70.MM2, RHO=0.0007, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   37./  172. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    14415.4 cm^4

  2        336.     2 - 12MM         0.          838.        YES  NO
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=      6.08 KN-MET  AT     0.MM, LOAD  402|
|   REQD STEEL=    64.MM2, RHO=0.0006, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   37./  172. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|--------------------------------------------------------------|

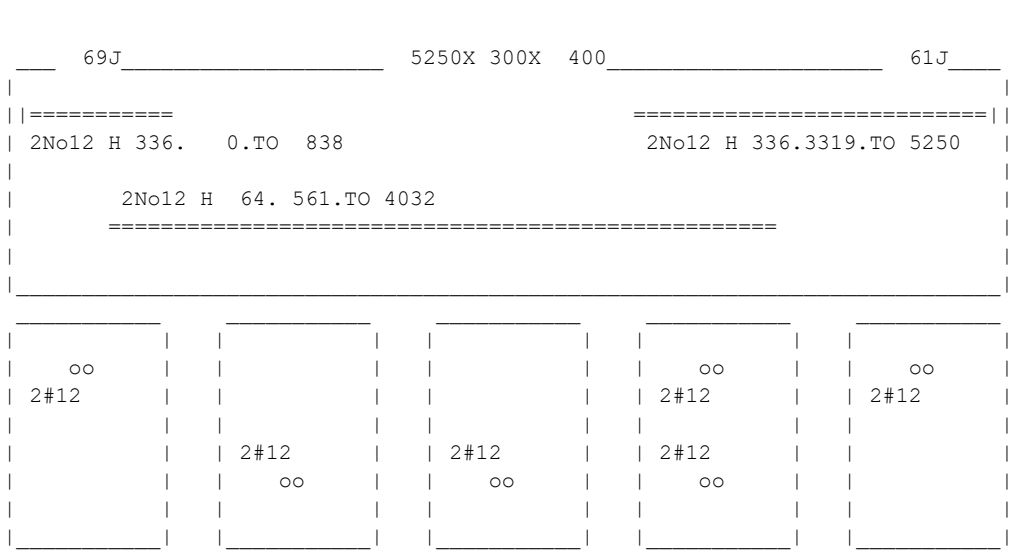
  Cracked Moment of Inertia Iz at above location =    14415.4 cm^4

  3        336.     2 - 12MM      3319.         5250.        NO   YES
|--------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     14.92 KN-MET  AT  5250.MM, LOAD  402|
|   REQD STEEL=   158.MM2, RHO=0.0016, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   37./  172. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|--------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    14415.4 cm^4
```

B E A M N O. 340 D E S I G N R E S U L T S - SHEAR

```
AT START SUPPORT - Vu=    8.27 KNS  Vc=   95.84 KNS  Vs=    0.00 KNS
Tu=    0.12 KN-MET  Tc=    3.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
AT END   SUPPORT - Vu=   11.68 KNS  Vc=   94.57 KNS  Vs=    0.00 KNS
Tu=    0.12 KN-MET  Tc=    3.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```



BEAM NO. 341 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 5250. MM FY - 420. FC - 35. MPA, SIZE - 300. X 400. MMS

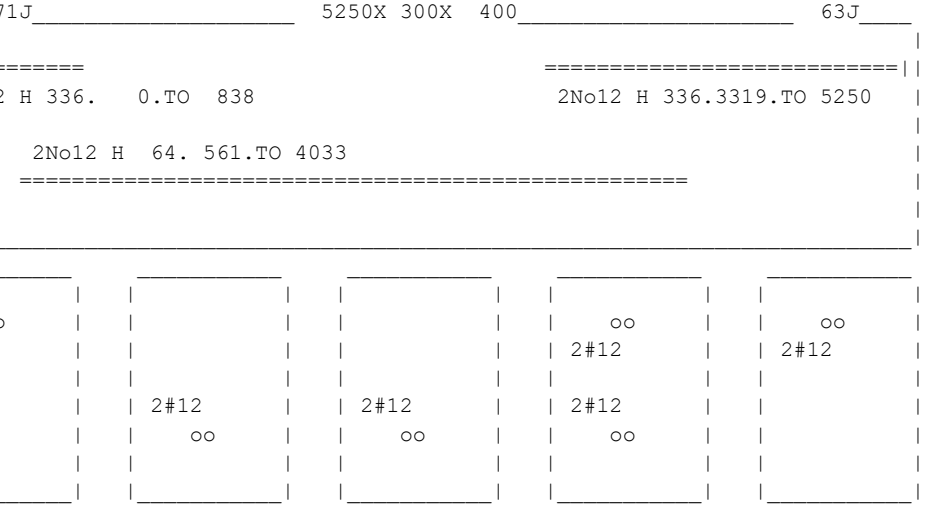
LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	561.	4033.	NO	NO
	CRITICAL POS MOMENT= 6.66 KN-MET AT 2188.MM, LOAD 403					
	REQD STEEL= 70.MM2, RHO=0.0007, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 172. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 14415.4 cm^4					
2	336.	2 - 12MM	0.	838.	YES	NO
	CRITICAL NEG MOMENT= 6.03 KN-MET AT 0.MM, LOAD 402					
	REQD STEEL= 63.MM2, RHO=0.0006, RHOMX=0.0249 RHOMN=0.0035					
	MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 172. MMS					
	REQD. DEVELOPMENT LENGTH = 308. MMS					
	Cracked Moment of Inertia Iz at above location = 14415.4 cm^4					
3	336.	2 - 12MM	3319.	5250.	NO	YES

```
|-------------------------------------------------------------|
|   CRITICAL NEG MOMENT=     14.90 KN-MET  AT  5250.MM, LOAD  402|
|   REQD STEEL=   157.MM2, RHO=0.0016, RHOMX=0.0249 RHOMN=0.0035 |
|   MAX/MIN/ACTUAL BAR SPACING=   230./   37./  172. MMS         |
|   REQD. DEVELOPMENT LENGTH =   308. MMS                       |
|-------------------------------------------------------------|

  Cracked Moment of Inertia Iz at above location =    14415.4 cm^4
```

B E A M N O. 341 D E S I G N R E S U L T S - SHEAR

```
 AT START SUPPORT - Vu=    8.30 KNS  Vc=   96.29 KNS  Vs=    0.00 KNS
 Tu=    0.06 KN-MET  Tc=    3.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
 AT END   SUPPORT - Vu=   11.65 KNS  Vc=   94.60 KNS  Vs=    0.00 KNS
 Tu=    0.06 KN-MET  Tc=    3.8 KN-MET  Ts=    0.0 KN-MET  LOAD   401
                  STIRRUPS ARE NOT REQUIRED.
```




```
 ******************END OF BEAM DESIGN*************************


 401. END CONCRETE DESIGN
 402. FINISH
```

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 4,2020 TIME= 12:33:57 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
* *
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****

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2		
Date	04-04-2020	29-04-2020	01-11-2023				Date	04-04-2020	29-04-2020	01-11-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	APL	APL	APL				Prepared By	APL	APL	APL		
Checked	SBY	SBY	SBY				Checked	SBY	SBY	SBY		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 4 **STAAD INPUT & OUTPUT FILES OF COMBINED FOOTING MKD. F2 FOR TR1 & TR2**

```
STAAD SPACE
START JOB INFORMATION
ENGINEER DATE 03-Apr-20
JOB NAME MMGP-SAUDI ARABIA
JOB CLIENT MA'ADEN
JOB NO O19045/900707
JOB REV 0
JOB PART TRESTLE TR2 FOUNDATION
JOB REF FOUNDATIONS FOR CONVEYOR 3130-FDR-004
ENGINEER NAME APL
CHECKER NAME SBY
APPROVED NAME PRI
END JOB INFORMATION
INPUT WIDTH 79
UNIT METER KN
JOINT COORDINATES
1 0 -3 0; 2 0 -3 2; 3 0 -1 0; 4 0 -1 2; 5 0 -0.65 0; 6 0 -0.65 2;
14 -0.75 -3 2.75; 17 -0.5 -3 2.75; 19 -0.25 -3 2.75; 20 0 -3 2.75;
22 0.25 -3 2.75; 24 0.5 -3 2.75; 26 0.75 -3 2.75; 28 -0.75 -3 2.5;
30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5; 33 0.25 -3 2.5; 34 0.5 -3 2.5;
35 0.75 -3 2.5; 37 -0.75 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25;
41 0 -3 2.25; 42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 46 -0.75 -3 2;
48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2; 51 0.5 -3 2; 52 0.75 -3 2;
54 -0.75 -3 1.75; 56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75;
59 0.25 -3 1.75; 60 0.5 -3 1.75; 61 0.75 -3 1.75; 63 -0.75 -3 1.5;
65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5; 69 0.5 -3 1.5;
70 0.75 -3 1.5; 72 -0.75 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25;
76 0 -3 1.25; 77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 81 -0.75 -3 1;
83 -0.5 -3 1; 84 -0.25 -3 1; 85 0 -3 1; 86 0.25 -3 1; 87 0.5 -3 1;
88 0.75 -3 1; 90 -0.75 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75;
94 0 -3 0.75; 95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75;
99 -0.75 -3 0.5; 101 -0.5 -3 0.5; 102 -0.25 -3 0.5; 103 0 -3 0.5;
104 0.25 -3 0.5; 105 0.5 -3 0.5; 106 0.75 -3 0.5; 108 -0.75 -3 0.25;
110 -0.5 -3 0.25; 111 -0.25 -3 0.25; 112 0 -3 0.25; 113 0.25 -3 0.25;
114 0.5 -3 0.25; 115 0.75 -3 0.25; 117 -0.75 -3 0; 119 -0.5 -3 0;
120 -0.25 -3 0; 121 0.25 -3 0; 122 0.5 -3 0; 123 0.75 -3 0; 125 -0.75 -3 -0.25;
127 -0.5 -3 -0.25; 128 -0.25 -3 -0.25; 129 0 -3 -0.25; 130 0.25 -3 -0.25;
131 0.5 -3 -0.25; 132 0.75 -3 -0.25; 134 -0.75 -3 -0.5; 136 -0.5 -3 -0.5;
137 -0.25 -3 -0.5; 138 0 -3 -0.5; 139 0.25 -3 -0.5; 140 0.5 -3 -0.5;
141 0.75 -3 -0.5; 143 -0.75 -3 -0.75; 145 -0.5 -3 -0.75; 146 -0.25 -3 -0.75;
147 0 -3 -0.75; 148 0.25 -3 -0.75; 149 0.5 -3 -0.75; 150 0.75 -3 -0.75;
158 -0.75 -3 -1; 162 -0.5 -3 -1; 164 -0.25 -3 -1; 166 0 -3 -1; 168 0.25 -3 -1;
170 0.5 -3 -1; 172 0.75 -3 -1; 178 -0.75 -3 3; 181 -0.5 -3 3; 183 -0.25 -3 3;
185 0 -3 3; 187 0.25 -3 3; 189 0.5 -3 3; 191 0.75 -3 3; 212 1.00033 -3 2.75;
213 1.00033 -3 2.5; 216 1.00033 -3 2.25; 218 1.00033 -3 2; 220 1.00033 -3 1.75;
222 1.00033 -3 1.5; 224 1.00033 -3 1.25; 226 1.00033 -3 1; 228 1.00033 -3 0.75;
230 1.00033 -3 0.5; 232 1.00033 -3 0.25; 234 1.00033 -3 0;
236 1.00033 -3 -0.25; 238 1.00033 -3 -0.5; 240 1.00033 -3 -0.75;
242 1.00033 -3 -1; 247 1.00033 -3 3; 260 -1.00033 -3 2.75; 261 -1.00033 -3 2.5;
263 -1.00033 -3 2.25; 265 -1.00033 -3 2; 267 -1.00033 -3 1.75;
269 -1.00033 -3 1.5; 271 -1.00033 -3 1.25; 273 -1.00033 -3 1;
275 -1.00033 -3 0.75; 277 -1.00033 -3 0.5; 279 -1.00033 -3 0.25;
281 -1.00033 -3 0; 283 -1.00033 -3 -0.25; 285 -1.00033 -3 -0.5;
287 -1.00033 -3 -0.75; 289 -1.00033 -3 -1; 295 -1.00033 -3 3;
```

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MEMBER INCIDENCES
1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4;
ELEMENT INCIDENCES SHELL
15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33;
20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41;
27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49;
34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56;
41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63;
48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69;
55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77;
60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85;
67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93;
74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99;
80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104;
84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111;
90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117;
96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121;
100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127;
105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131;
111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137;
114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140;
119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146;
122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149;
137 143 145 162 158; 140 145 146 164 162; 143 146 147 166 164;
146 147 148 168 166; 149 148 149 170 168; 152 149 150 172 170;
163 178 181 17 14; 166 181 183 19 17; 169 183 185 20 19; 172 185 187 22 20;
175 187 189 24 22; 178 189 191 26 24; 214 26 212 213 35; 217 35 213 216 44;
220 44 216 218 52; 223 52 218 220 61; 226 61 220 222 70; 229 70 222 224 79;
232 79 224 226 88; 235 88 226 228 97; 238 97 228 230 106; 241 106 230 232 115;
244 115 232 234 123; 247 123 234 236 132; 250 132 236 238 141;
253 141 238 240 150; 256 150 240 242 172; 265 191 247 212 26;
282 260 14 28 261; 285 261 28 37 263; 288 263 37 46 265; 291 265 46 54 267;
294 267 54 63 269; 297 269 63 72 271; 300 271 72 81 273; 303 273 81 90 275;
306 275 90 99 277; 309 277 99 108 279; 312 279 108 117 281;
315 281 117 125 283; 318 283 125 134 285; 321 285 134 143 287;
324 287 143 158 289; 333 295 178 14 260;
START GROUP DEFINITION
ELEMENT
_MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
108 111 112 115 116 119 120 123 124 137 140 149 152 163 166 175 178 214 217 -
220 223 226 229 232 235 238 241 244 247 250 253 256 265 282 285 288 291 294 -
297 300 303 306 309 312 315 318 321 324 333
_MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
111 TO 116 119 TO 124 137 140 143 146 149 152 163 166 169 172 175 178 214 -
217 226 229 232 235 238 241 250 253 256 265 282 285 294 297 300 303 306 309 -
318 321 324 333
_SHEAR 63 TO 68 71 TO 76 137 140 143 146 149 152 163 166 169 172 175 178 214 -
217 220 223 226 229 232 235 238 241 244 247 250 253 256 265 282 285 288 291 -
294 297 300 303 306 309 312 315 318 321 324 333
END GROUP DEFINITION
ELEMENT PROPERTY
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 140 143 146 -
```

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149 152 163 166 169 172 175 178 214 217 220 223 226 229 232 235 238 241 244 -
247 250 253 256 265 282 285 288 291 294 297 300 303 306 309 312 315 318 321 -
324 333 THICKNESS 0.4
DEFINE MATERIAL START
ISOTROPIC CONCRETE
E 2.17185e+007
POISSON 0.17
DENSITY 23.5616
ALPHA 1e-005
DAMP 0.05
TYPE CONCRETE
STRENGTH FCU 27579
END DEFINE MATERIAL
MEMBER PROPERTY AMERICAN
1 TO 4 PRIS YD 0.6 ZD 0.6
5 PRIS YD 0.5 ZD 0.3
CONSTANTS
MATERIAL CONCRETE ALL
*****************************************
SUPPORTS
1 2 14 17 19 20 22 24 26 28 30 TO 35 37 39 TO 44 46 48 TO 52 54 56 TO 61 63 -
65 TO 70 72 74 TO 79 81 83 TO 88 90 92 TO 97 99 101 TO 106 108 110 TO 115 -
117 119 TO 123 125 127 TO 132 134 136 TO 141 143 145 TO 150 158 162 164 166 -
168 170 172 178 181 183 185 187 189 191 212 213 216 218 220 222 224 226 228 -
230 232 234 236 238 240 242 247 260 261 263 265 267 269 271 273 275 277 279 -
281 283 285 287 289 295 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
*****************************************
MEMBER OFFSET
5 START 0 -0.25 0
5 END 0 -0.25 0
LOAD 1 DEAD LOAD    DL
SELFWEIGHT Y -1
JOINT LOAD
5 FX -0.289 FY -70.074 FZ -2.545
6 FX -0.339 FY -57.696 FZ 9.545
ELEMENT LOAD
15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 140 143 146 -
149 152 163 166 169 172 175 178 214 217 220 223 226 229 232 235 238 241 244 -
247 250 253 256 265 282 285 288 291 294 297 300 303 306 309 312 315 318 321 -
324 333 PR GY -28.8
*****************************************
LOAD 2 LIVE LOAD    LL
JOINT LOAD
5 FX -0.217 FY -32.1 FZ -2.458
6 FX -0.228 FY -31.9 FZ 2.458
*****************************************
LOAD 3 MATERIAL LOAD    LL
JOINT LOAD
5 FX -0.134 FY -25.6 FZ -1.549
6 FX -0.145 FY -25.4 FZ 1.549
*****************************************
LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
JOINT LOAD
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5 FX 0.033 FY 46.654 FZ 13.252
6 FX -0.033 FY -46.654 FZ 13.523
*****************************************
LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
JOINT LOAD
5 FX -0.033 FY -47.674 FZ -14.104
6 FX 0.033 FY 47.674 FZ -14.388
*****************************************
LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
JOINT LOAD
5 FX 0.562 FY 0 FZ 0
6 FX 0.562 FY 0 FZ 0
*****************************************
LOAD 7 WIND LOAD ACROSS THE CONV.   WL (-X)
JOINT LOAD
5 FX -0.562 FY 0 FZ 0
6 FX -0.562 FY 0 FZ 0
*****************************************
LOAD 8 TEMP LOAD    TL1
JOINT LOAD
5 FX 0.028 FY 0.5 FZ -6.391
6 FX -0.028 FY -0.5 FZ 6.391
*****************************************
LOAD 9 TEMP LOAD    TL2
JOINT LOAD
5 FX -0.028 FY -0.5 FZ 6.391
6 FX 0.028 FY 0.5 FZ -6.391
*****************************************
LOAD 10 SEISMIC LOAD ALONG THE CONV.    SL (+X)
JOINT LOAD
5 FX 0.061 FY -0.5 FZ -0.035
6 FX 0.061 FY -0.5 FZ 0.035
*****************************************
LOAD 11 SEISMIC LOAD ACROSS THE CONV.   SL (+Z)
JOINT LOAD
5 FX 0 FY 8.882 FZ 2.661
6 FX 0 FY -8.882 FZ 2.755
*****************************************
*****Load combination for bearing capacity check
LOAD 301 1.0 DL
REPEAT LOAD
1 1.0
LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 8 1.0 2 1.0 3 1.0
LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 9 1.0 2 1.0 3 1.0
LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 4 0.75 2 0.75 3 0.75
LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 5 0.75 2 0.75 3 0.75
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LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 6 0.75 2 0.75 3 0.75
LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 7 0.75 2 0.75 3 0.75
LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 0.525 2 0.75 3 0.75
LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 10 -0.525 2 0.75 3 0.75
LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 0.525 2 0.75 3 0.75
LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 11 -0.525 2 0.75 3 0.75
LOAD 312 0.6 DL + 1.0 WL(+Z)
REPEAT LOAD
1 0.6 4 1.0
LOAD 313 0.6 DL + 1.0 WL(-Z)
REPEAT LOAD
1 0.6 5 1.0
LOAD 314 0.6 DL + 1.0 WL(+X)
REPEAT LOAD
1 0.6 6 1.0
LOAD 315 0.6 DL + 1.0 WL(-X)
REPEAT LOAD
1 0.6 7 1.0
LOAD 316 0.6 DL + 0.7 EL(+X)
REPEAT LOAD
1 0.6 10 0.7
LOAD 317 0.6 DL + 0.7 EL(-X)
REPEAT LOAD
1 0.6 10 -0.7
LOAD 318 0.6 DL + 0.7 EL(+Z)
REPEAT LOAD
1 0.6 11 0.7
LOAD 319 0.6 DL + 0.7 EL(-Z)
REPEAT LOAD
1 0.6 11 -0.7
*****************************************
*****Load combination for foundation design
LOAD 401 1.4 DL
REPEAT LOAD
1 1.4
LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 8 1.4 2 1.7 3 1.7
LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
REPEAT LOAD
1 1.4 9 1.4 2 1.7 3 1.7
LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
```

```
REPEAT LOAD
1 1.2 4 1.6 2 0.5 3 0.5
LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 5 1.6 2 0.5 3 0.5
LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 6 1.6 2 0.5 3 0.5
LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 7 1.6 2 0.5 3 0.5
LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 1.0 2 0.5 3 0.5
LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 10 -1.0 2 0.5 3 0.5
LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 1.0 2 0.5 3 0.5
LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
REPEAT LOAD
1 1.2 11 -1.0 2 0.5 3 0.5
LOAD 412 0.9 DL + 1.6 WL(+Z)
REPEAT LOAD
1 0.9 4 1.6
LOAD 413 0.9 DL + 1.6 WL(-Z)
REPEAT LOAD
1 0.9 5 1.6
LOAD 414 0.9 DL + 1.6 WL(+X)
REPEAT LOAD
1 0.9 6 1.6
LOAD 415 0.9 DL + 1.6 WL(-X)
REPEAT LOAD
1 0.9 7 1.6
LOAD 416 0.9 DL + 1.0 EL(+X)
REPEAT LOAD
1 0.9 10 1.0
LOAD 417 0.9 DL + 1.0 EL(-X)
REPEAT LOAD
1 0.9 10 -1.0
LOAD 418 0.9 DL + 1.0 EL(+Z)
REPEAT LOAD
1 0.9 11 1.0
LOAD 419 0.9 DL + 1.0 EL(-Z)
REPEAT LOAD
1 0.9 11 -1.0
****************************************
PERFORM ANALYSIS
****************************************
LOAD LIST 401 TO 419
START CONCRETE DESIGN
CODE ACI 2008
FC 35000 MEMB 1 TO 5
```

```
FYMAIN 420000 MEMB 1 TO 5
TRACK 1 ALL
CLB 0.05 MEMB 1 TO 5
CLT 0.05 MEMB 1 TO 5
CLS 0.05 ALL
MAXMAIN 20 MEMB 1 TO 4
MAXMAIN 12 MEMB 5
MINMAIN 16 MEMB 1 TO 4
MINMAIN 12 MEMB 5
MINSEC 8 ALL
REINF 0 MEMB 1 TO 4
DESIGN COLUMN 1 TO 4
DESIGN BEAM 5
END CONCRETE DESIGN
FINISH
```

```
**************************************************
*                                                *
*           STAAD.Pro V8i SELECTseries6          *
*           Version  20.07.11.70                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    APR  4, 2020                *
*           Time=    12:22:25                    *
*                                                *
*      USER ID: lnt                              *
**************************************************
```

```
    1. STAAD SPACE
INPUT FILE: D:\APL\Assignments\13-Maaden-Crushing_n_Screening\Conv-3130-FDR-003 _004\Staad\3130-FDR-0.. .STD
    2. START JOB INFORMATION
    3. ENGINEER DATE 03-APR-20
    4. JOB NAME MMGP-SAUDI ARABIA
    5. JOB CLIENT MA'ADEN
    6. JOB NO O19045/900707
    7. JOB REV 0
    8. JOB PART TRESTLE TR2 FOUNDATION
    9. JOB REF FOUNDATIONS FOR CONVEYOR 3130-FDR-004
   10. ENGINEER NAME APL
   11. CHECKER NAME SBY
   12. APPROVED NAME PRI
   13. END JOB INFORMATION
   14. INPUT WIDTH 79
   15. UNIT METER KN
   16. JOINT COORDINATES
   17. 1 0 -3 0; 2 0 -3 2; 3 0 -1 0; 4 0 -1 2; 5 0 -0.65 0; 6 0 -0.65 2
   18. 14 -0.75 -3 2.75; 17 -0.5 -3 2.75; 19 -0.25 -3 2.75; 20 0 -3 2.75
   19. 22 0.25 -3 2.75; 24 0.5 -3 2.75; 26 0.75 -3 2.75; 28 -0.75 -3 2.5
   20. 30 -0.5 -3 2.5; 31 -0.25 -3 2.5; 32 0 -3 2.5; 33 0.25 -3 2.5; 34 0.5 -3 2.5
   21. 35 0.75 -3 2.5; 37 -0.75 -3 2.25; 39 -0.5 -3 2.25; 40 -0.25 -3 2.25
   22. 41 0 -3 2.25; 42 0.25 -3 2.25; 43 0.5 -3 2.25; 44 0.75 -3 2.25; 46 -0.75 -3 2
   23. 48 -0.5 -3 2; 49 -0.25 -3 2; 50 0.25 -3 2; 51 0.5 -3 2; 52 0.75 -3 2
   24. 54 -0.75 -3 1.75; 56 -0.5 -3 1.75; 57 -0.25 -3 1.75; 58 0 -3 1.75
   25. 59 0.25 -3 1.75; 60 0.5 -3 1.75; 61 0.75 -3 1.75; 63 -0.75 -3 1.5
   26. 65 -0.5 -3 1.5; 66 -0.25 -3 1.5; 67 0 -3 1.5; 68 0.25 -3 1.5; 69 0.5 -3 1.5
   27. 70 0.75 -3 1.5; 72 -0.75 -3 1.25; 74 -0.5 -3 1.25; 75 -0.25 -3 1.25
   28. 76 0 -3 1.25; 77 0.25 -3 1.25; 78 0.5 -3 1.25; 79 0.75 -3 1.25; 81 -0.75 -3 1
   29. 83 -0.5 -3 1; 84 -0.25 -3 1; 85 0 -3 1; 86 0.25 -3 1; 87 0.5 -3 1
   30. 88 0.75 -3 1; 90 -0.75 -3 0.75; 92 -0.5 -3 0.75; 93 -0.25 -3 0.75
   31. 94 0 -3 0.75; 95 0.25 -3 0.75; 96 0.5 -3 0.75; 97 0.75 -3 0.75
   32. 99 -0.75 -3 0.5; 101 -0.5 -3 0.5; 102 -0.25 -3 0.5; 103 0 -3 0.5
   33. 104 0.25 -3 0.5; 105 0.5 -3 0.5; 106 0.75 -3 0.5; 108 -0.75 -3 0.25
   34. 110 -0.5 -3 0.25; 111 -0.25 -3 0.25; 112 0 -3 0.25; 113 0.25 -3 0.25
   35. 114 0.5 -3 0.25; 115 0.75 -3 0.25; 117 -0.75 -3 0; 119 -0.5 -3 0
   36. 120 -0.25 -3 0; 121 0.25 -3 0; 122 0.5 -3 0; 123 0.75 -3 0; 125 -0.75 -3 -0.25
   37. 127 -0.5 -3 -0.25; 128 -0.25 -3 -0.25; 129 0 -3 -0.25; 130 0.25 -3 -0.25
   38. 131 0.5 -3 -0.25; 132 0.75 -3 -0.25; 134 -0.75 -3 -0.5; 136 -0.5 -3 -0.5
```

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39. 137 -0.25 -3 -0.5; 138 0 -3 -0.5; 139 0.25 -3 -0.5; 140 0.5 -3 -0.5
40. 141 0.75 -3 -0.5; 143 -0.75 -3 -0.75; 145 -0.5 -3 -0.75; 146 -0.25 -3 -0.75
41. 147 0 -3 -0.75; 148 0.25 -3 -0.75; 149 0.5 -3 -0.75; 150 0.75 -3 -0.75
42. 158 -0.75 -3 -1; 162 -0.5 -3 -1; 164 -0.25 -3 -1; 166 0 -3 -1; 168 0.25 -3 -1.
43. 170 0.5 -3 -1; 172 0.75 -3 -1; 178 -0.75 -3 3; 181 -0.5 -3 3; 183 -0.25 -3 3
44. 185 0 -3 3; 187 0.25 -3 3; 189 0.5 -3 3; 191 0.75 -3 3; 212 1.00033 -3 2.75
45. 213 1.00033 -3 2.5; 216 1.00033 -3 2.25; 218 1.00033 -3 2; 220 1.00033 -3 1.75
46. 222 1.00033 -3 1.5; 224 1.00033 -3 1.25; 226 1.00033 -3 1; 228 1.00033 -3 0.75
47. 230 1.00033 -3 0.5; 232 1.00033 -3 0.25; 234 1.00033 -3 0
48. 236 1.00033 -3 -0.25; 238 1.00033 -3 -0.5; 240 1.00033 -3 -0.75
49. 242 1.00033 -3 -1; 247 1.00033 -3 3; 260 -1.00033 -3 2.75; 261 -1.00033 -3 2.5
50. 263 -1.00033 -3 2.25; 265 -1.00033 -3 2; 267 -1.00033 -3 1.75
51. 269 -1.00033 -3 1.5; 271 -1.00033 -3 1.25; 273 -1.00033 -3 1
52. 275 -1.00033 -3 0.75; 277 -1.00033 -3 0.5; 279 -1.00033 -3 0.25
53. 281 -1.00033 -3 0; 283 -1.00033 -3 -0.25; 285 -1.00033 -3 -0.5
54. 287 -1.00033 -3 -0.75; 289 -1.00033 -3 -1; 295 -1.00033 -3 3
55. MEMBER INCIDENCES
56. 1 1 3; 2 3 5; 3 2 4; 4 4 6; 5 3 4
57. ELEMENT INCIDENCES SHELL
58. 15 14 17 30 28; 16 17 19 31 30; 17 19 20 32 31; 18 20 22 33 32; 19 22 24 34 33
59. 20 24 26 35 34; 23 28 30 39 37; 24 30 31 40 39; 25 31 32 41 40; 26 32 33 42 41
60. 27 33 34 43 42; 28 34 35 44 43; 31 37 39 48 46; 32 39 40 49 48; 33 40 41 2 49
61. 34 41 42 50 2; 35 42 43 51 50; 36 43 44 52 51; 39 46 48 56 54; 40 48 49 57 56
62. 41 49 2 58 57; 42 2 50 59 58; 43 50 51 60 59; 44 51 52 61 60; 47 54 56 65 63
63. 48 56 57 66 65; 49 57 58 67 66; 50 58 59 68 67; 51 59 60 69 68; 52 60 61 70 69
64. 55 63 65 74 72; 56 65 66 75 74; 57 66 67 76 75; 58 67 68 77 76; 59 68 69 78 77
65. 60 69 70 79 78; 63 72 74 83 81; 64 74 75 84 83; 65 75 76 85 84; 66 76 77 86 85
66. 67 77 78 87 86; 68 78 79 88 87; 71 81 83 92 90; 72 83 84 93 92; 73 84 85 94 93
67. 74 85 86 95 94; 75 86 87 96 95; 76 87 88 97 96; 79 90 92 101 99
68. 80 92 93 102 101; 81 93 94 103 102; 82 94 95 104 103; 83 95 96 105 104
69. 84 96 97 106 105; 87 99 101 110 108; 88 101 102 111 110; 89 102 103 112 111
70. 90 103 104 113 112; 91 104 105 114 113; 92 105 106 115 114; 95 108 110 119 117
71. 96 110 111 120 119; 97 111 112 1 120; 98 112 113 121 1; 99 113 114 122 121
72. 100 114 115 123 122; 103 117 119 127 125; 104 119 120 128 127
73. 105 120 1 129 128; 106 1 121 130 129; 107 121 122 131 130; 108 122 123 132 131
74. 111 125 127 136 134; 112 127 128 137 136; 113 128 129 138 137
75. 114 129 130 139 138; 115 130 131 140 139; 116 131 132 141 140
76. 119 134 136 145 143; 120 136 137 146 145; 121 137 138 147 146
77. 122 138 139 148 147; 123 139 140 149 148; 124 140 141 150 149
78. 137 143 145 162 158; 140 145 146 164 162; 143 146 147 166 164
79. 146 147 148 168 166; 149 148 149 170 168; 152 149 150 172 170
80. 163 178 181 17 14; 166 181 183 19 17; 169 183 185 20 19; 172 185 187 22 20
81. 175 187 189 24 22; 178 189 191 26 24; 214 26 212 213 35; 217 35 213 216 44
82. 220 44 216 218 52; 223 52 218 220 61; 226 61 220 222 70; 229 70 222 224 79
83. 232 79 224 226 88; 235 88 226 228 97; 238 97 228 230 106; 241 106 230 232 115
84. 244 115 232 234 123; 247 123 234 236 132; 250 132 236 238 141
85. 253 141 238 240 150; 256 150 240 242 172; 265 191 247 212 26
86. 282 260 14 28 261; 285 261 28 37 263; 288 263 37 46 265; 291 265 46 54 267
87. 294 267 54 63 269; 297 269 63 72 271; 300 271 72 81 273; 303 273 81 90 275
88. 306 275 90 99 277; 309 277 99 108 279; 312 279 108 117 281
89. 315 281 117 125 283; 318 283 125 134 285; 321 285 134 143 287
90. 324 287 143 158 289; 333 295 178 14 260
91. START GROUP DEFINITION
92. ELEMENT
93. _MX 15 16 19 20 23 24 27 28 31 32 35 36 39 40 43 44 47 48 51 52 55 56 59 60 -
94. 63 64 67 68 71 72 75 76 79 80 83 84 87 88 91 92 95 96 99 100 103 104 107 -
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 95. 108 111 112 115 116 119 120 123 124 137 140 149 152 163 166 175 178 214 217 -
 96. 220 223 226 229 232 235 238 241 244 247 250 253 256 265 282 285 288 291 294 -
 97. 297 300 303 306 309 312 315 318 321 324 333
 98. _MY 15 TO 20 23 TO 28 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 TO 84 87 TO 92 -
 99. 111 TO 116 119 TO 124 137 140 143 146 149 152 163 166 169 172 175 178 214 -
100. 217 226 229 232 235 238 241 250 253 256 265 282 285 294 297 300 303 306 309 -
101. 318 321 324 333
102. _SHEAR 63 TO 68 71 TO 76 137 140 143 146 149 152 163 166 169 172 175 178 214 -
103. 217 220 223 226 229 232 235 238 241 244 247 250 253 256 265 282 285 288 291 -
104. 294 297 300 303 306 309 312 315 318 321 324 333
105. END GROUP DEFINITION
106. ELEMENT PROPERTY
107. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
108. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 140 143 146 -
109. 149 152 163 166 169 172 175 178 214 217 220 223 226 229 232 235 238 241 244 -
110. 247 250 253 256 265 282 285 288 291 294 297 300 303 306 309 312 315 318 321 -
111. 324 333 THICKNESS 0.4
112. DEFINE MATERIAL START
113. ISOTROPIC CONCRETE
114. E 2.17185E+007
115. POISSON 0.17
116. DENSITY 23.5616
117. ALPHA 1E-005
118. DAMP 0.05
119. TYPE CONCRETE
120. STRENGTH FCU 27579
121. END DEFINE MATERIAL
122. MEMBER PROPERTY AMERICAN
123. 1 TO 4 PRIS YD 0.6 ZD 0.6
124. 5 PRIS YD 0.5 ZD 0.3
125. CONSTANTS
126. MATERIAL CONCRETE ALL
127. ****************************************
128. SUPPORTS
129. 1 2 14 17 19 20 22 24 26 28 30 TO 35 37 39 TO 44 46 48 TO 52 54 56 TO 61 63 -
130. 65 TO 70 72 74 TO 79 81 83 TO 88 90 92 TO 97 99 101 TO 106 108 110 TO 115 -
131. 117 119 TO 123 125 127 TO 132 134 136 TO 141 143 145 TO 150 158 162 164 166 -
132. 168 170 172 178 181 183 185 187 189 191 212 213 216 218 220 222 224 226 228 -
133. 230 232 234 236 238 240 242 247 260 261 263 265 267 269 271 273 275 277 279 -
134. 281 283 285 287 289 295 ELASTIC MAT DIRECT Y SUBGRADE 72000 PRINT
```

JOINT	INFLUENCE AREA
1	0.062500
2	0.062500
14	0.062541
17	0.062500
19	0.062500
20	0.062500
22	0.062500
24	0.062500
26	0.062541
28	0.062541
30	0.062500
31	0.062500

32	0.062500
33	0.062500
34	0.062500
35	0.062541
37	0.062541
39	0.062500
40	0.062500
41	0.062500
42	0.062500
43	0.062500
44	0.062541
46	0.062541
48	0.062500
49	0.062500
50	0.062500
51	0.062500
52	0.062541
54	0.062541
56	0.062500
57	0.062500
58	0.062500
59	0.062500
60	0.062500
61	0.062541
63	0.062541
65	0.062500
66	0.062500
67	0.062500
68	0.062500
69	0.062500
70	0.062541
72	0.062541
74	0.062500
75	0.062500
76	0.062500
77	0.062500
78	0.062500
79	0.062541
81	0.062541
83	0.062500
84	0.062500
85	0.062500
86	0.062500
87	0.062500
88	0.062541
90	0.062541
92	0.062500
93	0.062500
94	0.062500
95	0.062500
96	0.062500
97	0.062541
99	0.062541
101	0.062500
102	0.062500
103	0.062500

104	0.062500
105	0.062500
106	0.062541
108	0.062541
110	0.062500
111	0.062500
112	0.062500
113	0.062500
114	0.062500
115	0.062541
117	0.062541
119	0.062500
120	0.062500
121	0.062500
122	0.062500
123	0.062541
125	0.062541
127	0.062500
128	0.062500
129	0.062500
130	0.062500
131	0.062500
132	0.062541
134	0.062541
136	0.062500
137	0.062500
138	0.062500
139	0.062500
140	0.062500
141	0.062541
143	0.062541
145	0.062500
146	0.062500
147	0.062500
148	0.062500
149	0.062500
150	0.062541
158	0.031271
162	0.031250
164	0.031250
166	0.031250
168	0.031250
170	0.031250
172	0.031271
178	0.031271
181	0.031250
183	0.031250
185	0.031250
187	0.031250
189	0.031250
191	0.031271
212	0.031291
213	0.031291
216	0.031291
218	0.031291
220	0.031291

222	0.031291
224	0.031291
226	0.031291
228	0.031291
230	0.031291
232	0.031291
234	0.031291
236	0.031291
238	0.031291
240	0.031291
242	0.015646
247	0.015646
260	0.031291
261	0.031291
263	0.031291
265	0.031291
267	0.031291
269	0.031291
271	0.031291
273	0.031291
275	0.031291
277	0.031291
279	0.031291
281	0.031291
283	0.031291
285	0.031291
287	0.031291
289	0.015646
295	0.015646
TOTAL	8.002645

```
135. ****************************************
136. MEMBER OFFSET
137. 5 START 0 -0.25 0
138. 5 END 0 -0.25 0
139. LOAD 1 DEAD LOAD DL
140. SELFWEIGHT Y -1
141. JOINT LOAD
142. 5 FX -0.289 FY -70.074 FZ -2.545
143. 6 FX -0.339 FY -57.696 FZ 9.545
144. ELEMENT LOAD
145. 15 TO 20 23 TO 28 31 TO 36 39 TO 44 47 TO 52 55 TO 60 63 TO 68 71 TO 76 79 -
146. 80 TO 84 87 TO 92 95 TO 100 103 TO 108 111 TO 116 119 TO 124 137 140 143 146 -
147. 149 152 163 166 169 172 175 178 214 217 220 223 226 229 232 235 238 241 244 -
148. 247 250 253 256 265 282 285 288 291 294 297 300 303 306 309 312 315 318 321 -
149. 324 333 PR GY -28.8
150. ****************************************
151. LOAD 2 LIVE LOAD LL
152. JOINT LOAD
153. 5 FX -0.217 FY -32.1 FZ -2.458
154. 6 FX -0.228 FY -31.9 FZ 2.458
155. ****************************************
156. LOAD 3 MATERIAL LOAD LL
157. JOINT LOAD
158. 5 FX -0.134 FY -25.6 FZ -1.549
159. 6 FX -0.145 FY -25.4 FZ 1.549
160. ****************************************
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LOAD 4 WIND LOAD ALONG CONVEYOR WL (+Z)
JOINT LOAD
5 FX 0.033 FY 46.654 FZ 13.252
6 FX -0.033 FY -46.654 FZ 13.523
*****************************************
LOAD 5 WIND LOAD ALONG CONVEYOR WL (-Z)
JOINT LOAD
5 FX -0.033 FY -47.674 FZ -14.104
6 FX 0.033 FY 47.674 FZ -14.388
*****************************************
LOAD 6 WIND LOAD ACROSS THE CONV.  WL (+X)
JOINT LOAD
5 FX 0.562 FY 0 FZ 0
6 FX 0.562 FY 0 FZ 0
*****************************************
LOAD 7 WIND LOAD ACROSS THE CONV.  WL (-X)
JOINT LOAD
5 FX -0.562 FY 0 FZ 0
6 FX -0.562 FY 0 FZ 0
*****************************************
LOAD 8 TEMP LOAD   TL1
JOINT LOAD
5 FX 0.028 FY 0.5 FZ -6.391
6 FX -0.028 FY -0.5 FZ 6.391
*****************************************
LOAD 9 TEMP LOAD   TL2
JOINT LOAD
5 FX -0.028 FY -0.5 FZ 6.391
6 FX 0.028 FY 0.5 FZ -6.391
*****************************************
LOAD 10 SEISMIC LOAD ALONG THE CONV. SL (+X)
JOINT LOAD
5 FX 0.061 FY -0.5 FZ -0.035
6 FX 0.061 FY -0.5 FZ 0.035
*****************************************
LOAD 11 SEISMIC LOAD ACROSS THE CONV. SL (+Z)
JOINT LOAD
5 FX 0 FY 8.882 FZ 2.661
6 FX 0 FY -8.882 FZ 2.755
*****************************************
*****LOAD COMBINATION FOR BEARING CAPACITY CHECK
LOAD 301 1.0 DL
REPEAT LOAD
1 1.0
LOAD 302 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 8 1.0 2 1.0 3 1.0
LOAD 303 1.0 (DL + TL1) + 1.0  (LL+ML)
REPEAT LOAD
1 1.0 9 1.0 2 1.0 3 1.0
LOAD 304 1.0 DL+ 0.75 WL(+Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 4 0.75 2 0.75 3 0.75
LOAD 305 1.0 DL+ 0.75 WL(-Z) + 0.75  (LL+ML)
REPEAT LOAD
1 1.0 5 0.75 2 0.75 3 0.75
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217. LOAD 306 1.0 DL+ 0.75 WL(+X) + 0.75  (LL+ML)
218. REPEAT LOAD
219. 1 1.0 6 0.75 2 0.75 3 0.75
220. LOAD 307 1.0 DL+ 0.75 WL(-X) + 0.75  (LL+ML)
221. REPEAT LOAD
222. 1 1.0 7 0.75 2 0.75 3 0.75
223. LOAD 308 1.0 DL + 0.525 EL(+X) + 0.75  (LL+ML)
224. REPEAT LOAD
225. 1 1.0 10 0.525 2 0.75 3 0.75
226. LOAD 309 1.0 DL + 0.525 EL(-X) + 0.75  (LL+ML)
227. REPEAT LOAD
228. 1 1.0 10 -0.525 2 0.75 3 0.75
229. LOAD 310 1.0 DL + 0.525 EL(+Z) + 0.75  (LL+ML)
230. REPEAT LOAD
231. 1 1.0 11 0.525 2 0.75 3 0.75
232. LOAD 311 1.0 DL + 0.525 EL(-Z) + 0.75  (LL+ML)
233. REPEAT LOAD
234. 1 1.0 11 -0.525 2 0.75 3 0.75
235. LOAD 312 0.6 DL + 1.0 WL(+Z)
236. REPEAT LOAD
237. 1 0.6 4 1.0
238. LOAD 313 0.6 DL + 1.0 WL(-Z)
239. REPEAT LOAD
240. 1 0.6 5 1.0
241. LOAD 314 0.6 DL + 1.0 WL(+X)
242. REPEAT LOAD
243. 1 0.6 6 1.0
244. LOAD 315 0.6 DL + 1.0 WL(-X)
245. REPEAT LOAD
246. 1 0.6 7 1.0
247. LOAD 316 0.6 DL + 0.7 EL(+X)
248. REPEAT LOAD
249. 1 0.6 10 0.7
250. LOAD 317 0.6 DL + 0.7 EL(-X)
251. REPEAT LOAD
252. 1 0.6 10 -0.7
253. LOAD 318 0.6 DL + 0.7 EL(+Z)
254. REPEAT LOAD
255. 1 0.6 11 0.7
256. LOAD 319 0.6 DL + 0.7 EL(-Z)
257. REPEAT LOAD
258. 1 0.6 11 -0.7
259. ****************************************
260. *****LOAD COMBINATION FOR FOUNDATION DESIGN
261. LOAD 401 1.4 DL
262. REPEAT LOAD
263. 1 1.4
264. LOAD 402 1.4 (DL + TL1) + 1.7  (LL+ML)
265. REPEAT LOAD
266. 1 1.4 8 1.4 2 1.7 3 1.7
267. LOAD 403 1.4 (DL + TL1) + 1.7  (LL+ML)
268. REPEAT LOAD
269. 1 1.4 9 1.4 2 1.7 3 1.7
270. LOAD 404 1.2 DL+ 1.6 WL(+Z) + 0.5  (LL+ML)
271. REPEAT LOAD
272. 1 1.2 4 1.6 2 0.5 3 0.5
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273. LOAD 405 1.2 DL+ 1.6 WL(-Z) + 0.5  (LL+ML)
274. REPEAT LOAD
275. 1 1.2 5 1.6 2 0.5 3 0.5
276. LOAD 406 1.2 DL+ 1.6 WL(+X) + 0.5  (LL+ML)
277. REPEAT LOAD
278. 1 1.2 6 1.6 2 0.5 3 0.5
279. LOAD 407 1.2 DL+ 1.6 WL(-X) + 0.5  (LL+ML)
280. REPEAT LOAD
281. 1 1.2 7 1.6 2 0.5 3 0.5
282. LOAD 408 1.2 DL + 1.0 EL(+X) + 0.5  (LL+ML)
283. REPEAT LOAD
284. 1 1.2 10 1.0 2 0.5 3 0.5
285. LOAD 409 1.2 DL + 1.0 EL(-X) + 0.5  (LL+ML)
286. REPEAT LOAD
287. 1 1.2 10 -1.0 2 0.5 3 0.5
288. LOAD 410 1.2 DL + 1.0 EL(+Z) + 0.5  (LL+ML)
289. REPEAT LOAD
290. 1 1.2 11 1.0 2 0.5 3 0.5
291. LOAD 411 1.2 DL + 1.0 EL(-Z) + 0.5  (LL+ML)
292. REPEAT LOAD
293. 1 1.2 11 -1.0 2 0.5 3 0.5
294. LOAD 412 0.9 DL + 1.6 WL(+Z)
295. REPEAT LOAD
296. 1 0.9 4 1.6
297. LOAD 413 0.9 DL + 1.6 WL(-Z)
298. REPEAT LOAD
299. 1 0.9 5 1.6
300. LOAD 414 0.9 DL + 1.6 WL(+X)
301. REPEAT LOAD
302. 1 0.9 6 1.6
303. LOAD 415 0.9 DL + 1.6 WL(-X)
304. REPEAT LOAD
305. 1 0.9 7 1.6
306. LOAD 416 0.9 DL + 1.0 EL(+X)
307. REPEAT LOAD
308. 1 0.9 10 1.0
309. LOAD 417 0.9 DL + 1.0 EL(-X)
310. REPEAT LOAD
311. 1 0.9 10 -1.0
312. LOAD 418 0.9 DL + 1.0 EL(+Z)
313. REPEAT LOAD
314. 1 0.9 11 1.0
315. LOAD 419 0.9 DL + 1.0 EL(-Z)
316. REPEAT LOAD
317. 1 0.9 11 -1.0
318. ****************************************
319. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**157**	**NUMBER OF MEMBERS**	**5**
NUMBER OF PLATES	**128**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**153**

SOLVER USED IS THE IN-CORE ADVANCED MATH SOLVER

TOTAL PRIMARY LOAD CASES = 49, TOTAL DEGREES OF FREEDOM = 483
TOTAL LOAD COMBINATION CASES = 0 SO FAR.

320. **
321. LOAD LIST 401 TO 419
322. START CONCRETE DESIGN

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323. CODE ACI 2008
324. FC 35000 MEMB 1 TO 5
325. FYMAIN 420000 MEMB 1 TO 5
326. TRACK 1 ALL
327. CLB 0.05 MEMB 1 TO 5
328. CLT 0.05 MEMB 1 TO 5
329. CLS 0.05 ALL
330. MAXMAIN 20 MEMB 1 TO 4
331. MAXMAIN 12 MEMB 5
332. MINMAIN 16 MEMB 1 TO 4
333. MINMAIN 12 MEMB 5
334. MINSEC 8 ALL
335. REINF 0 MEMB 1 TO 4
336. DESIGN COLUMN 1 TO 4
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COLUMN NO. 1 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	155.91	0.23	0.00072

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	155.91	10.14	0.03252

==========

COLUMN NO. 2 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	150.93	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	150.93	0.00	0.00000

==========

COLUMN NO. 3 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	147.30	0.25	0.00084

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	147.30	2.63	0.00892

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COLUMN NO. 4 DESIGN PER ACI 318-08 - AXIAL + BENDING

FY - 420.0 FC - 35.0 MPA, SQRE SIZE - 600.0 X 600.0 MMS, TIED
ONLY MINIMUM STEEL IS REQUIRED.
AREA OF STEEL REQUIRED = 3600.0 SQ. MM

BAR CONFIGURATION	REINF PCT.	LOAD	LOCATION	PHI
12 - 20 MM	1.047	401	END	0.650

(PROVIDE EQUAL NUMBER OF BARS ON EACH FACE)
TIE BAR NUMBER 8 SPACING 320.00 MM

COLUMN INTERACTION: MOMENT ABOUT Z -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	124.27	0.00	0.00000

COLUMN INTERACTION: MOMENT ABOUT Y -AXIS (KN-MET)

P0	Pn max	P-bal.	M-bal.	e-bal. (MM)
12181.21	9744.97	4499.60	1041.49	231.5
M0	P-tens.	Des.Pn	Des.Mn	e/h
411.95	-1583.37	124.27	0.00	0.00000

**********END OF COLUMN DESIGN RESULTS**********

337. DESIGN BEAM 5

BEAM NO. 5 DESIGN RESULTS - FLEXURE PER CODE ACI 318-08

LEN - 2000. MM FY - 420. FC - 35. MPA, SIZE - 300. X 500. MMS

LEVEL	HEIGHT (MM)	BAR INFO	FROM (MM)	TO (MM)	ANCHOR STA	ANCHOR END
1	64.	2 - 12MM	0.	2000.	YES	YES

CRITICAL POS MOMENT= 23.16 KN-MET AT 0.MM, LOAD 404
REQD STEEL= 188.MM2, RHO=0.0014, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 172. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 24968.6 cm^4

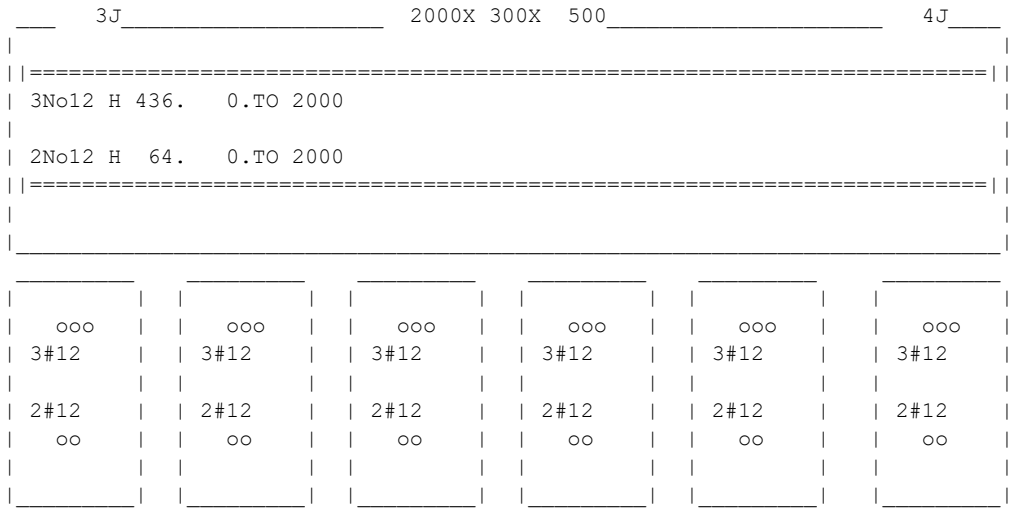
2	436.	3 - 12MM	0.	2000.	YES	YES

CRITICAL NEG MOMENT= 28.27 KN-MET AT 2000.MM, LOAD 404
REQD STEEL= 230.MM2, RHO=0.0018, RHOMX=0.0249 RHOMN=0.0035
MAX/MIN/ACTUAL BAR SPACING= 230./ 37./ 86. MMS
REQD. DEVELOPMENT LENGTH = 308. MMS

Cracked Moment of Inertia Iz at above location = 35784.0 cm^4

B E A M N O. 5 D E S I G N R E S U L T S - SHEAR

AT START SUPPORT - Vu= 3.06 KNS Vc= 126.45 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.
AT END SUPPORT - Vu= 8.67 KNS Vc= 126.45 KNS Vs= 0.00 KNS
Tu= 0.01 KN-MET Tc= 5.1 KN-MET Ts= 0.0 KN-MET LOAD 401
STIRRUPS ARE NOT REQUIRED.



**********END OF BEAM DESIGN**********

338. END CONCRETE DESIGN
339. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= APR 4,2020 TIME= 12:22:29 ****

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*   For technical assistance on STAAD.Pro, please visit    *
*   http://selectservices.bentley.com/en-US/               *
*                                                          *
*   Details about additional assistance from               *
*   Bentley and Partners can be found at program menu      *
*   Help->Technical Support                                *
*                                                          *
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*                http://www.bentley.com                    *
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